# Kinetic Theory of the Thermal Farley-Buneman Instability in the E-Region Ionosphere

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**Key Points:**

- A fully kinetic linear theory of the thermal Farley-Buneman instability (TFBI) in the E-region ionosphere has been developed.
- For the first time in the E-region instability research, the driving electric field has been included in the kinetic description of ions.
- The analytical results can enable observers to more accurately interpret observations of electrojet turbulence in the lower E region.

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**Abstract**

This paper develops a fully kinetic linear theory of the thermal Farley-Buneman instability (TFBI) in the Earth's E-region ionosphere assuming unmagnetized ions. The TFBI combines the Farley-Buneman instability (FBI), ion thermal instability (ITI), and electron thermal instability (ETI). Similar collision-dominated plasma instabilities can also occur in stellar and other planetary ionospheres. This new theory adds a kinetic description of ions that includes a driving electric field, resulting in the automatic inclusion of the ITI. This analytic theory has produced a comprehensive linear wave dispersion relation. It is remarkable that – similarly to the oversimplified earlier ion-kinetic studies – this much more general kinetic dispersion relation involves only elementary functions and the standard plasma dispersion function (albeit of several different arguments). This new theory is limited to plasma waves with the frequencies of the order, or larger than, the ion-neutral collision frequency. This kinetic frequency range allows for accurate interpretations of radar signals scattered from relatively high E-region altitudes, but at altitudes where ions are unmagnetized (mostly, below 110 km).

## 1 Introduction

In the lower E-region ionosphere, mostly at the magnetic equator and high latitudes, DC electric fields $\mathbf{E}_0$ perpendicular to the geomagnetic field $\mathbf{B}$ create intense currents called electrojets. When these fields reach sufficient strength, they initiate low-frequency cross-field plasma instabilities that develop into plasma turbulence. Radars, such as the Jicamarca Radio Observatory (JRO, Peru) at the magnetic equator (Hysell & Chau, 2002; Hysell et al., 2007), EISCAT (Schlegel et al., 1990) and ICEBEAR (Ivarsen et al., 2024) at high latitudes, and many others frequently detect this plasma turbulence. At high latitudes, during big geomagnetic storms or under other circumstances, the driving field magnitude, $E_0$, may reach hundreds of mV/m, which is many times the typical instability threshold. The stronger the driving field, the more intense the observed plasma turbulence and its macroscopic effects [about the latter, see Dimant and Oppenheim (2011); Wiltberger et al. (2017); St-Maurice and Goodwin (2021); Liang et al. (2021) and references therein]. The analysis of this paper mostly applies to these strong electric field conditions.

These electrojet instabilities occur in the lower E region and the top D region (Blix et al., 1996), mostly within the altitude range between 70 and 130 km, where electrons

are highly magnetized, while ions are fully or partially unmagnetized due to frequent ion-neutral collisions. Under these conditions, the strong driving field $\mathbf{E}_0$ causes a fast $\mathbf{E}_0 \times \mathbf{B}$ drift of electrons, while ions travel mostly with the surrounding neutral gas. This creates the two-stream condition crucial for driving the instability.

The most well-known electrojet instabilities are the Farley-Buneman instability (FBI) (Farley, 1963b; Buneman, 1963) and the gradient drift instability (GDI) (Simon, 1963; Maeda et al., 1963; Hoh, 1963). More recently, two more instabilities have been predicted theoretically: the electron thermal instability (ETI) (Dimant & Sudan, 1995b, 1997; Robinson, 1998; St.-Maurice & Kissack, 2000), with possible observational evidences (Blix et al., 1996; Tsunoda et al., 1997), and the ion thermal instability (ITI) (Kagan & Kelley, 2000; Dimant & Oppenheim, 2004; Makarevich, 2021) (note that this terminology had not been established until recently). Signatures of these thermal instabilities have been also detected in fully kinetic particle-in-cell (PIC) simulations (Janhunen, 1994, 1995; M. Oppenheim et al., 1995, 1996; M. M. Oppenheim & Dimant, 2004; M. M. Oppenheim et al., 2008; M. M. Oppenheim & Dimant, 2013). All these instabilities differ by their driving mechanisms: the FBI is caused by the ion inertia, while the two thermal instabilities are caused by wave-modulated frictional heating of the corresponding plasma particles (electrons or ions). Such processes can also occur in the solar chromosphere, where conditions similar to the E-region ionosphere exist; for the corresponding literature, see M. M. Oppenheim et al. (2020); Dimant et al. (2023); Evans et al. (2023) and references therein. The two last papers, based on a multi-fluid theoretical approach, coined the term TFBI for the combination of the FBI, ETI, and ITI. This paper develops the linear kinetic theory of this combined instability.

Starting from the pioneering works by Farley (1963a, 1963b) and Buneman (1963), the linear theory of the FBI has been developed using fluid equations or kinetically (or using a hybrid approach when one plasma component was described using a fluid approach, while the other used a kinetic theory); for an early review on the theory, see Fejer et al. (1984); Farley (1985). The fluid approach is simpler, but the kinetic approach is more general and accurate. The validity of the fluid equations for both electrons and ions is restricted by the conditions of strong collisionality. Essentially, this means that the absolute value of the cyclic wave frequency, $\omega = \omega_r + i\gamma$, must be much less than the ion-neutral mean collision frequency, $\nu_{in}$ (here $\omega_r$ is the real frequency, while $\gamma$ is the instability growth or damping rate, depending on the sign). A proper theoretical description

of the weakly collisional, medium-to-short wavelengths, corresponding to frequencies $|\omega| \gtrsim \nu_{in}$, necessarily requires the kinetic theory.

The FBI-related collisional kinetic theory had been developed or employed by a number of researchers (Farley, 1963a; Lee et al., 1971; Schmidt & Gary, 1973; Ossakow et al., 1975; Schlegel & St.-Maurice, 1983; Basu, 1995, and others) using for both ions and electrons kinetic equations with the simplified BGK collisional model (Bhatnagar et al., 1954). None of these models, however, included the driving electric field, $\mathbf{E}_0$, in the kinetic description of ions. As a result, the employed ion kinetic theory could only describe the pure FBI and no thermal processes like the ITI could have been covered. The kinetic description of electrons included $\mathbf{E}_0$, but only through the $\mathbf{E}_0 \times \mathbf{B}$ drift velocity (this is usually sufficient for the pure FBI description).

For electrons, a simplified kinetic equation with the BGK collision term is not appropriate, as we discuss in Sec. 3 below. Dimant and Sudan (1995a) developed a consistent kinetic theory for the electron description of the FBI. The corresponding kinetic treatment had also automatically predicted the ETI (Dimant & Sudan, 1995b, 1995c). That theory, however, was restricted to the long-wavelength range, where a simpler fluid model is also approximately applicable (Dimant & Sudan, 1997).

There are a number of reasons why a theoretical description of the electrojet instabilities requires kinetic theory. First, this theory is the most general approach to describing any wave plasma processes, so that without such theory no description is complete. Second, for weakly collisional short-wavelength waves ($|\omega| \gtrsim \nu_{in}$), such purely kinetic factor as the collisionless Landau damping plays a crucial role for the wave dissipation. Unfortunately, due to a significant complexity, the analytic kinetic approach can be successfully developed mostly for the linear instability theory with limited applications. To determine full characteristics of developed wave turbulence, one needs the complete nonlinear theory (still looks unfeasible) or can use advanced fully kinetic simulations (M. M. Oppenheim & Dimant, 2004; M. M. Oppenheim et al., 2008; M. M. Oppenheim & Dimant, 2013; M. M. Oppenheim et al., 2020).

Fully kinetic PIC simulations usually include the entire kinetic physics and, automatically, all thermal-driven processes. At present time, this approach is the most powerful tool for modeling the electrojet instabilities, especially in the highly nonlinear stage of instability-driven plasma turbulence for which no comprehensive analytical descrip-

tion even exists. PIC simulations, however, also have limitations: the restricted box sizes, a higher noise level due to a limited number of particles, often the artificially increased electron-to-ion mass ratio. Besides, typical multi-processor, highly parallelized, PIC simulations require significant supercomputer resources and often a long computational time.

In general, it is unfeasible to fully understand nonlinear processes without proper understanding of the linear theory. Also, PIC simulations cannot provide parameter dependencies and proper scaling. To obtain those, one needs analytical theory. A primary benefit of linear theory is to reveal the ionospheric conditions where radars will observe plasma irregularities. However, the main goal of the linear kinetic theory developed in this paper is to help interpret radar signals, which are usually detected during the nonlinearly saturated stage of instability. Applying a simplified linear theory should help understand whether the detected radar signals were scattered from linearly stable or linearly unstable plasma waves. A primary benefit of linear theory is to reveal the ionospheric conditions where radars will observe plasma irregularities. Furthermore, for a given wavevector $\mathbf{k}$ determined by the radar frequency and the line-of-sight (LoS), the accurate linear theory will also determine the corresponding linear instability growth/damping rates and the wave phase velocities. The latter should help explain the observed Doppler frequency shifts. Of course, the linear theory alone cannot predict such energy-related wave characteristics as the amplitudes of the turbulent waves and the turbulence spectra, ion line widths, etc. Without nonlinear theory or simulations, the linear theory remains incomplete.

In order to properly interpret the radar signals, one may or may not need the kinetic theory. This depends upon a specific radar beam frequency and the LoS. If the detected backscattered wavelength, which equals the half of the transmitted radar wavelength along the given LoS, falls into the predominantly collisional long-wavelength range, $|\omega| \ll \nu_{in}$, then one may apply the much simpler fluid description. However, VHF radars can receive such signals only from relatively low altitudes (see below Figure 1 and the corresponding discussion); from higher altitudes, only HF radars operate at such low frequencies, but their waves experience appreciable refraction by the ambient plasma. If the detected wavelength is within the weakly collisional, medium-to-short-wavelength range of $|\omega| \gtrsim \nu_{in}$, then it necessarily requires the more general kinetic theory.

Furthermore, for typical VHF radars (e.g., 50 MHz, as in the JRO and, approximately, in the ICEBEAR), which detect $\simeq 3$ m ionospheric irregularities, scattering of the radar signal from different altitudes may require a different interpretation: one based on a fluid model if the signal arrives from lower altitudes and another based on the kinetic theory if the signal arrives from higher altitudes. The reason is that the collision frequencies, proportional to the neutral atmosphere density, exponentially decrease with increasing altitude. We may loosely define an interface between the two regimes as $|\omega| = \nu_{in}(h)$, where $h$ is the altitude. Given the $\omega(\mathbf{k})$-dependence [usually, $\omega(\mathbf{k}) \propto k \equiv |\mathbf{k}|$, with the proportionality coefficient varying between the $\mathbf{E}_0 \times \mathbf{B}$ drift speed and the ion-acoustic speed $C_s \simeq [(T_e + T_i)/m_i]^{1/2}$, where $T_{e,i}$ are the electron and ion temperatures and $m_i$ is the ion mass, see, e.g., Sahr and Fejer (1996)], we may relate the wave frequency $\omega$ to the radar frequency $\Omega_{\rm Radar}$. Figure 1 illustrates this by plotting the corresponding loosely defined curve $h(\Omega_{\rm Radar})$. Given $\Omega_{\rm Radar}$ and the corresponding vertical line, we find the corresponding intersection $h_0 = h(\Omega_{\rm Radar})$. For the altitudes well below $h_0$, one may, to a reasonable accuracy, use a fluid model, while for altitudes $h \gtrsim h_0$ one has to employ the more general, and hence much more accurate results of the kinetic theory, even if they are approximate. Rigorously speaking, the fluid model is only applicable at altitudes well below $h_0$; around this altitude (as well as above it), the accurate description requires kinetic theory.

Earlier works employing the ion kinetics ignored the driving field $\mathbf{E}_0$ and hence completely ignored any thermal processes. Dimant and Sudan (1995a, 1995b) explicitly included $\mathbf{E}_0$ in their kinetic description of electrons (although for ions they also used an isothermal fluid model). This had allowed those authors to predict the ETI (Dimant & Sudan, 1995c, 1997). However, Dimant and Sudan (1995a, 1995b) kinetic theory has only been developed for sufficiently long-wavelength waves, for which a fluid model was also applicable, albeit with less accurate results (Dimant & Sudan, 1997; Robinson, 1998; St.-Maurice & Kissack, 2000; Dimant et al., 2023).

Most probably, the main reason why no previous kinetic studies for ions explicitly included the driving electric field in their treatment was that including $\mathbf{E}_0$ in the kinetic theory creates significant mathematical difficulties. Indeed, without $\mathbf{E}_0$, the linear wave perturbations of the particle distribution function relate to the perturbations of the electrostatic potential, $\Phi_{\omega\mathbf{k}}$, by a simple algebraic expression. The resultant relation between the corresponding density perturbations and $\Phi_{\omega\mathbf{k}}$ can be expressed in terms of the well-

known plasma dispersion function $Z(x)$ (Fried & Conte, 1961). However, if $\mathbf{E}_0$ is explicitly included in the kinetic equations, then one obtains a much less tractable differential equation. Under reasonably justified approximations, the ion kinetic equation leads to a first-order differential equation, while the electron kinetic equation results in a second-order differential equation. The real problem, however, is not solving these differential equations, but rather is the integration of the solution over the corresponding particle velocities in order to obtain the density perturbations. For practical applications, it is of paramount importance to have the final dispersion relation in a form of an explicit functional expression, rather than in terms of tricky integrals.

In this paper, we have overcome these mathematical difficulties and succeeded in including the driving electric field into the kinetic description of ions in the E-region plasma instabilities. For electrons, the driving field was included in Dimant and Sudan (1995a, 1995b), but in this paper we have shifted that kinetic description into the short-wavelength range. Within the framework of this fully kinetic approach, we have derived the wave dispersion relation that describes the combined TFBI. Some derivations (mostly for electrons) have required approximations, but those are based on the actual physical conditions in the E-region ionosphere. It is remarkable that the resultant kinetic wave dispersion relation has a form similar to those in the previous studies: this relation includes only elementary functions combined with a few $Z$-functions of different arguments (for both ions and electrons).

The main restriction of this paper (as well as of the earlier ion kinetic studies) is that in this paper we have not included (yet) the finite ion magnetization. This limits our results to the conditions of $\Omega_i \ll \nu_{in}$, which are largely valid at altitudes below 110 km (here $\Omega_i$ is the ion gyrofrequency and $\nu_{in}$ is the ion-neutral collision frequency). Usually, this lower altitude range covers most of the turbulent E region, but some E-region plasma waves exist at higher altitudes as well. There are questions that still cannot be answered in the framework of this limited study. For instance, it is not clear yet how medium-to-short wavelength plasma waves behave near the ion magnetization boundary, $\Omega_i = \nu_{in}$. Above the ion magnetization boundary, where $\nu_{in} < \Omega_i$, in the long-wavelength limit the FBI destabilizing mechanism, i.e., the ion inertia, becomes, on the contrary, stabilizing, while the ITI thermal mechanism extends the unstable altitude range up to a few kilometers over the ion magnetization boundary (Dimant & Oppenheim, 2004). For the VHF and UHF radars, at those altitudes such waves are deeply within the kinetic

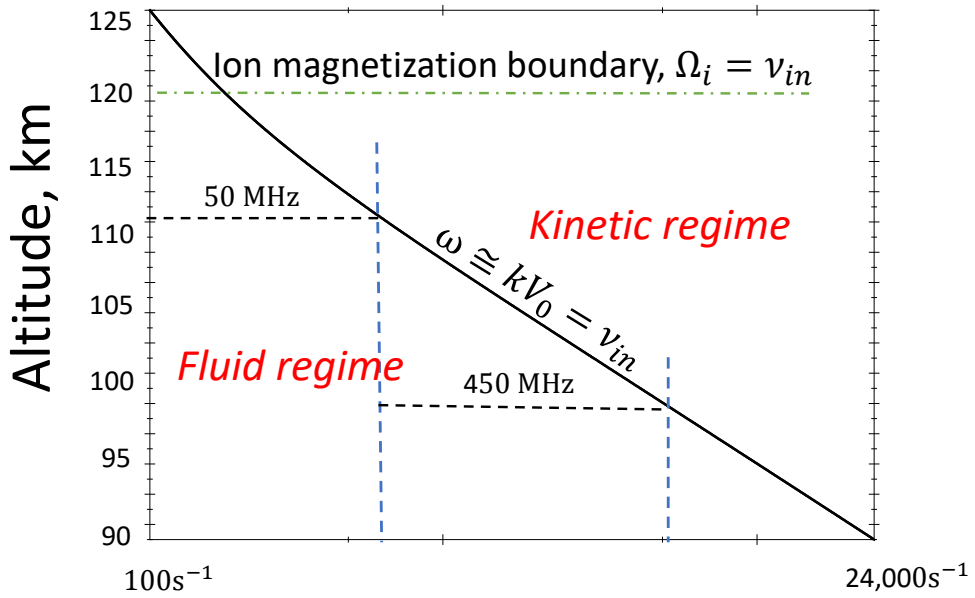



**Figure 1.** Schematic diagram illustrating the areas of applicability for two different theoretical approaches of the plasma wave description (fluid vs. kinetic). The solid curve corresponds to $\omega = \nu_{in}$, where the cyclic wave frequency, $\omega$, is shown on the horizontal axis (in the logarithmic scale) and the $i$-$n$ collision frequency, $\nu_{in}$, is shown implicitly on the vertical axis using a simplified model of the altitudinal dependence of $\nu_{in}$. The two dashed lines show loosely defined altitude boundaries for two different radars with the corresponding frequencies shown above each line. The fluid model is rigorously applicable only well below the solid curve, while around the curve and above it the more general kinetic model should be employed. In this paper, we still assume unmagnetized ions, so that the present kinetic description of ions is applicable well below the magnetization boundary (shown by the green dash-dotted horizontal line).

range, see Fig. 1, so it is important to understand the characteristics of the shorter-wavelength waves above the magnetization boundary. Inclusion of the finite ion magnetization in the collisional kinetic theory creates additional mathematical difficulties, which we hope will be overcome in the future. This would complete the full linear kinetic theory of the E-region plasma instabilities for the entire E region.

The theoretical analysis of this paper is quite lengthy and cumbersome, so that it is restricted only to analytically deriving the kinetic wave dispersion relation and dis-

cussing its limitations. We are planning to perform a specific analysis of this dispersion relation with application to radar observations in the near future.

The paper is organized as follows. Section 2 derives the kinetic relation between the linear wave perturbations of the ion particle density and the electrostatic potential. Section 3 derives a similar kinetic relation between the linear wave perturbations of the electron particle density and the electrostatic potential (unlike Dimant and Sudan (1995a, 1995b). The treatment of this paper is restricted to plasma waves with the frequency of the order of, or larger than, the ion-neutral collision frequency, so that the local collisional cooling can be neglected). Section 4 closes the theory and presents the fully kinetic wave dispersion relation. Section 5 discusses the major limitations of our theory. Section 6 concludes the paper. The appendices review a simplified analytic model for the background ion distribution function (Koontaweepunya et al., 2024) and present the most complicated mathematical details of our analytic theory.

## 2 Ion Contribution to the Wave Dispersion Relation

For the ion component of the wave dispersion relation, similarly to the previous ion kinetic studies, we will use the kinetic equation with the BGK collision operator:

$$\frac{\partial f_i}{\partial t} + \frac{q_i}{m_i}(\mathbf{E} + \mathbf{V}_i \times \mathbf{B}) \cdot \frac{\partial f_i}{\partial \mathbf{V}_i} + \mathbf{V}_i \cdot \frac{\partial f_i}{\partial \mathbf{r}} = -\nu_{in} \left( f_i - \frac{n_i(\mathbf{r},t)}{n_0} \; f_{i0}^{\mathrm{Coll}} \right). \tag{1}$$

This equation is written in the rest frame of reference for the neutral particles. Here $f_i(\mathbf{V}_i, \mathbf{r}, t)$ is the space ($\mathbf{r}$) and time ($t$) dependent ion velocity ($\mathbf{V}_i$) distribution; $\mathbf{E}$ and $\mathbf{B}$ are the total electric and magnetic fields, respectively; $m_i$ and $q_i$ are the mean ion mass and the electric charge, respectively (in the E-region ionosphere with the vastly predominant singly charged positive ions, $q_i = e$, where $e$ is the elementary charge); the ion-neutral collision frequency, $\nu_{in}$, is assumed constant (see below). Further, $n_i(\mathbf{r}, t)$ is the time and space-varying ion density, and $f_{i0}^{\mathrm{Coll}}(\mathbf{V}_i)$ is the Maxwellian velocity distribution normalized to the spatially uniform and constant background density of electrons and singly charged ions, $n_0 = n_{e0} = n_{i0}$,

$$f_{i0}^{\mathrm{Coll}}(\mathbf{V}_i) = n_0 \left( \frac{m_i}{2\pi T_n} \right)^{3/2} \exp\left( - \frac{m_i V_i^2}{2T_n} \right), \tag{2}$$

where $V_i = |\mathbf{V}_i|$ and $T_n$ is the equilibrium temperature of the neutral particles (here and across the entire text, all temperatures are given in the energy units). The two ion

distribution functions are normalized according to

$$n_i(\mathbf{r},t) = \int f_i(\mathbf{V}_i,\mathbf{r},t)d^3V_i, \qquad n_0 = \int f_{i0}^{\mathrm{Coll}}(\mathbf{V}_i)d^3V_i. \tag{3}$$

The left-hand side (LHS) of equation (1) represents the collisionless part of the ion kinetic equation, while the right-hand side (RHS) is the collisional term in its simplest BGK form.

The BGK collision term (Bhatnagar et al., 1954) does not follow from the accurate dual-collision integro-differential Boltzmann operator, but is rather an artificial construct convenient for analytical treatment. This model automatically provides thermal relaxation of arbitrarily unbalanced particles to their thermal equilibrium. The BGK model also conserves the local number of particles and reasonably well describes the balance of the average momentum and energy. In this paper, we set the dominant neutral particles ($N_2$ and $O_2$) and the major ions ($O_2^+$ and $NO^+$) to have approximately the same mass, $m_i \approx m_n$. In this approximation, the BGK model even provides accurate expressions for the mean ion frictional heating and cooling rates (Dimant, 2024). An important restriction of the BGK model is the assumption of constant, i.e., velocity independent, collision frequency $\nu_{in}$. For the E-region ions, this approximation called Maxwell molecule collisions (Schunk & Nagy, 2009) is valid if the ion energies are well below 1 eV. A more serious problem with the BGK collision model, however, is that it includes no angular scattering of ions through their collisions with the neutral gas particles. In reality, if the masses of the colliding particles are close to each other, the rates of the collisional kinetic redistribution of both the particle energies and velocity angles are comparable. Nevertheless, for the ion-neutral collisions [unlike the electron-neutral collisions (Dimant & Sudan, 1995a), see below], even the far-from-ideal BGK model is reasonably well applicable and still remains quite popular among researchers.

In this paper, we restrict our treatment to unmagnetized ions, such that we disregard the magnetic field $\mathbf{B}$. In the Earth's ionosphere, the conditions for such neglect are $\Omega_i \ll \nu_{in}$, where $\Omega_i = eB/m_i$ is the ion gyrofrequency. These conditions are reasonably well satisfied at altitudes below 110 km, see, e.g., Dimant and Oppenheim (2004, Fig. 2). Unlike the previous kinetic studies (Farley, 1963a; Lee et al., 1971; Schmidt & Gary, 1973; Ossakow et al., 1975; Schlegel & St.-Maurice, 1983), we include the background electric field, $\mathbf{E}_0$, in the ion kinetic equation. This inclusion will automatically

add ion thermal (i.e., heating and cooling) effects for plasma waves, and this is the central point of our ion-kinetic analysis.

### 2.1 The background ion velocity distribution

A strong driving electric field $\mathbf{E}_0$, perpendicular to the magnetic field $\mathbf{B}$, gives rise to plasma instabilities and, simultaneously, it modifies the background ion distribution function (BIDF), $f_{i0}(\mathbf{V}_i)$. If the driving field is strong, then it heats and shifts the entire ion velocity distribution $f_{i0}(\mathbf{V}_i)$ in such a way that $f_{i0}(\mathbf{V}_i)$ can deviate significantly from $f_0^{\rm Coll}(V_i)$ (Gurevich, 1978; Koontaweepunya et al., 2024). The earlier kinetic studies of the FBI have always assumed $f_{i0}(\mathbf{V}_i) = f_0^{\rm Coll}(V_i)$. More accurate instability theory requires better knowledge of the modified background ion distribution function.

Recently, using both an approximate analytic theory and more accurate PIC simulations, we studied the modification of the BIDF under the action of a strong external electric field (Koontaweepunya et al., 2024). For the analytic theory, we employed the above kinetic equation with the BGK collision term (1), which for the spatially uniform and stationary background with neglected magnetic field reduces to

$$\frac{e}{m_i}\,\mathbf{E}_0 \cdot \frac{\partial f_{i0}}{\partial \mathbf{V}_i} = -\,\nu_{in}\left(f_{i0} - \ f_{i0}^{\rm Coll}\right). \tag{4}$$

From this equation, it is clear that the imposed electric field $\mathbf{E}_0$ affects the BIDF only in the $\mathbf{E}_0$-direction (which we will denote the $V_{ix}$ axis), while along the two perpendicular components, $V_{iy}$ and $V_{iz}$, the velocity distribution contains the same Maxwellian form as does $f_{i0}^{\rm Coll}$. This is entirely due to the fact that the simple BGK collision term includes no angular redistribution of the colliding ions in the velocity space.

We briefly review the analytic theory based on equation (4) in Appendix A1, but for the wave treatment we approximate the actual ion background distribution function by a simpler bi-Maxwellian distribution

$$f_{i0}(\mathbf{V}_i) = \frac{1}{(2\pi)^{3/2}}\frac{n_0}{V_{Tx}V_{Ty}^2}\exp\left[-\frac{(V_{ix}-U_0)^2}{2V_{Tx}^2} - \frac{V_{iy}^2+V_{iz}^2}{2V_{Ty}^2}\right], \tag{5}$$

where $V_{Tx} = (T_x/m_i)^{1/2}$ and $V_{Ty} = (T_y/m_i)^{1/2} = (T_z/m_i)^{1/2}$, so that

$$\int_{-\infty}^{\infty} f_{i0}\,dV_{iz} = \frac{n_0}{2\pi V_{Tx}V_{Ty}}\exp\left[-\frac{(V_{ix}-U_0)^2}{2V_{Tx}^2} - \frac{V_{iy}^2}{2V_{Tx}^2}\right]. \tag{6}$$

Here $U_0 = eE_0/(m_i\nu_{in})$ is the magnitude of the total Pedersen drift in the $x$ direction. This approximate distribution function accurately describes the total background ion Pedersen drift and frictional heating, but fully ignores any distribution function skewness.

The BGK-based analytic model predicts this skewness, see Appendix A, Sec. A1, but this model neglects the angular redistribution of ions in the velocity space and, as a result, overestimates the BIDF skewness. We have observed the BIDF skewness in our PIC simulations (Koontaweepunya et al., 2024), albeit to a smaller degree than that predicted by the BGK-based model. The actual $V_{iz}$-integrated BIDF varies between those described by equations (A73) and (6) with no clear preference to the former. Even in spite of the $f_{i0}$-skewness absence, Eq. (6) is much more accurate than the conventional assumption of $f_{i0} = f_{i0}^{\rm Coll}$ (see Sec. 2.1). Note also that for the highly dynamic harmonic wave perturbations with $|\omega| \gtrsim \nu_{in}$, the effect of the angular scattering should be much less pronounced than it is for the quasi-stationary background ion distribution. This means that for the analytic studies of linear wave perturbations, see Sec. 2.2, the use of the simple BGK approximation is justified much better than that for the BIDF.

The quantities $V_{Tx}$ and $V_{Ty}$ are not totally free parameters because they are related by the general thermal balance. Indeed, the effective ion temperature for the BIDF given by equation (5) is $T_{\rm eff} = m_i(V_{Tx}^2 + 2V_{Ty}^2)$. On the other hand, using the general ion energy balance equation, see, e.g., equation (1c) in Dimant (2024) where $s = i$, $m_i = m_n$, $\delta_{in} = 1$, $M_{sn} = m_i/2$, and the LHS equals zero, for the same quantity we obtain $T_{\rm eff} = T_n + (1/3)m_i U_0^2$, where $U_0 = eE_0/(m_i\nu_{in})$ is the mean ion drift speed. If we also know the temperature anisotropy, $T_y/T_x = V_{Ty}^2/V_{Tx}^2$, then we can easily find the explicit value for each parameter $V_{Tx}$ or $V_{Ty}$. In this paper, however, we will use approximate equation (6) without specifying the ratio $V_{Ty}^2/V_{Tx}^2$, because the temperature anisotropy has a still unknown dependence on the electric field strength. We leave such analysis for future studies, but currently we include the temperature anisotropy as a free parameter and will investigate in future its possible effect on the linear instability characteristics, such as the growth/damping rates, wave phase velocity, and the instability threshold field.

### 2.2 Linear wave perturbations: the ion kinetics

To find the linear wave dispersion relation, we need to determine the wave perturbations of the ion and electron densities in terms of a wave perturbation of the corresponding electrostatic potential and then apply Poisson's equation. This will allow us to eliminate the electrostatic-potential perturbation and obtain the unique relation for the complex wave frequency in terms of the real wavevector. In this section, we deal only

with the ion density perturbation. To find the ion-density wave perturbation, we need to solve the corresponding linearized kinetic equation for the wave perturbation of the ion distribution function and integrate this perturbation over velocities.

Here we derive the kinetic relationship that relates the wave perturbations of the ion density, $n_i$, to the corresponding perturbations of the electrostatic potential, $\Phi$. In Appendix A, Sec. A2, we have reproduced the corresponding no-field ($\mathbf{E}_0 = 0$) relationship for unmagnetized ions (Lee et al., 1971; Schmidt & Gary, 1973) by using two slightly different approaches, see Sec. A21. The comparison of the two results have produced analytic formulas which become crucial for our general analytic treatment of ions, see equations (A41) [or (A50)] and (A57). The formulas derived provide exact analytic values for the integral convolutions of Gaussian functions with the error functions (or the corresponding $Z$-functions), whose complex arguments are, in general, different and arbitrary. Popular tables of integrals usually miss such formulas (Dwight, 1961; Jeffrey, 1995; Gradshteyn & Ryzhik, 2007) or express the results of such integrations in terms of more general special functions (Prudnikov et al., 1986) with no practical value for our treatment.

We start our linear theory from the full kinetic equation for totally unmagnetized ions with the BGK collision term,

$$\frac{\partial f_i}{\partial t} + \frac{e}{m_i}\,\mathbf{E}\cdot\frac{\partial f_i}{\partial \mathbf{V}_i} + \mathbf{V}_i\cdot\frac{\partial f_i}{\partial \mathbf{r}} = -\,\nu_{in}\left(f_i - \frac{n_i(\mathbf{r},t)}{n_0}\;f_{i0}^{\rm Coll}\right). \tag{7}$$

Using the standard approach, we will derive the relation between small linear perturbations of the ion density and the electrostatic potential for an isolated harmonic wave $\propto \exp[i(\mathbf{k}\cdot\mathbf{r} - \omega t)]$, where $\omega = \omega_r + i\gamma$ and $\mathbf{k}$ are the complex wave frequency and the real wavenumber, respectively. The real wave frequency, $\omega_r$, will always be assumed positive, while the wave growth/damping rate, $\gamma$, can be either positive (corresponding to the linear instability) or negative (corresponding to a stable situation when any finite initial wave perturbations will dampen and eventually disappear). From this point onward, such small harmonic wave perturbations of any scalar or vector quantity will be denoted by the subscript $\omega\mathbf{k}$.

For our further linear-wave analysis, it is useful to introduce dimensionless expressions for the harmonic wave perturbations of the ion particle density and the electrostatic potential, $\Phi_{\omega\mathbf{k}}$,

$$\phi_{\omega\mathbf{k}} \equiv \frac{e\Phi_{\omega\mathbf{k}}}{T_e}, \qquad \eta^i_{\omega\mathbf{k}} \equiv \frac{\int f^i_{\omega\mathbf{k}} d^3V_i}{n_0}, \tag{8}$$

where the electrostatic wave perturbations are normalized to the electron temperature, $T_e$, for definiteness (the notation $\phi_{\omega\mathbf{k}}$ will be used for both ions and electrons)

Now we start deriving the general kinetic relationship between $\phi_{\omega\mathbf{k}}$ and $\eta^i_{\omega\mathbf{k}}$ for the arbitrarily large external DC electric field $\mathbf{E}_0$. Linearizing the initial kinetic equation for ions (1), we obtain

$$(\mathbf{k}\cdot\mathbf{V}_i - \omega_i) f^i_{\omega\mathbf{k}} - i\mathbf{a}_{i0}\cdot\frac{\partial f^i_{\omega\mathbf{k}}}{\partial\mathbf{V}_i} = \frac{V^2_{Tn}}{\beta_T}\,\phi_{\omega\mathbf{k}}\left(\mathbf{k}\cdot\frac{\partial f_{i0}}{\partial\mathbf{V}_i}\right) - i\nu_{in} f^{\mathrm{Coll}}_{i0}\eta^i_{\omega\mathbf{k}}, \tag{9}$$

where $\mathbf{a}_{i0} = e\mathbf{E}_0/m_i$, $\beta_T = T_n/T_e$, $V^2_{Tn} = T_n/m_n$ ($m_n = m_i$), $\omega_i \equiv \omega + i\nu_{in}$, and the other notations were introduced by equations (2) and (A2). Note that, due to the normalization of $\phi_{\omega\mathbf{k}}$, in purely ion equation (9) there is no actual dependence on any electron parameters. There is also no actual dependence on the neutral temperature $T_n$. Instead of $T_n$, we might have used in $\beta_T$ any other constant temperature (with the corresponding change in the definition for the thermal velocity $V_{Tn}$), e.g., $T_x$ or $T_y$. We have picked $T_n$ because this temperature equals the equilibrium ion temperature in the total absence of an external electric field.

Unlike "classical" equation (A17) which represents a simple algebraic relationship between $f^i_{\omega\mathbf{k}}$, $\phi_{\omega\mathbf{k}}$, and $\eta^i_{\omega\mathbf{k}}$, equation (9) represents a linear differential equation. Finding its unique solution for $f^i_{\omega\mathbf{k}}(\mathbf{V}_i)$ and the corresponding normalized ion density, $\eta^i_{\omega\mathbf{k}} \propto \int f^i_{\omega\mathbf{k}} d^3V_i$, is the subject of this section with most of the mathematical details given in Appendix A, see Secs. A3 and A4.

In our treatment, since $\mathbf{a}_{i0} = e\mathbf{E}_0/m_i$ has only one directional component, we can reduce the complexity by introducing a coordinate system such that the wavevector $\mathbf{k}$ has only two components: $k_x$ and $k_y$, where the $x$-axis is directed along $\mathbf{a}_{i0}$. The absence of the third vector component, $k_z = 0$, suggests switching from $f_{\omega\mathbf{k}}(\mathbf{V}_i)$ to a two-dimensional ion distribution function,

$$F_{\omega\mathbf{k}}(V_{ix}, V_{iy}) \equiv \int_{-\infty}^{\infty} f^i_{\omega\mathbf{k}} dV_{iz}. \tag{10}$$

For this function, integrating equation (9) over $V_{iz}$ we reduce it to

$$\begin{aligned}&(k_x V_x + k_y V_y - \omega_i)\, F_{\omega\mathbf{k}} - i a_{i0}\,\frac{\partial F_{\omega\mathbf{k}}}{\partial V_{ix}}\\ &= \frac{V^2_{Ti}}{\beta_T}\,\phi_{\omega\mathbf{k}}\left(k_x\,\frac{\partial}{\partial V_{ix}} + k_y\,\frac{\partial}{\partial V_{iy}}\right)\int_{-\infty}^{\infty} f_{i0} dV_{iz} - i\nu_{in}\eta^i_{\omega\mathbf{k}}\int_{-\infty}^{\infty} f^{\mathrm{Coll}}_0 dV_{iz},\end{aligned} \tag{11}$$

where, in accord with equations (2) and (5), we have

$$\int_{-\infty}^{\infty} f_{i0} dV_{iz} = \frac{n_0}{2\pi V_{\parallel} V_{Ty}} \exp\left( - \frac{(V_{ix} - U_0)^2}{2V_{Tx}^2} - \frac{V_{iy}^2}{2V_{Ty}^2} \right), \tag{12a}$$

$$\int_{-\infty}^{\infty} f_0^{\mathrm{Coll}} dV_{iz} = \frac{n_0}{2\pi V_{Tn}^2} \exp\left( - \frac{V_{ix}^2 + V_{iy}^2}{2V_{Tn}^2} \right). \tag{12b}$$

Evidently, in the absence of the $\mathbf{E}_0$-field, i.e., when $U_0 = 0$ and all temperatures are equal, $V_{Ty} = V_{Tx} = V_{Tn}$, these two integrals coincide.

Since equation (11) is linear, it is convenient to separate $F_{\omega\mathbf{k}}$ into two parts: one proportional to $\eta^i_{\omega\mathbf{k}}$ and the other proportional to $\phi_{\omega\mathbf{k}}$,

$$F_{\omega\mathbf{k}} = F^{(1)}_{\omega\mathbf{k}}(V_{ix}, V_{iy}) + F^{(2)}_{\omega\mathbf{k}}(V_{ix}, V_{iy}),$$

so that each of them satisfies a separate linear differential equation:

$$(k_x V_{ix} + k_y V_{iy} - \omega_i) F^{(1)}_{\omega\mathbf{k}} - i a_{i0} \frac{\partial F^{(1)}_{\omega\mathbf{k}}}{\partial V_{ix}} = -i \frac{\nu_{in} n_0 \eta^i_{\omega\mathbf{k}}}{2\pi V_{Tn}^2} \exp\left( - \frac{V_{ix}^2 + V_{iy}^2}{2V_T^2} \right), \tag{13a}$$

$$(k_x V_{ix} + k_y V_{iy} - \omega_i) F^{(2)}_{\omega\mathbf{k}} - i a_{i0} \frac{\partial F^{(2)}_{\omega\mathbf{k}}}{\partial V_{ix}} = \frac{V_{Tn}^2}{\beta_T} \phi_{\omega\mathbf{k}} \left( k_x \frac{\partial}{\partial V_{ix}} + k_y \frac{\partial}{\partial V_{iy}} \right) \int_{-\infty}^{\infty} f_{i0} dV_{iz}, \tag{13b}$$

where $\int_{-\infty}^{\infty} f_{i0} dV_{iz}$ was given above by equation (12a). After finding $F^{(1)}_{\omega\mathbf{k}}$ and $F^{(2)}_{\omega\mathbf{k}}$, we will calculate $\eta^i_{\omega\mathbf{k}}$ as

$$\eta^i_{\omega\mathbf{k}} = \frac{1}{n_0} \iint_{-\infty}^{\infty} \left[ F^{(1)}_{\omega\mathbf{k}}(V_{ix}, V_{iy}) + F^{(2)}_{\omega\mathbf{k}}(V_{ix}, V_{iy}) \right] dV_{ix} dV_{iy}. \tag{14}$$

Equations (13a) and (13b) are ordinary differential equations for $F^{(1)}_{\omega\mathbf{k}}$ and $F^{(2)}_{\omega\mathbf{k}}$ with respect to the variable $V_{ix}$. Their formal solutions are rather straightforward, but evaluating the double integrals of these solutions in order to find $\eta^i_{\omega\mathbf{k}}$ in accord with equation (14) is much less trivial. The entire procedure of finding the $\eta^i_{\omega\mathbf{k}}$ is rather involved mathematically, so that we present the detailed solutions of equations (13a) and (13b), along with the ensuing integrations, in Appendix A, Secs. A3 and A4. Here, we only give the resultant linear relation between $\eta^i_{\omega\mathbf{k}}$ and $\phi_{\omega\mathbf{k}}$.

Combining equations (A69) and (A83) from Appendices A3 and A4, we obtain the full ion contribution to the wave dispersion relation:

$$\left[1-\frac{\sqrt{\pi}\;\nu_{in}}{\sqrt{2(k^2V_{Tn}^2+i\mathbf{k}\cdot\mathbf{a}_{i0})}}\;w\left(\frac{\omega_i}{\sqrt{2(k^2V_{Ty}^2+i\mathbf{k}\cdot\mathbf{a}_{i0})}}\right)\right]\eta^i_{\omega\mathbf{k}}$$
$$=-\frac{k^2V_{Tn}^2\phi_{\omega k}}{\beta_T(k_x^2V_{Tx}^2+k_y^2V_{Ty}^2+i\mathbf{k}\cdot\mathbf{a}_{i0})}$$
$$\times\left[1+\frac{i\sqrt{\pi}(\omega_i-\mathbf{k}\cdot\mathbf{U}_0)}{\sqrt{2(k_x^2V_{Tx}^2+k_y^2V_{Ty}^2+i\mathbf{k}\cdot\mathbf{a}_{i0})}}\;w\left(\frac{\omega_i-\mathbf{k}\cdot\mathbf{U}_0}{\sqrt{2(k_x^2V_{Tx}^2+k_y^2V_{Ty}^2+i\mathbf{k}\cdot\mathbf{a}_{i0})}}\right)\right]\phi_{\omega k}. \quad (15)$$

Equation (15) is given in terms of the standard error function $w(x)$ (Abramowitz & Stegun, 1972). Using the simple conversion relation

$$w(z)=-\frac{i}{\sqrt{\pi}}\,Z(z), \quad (16)$$

we can easily rewrite equation (15) in terms of the plasma dispersion function $Z(x)$ (Fried & Conte, 1961).

Equation (15) assumes that $\mathrm{Im}\,\omega_i > 0$, i.e., that $\gamma > -\nu_{in}$. This condition covers all linearly unstable and moderately damping waves. At the same time, a priori we cannot rule out the possibility that heavily damped waves corresponding to $\mathrm{Im}\,\omega_i < 0$, i.e., $\gamma < -\nu_{in}$, may also exist and be a part of an observed turbulence spectrum. Hypothetically, such heavily damped waves might occur, e.g., if the instability-driving mechanisms under certain conditions would become, on the contrary, stabilizing. For example, for sufficiently long-wavelength waves, above the magnetization boundary of $\nu_{in} = \Omega_i$ (about 120 km altitude) the FBI driving mechanism becomes stabilizing (Dimant & Oppenheim, 2004). The same pertains to the ETI for sufficiently short-wavelength waves (Dimant et al., 2023). If $\mathrm{Im}\,\omega_i < 0$, then in equation (15) the $w$-function should be replaced by $-w(-z) = w(z) - 2\exp(-z^2)$.

Equation (15) shows that the external electric field $\mathbf{E}_0$ contributes to the linear $\eta^i_{\omega\mathbf{k}}$–$\phi_{\omega k}$ relation only through the additional terms $i\mathbf{k}\cdot\mathbf{a}_{i0}$. Equation (15) also generalizes the corresponding conventional no-field relation by including the mean ion drift, $\mathbf{U}_0$, and through the elevated anisotropic ion temperatures, $T_x > T_y > T_n$, described by the combination $k_x^2V_{Tx}^2 + k_y^2V_{Ty}^2 > k^2V_{Tn}^2$ (recall that $V_{Tx,Ty,n}^2 = T_{x,y,n}/m_i$ and throughout the entire paper we set $m_n = m_i$).

To conclude this section, we emphasize that equation (15) is the exact and unique solution of differential equation (11), which satisfies the asymptotic boundary conditions

(the "good" behavior of the solution at $V_{ix} \to \pm\infty$). Obtaining this solution has required no other approximations except those which led to equation (7) itself.

## 3 Electron Contribution to the Wave Dispersion Relation

Under the E-region conditions, the accurate kinetic description of electrons differs significantly from that of ions. For electrons, the simple BGK collision operator is totally inapplicable for three major reasons: (1) unlike ions that collide with neutrals of essentially the same mass, the mean rate of the electron energy losses during one collision of a light electron with a heavy neutral particle is much less than the corresponding rate of the momentum losses; and (2) the electron-neutral ($e$-$n$) collision frequency is strongly velocity-dependent, $\nu_{en}(V_e)$, where $V_e \equiv |\mathbf{V}_e|$ and $\mathbf{V}_e$ is the velocity of an individual electron, and (3) e-n collisions rapidly isotropize the electron velocity distribution. None of these three factors can be properly described by the BGK collisional model.

Indeed, in the E-region ionosphere, the effective rate of collisional losses is given by $\delta_{en}\nu_{en}$, where the fraction of (predominantly inelastic) collisional energy losses is $\delta_{en} \sim (2\text{–}4)\times10^{-3}$ (while for the corresponding ion fraction we have $\delta_{in} \simeq 1$) (Gurevich, 1978). Furthermore, the $e$-$n$ kinetic frequency in the lower ionosphere is approximately $\nu_{en} \propto V_e^{2\alpha}$, where $\alpha \simeq 1$, while the $i$-$n$ collision frequency, $\nu_{in}$, is almost constant. These issues have been discussed in detail by Dimant and Sudan (1995a); Koontaweepunya et al. (2024), so that we will not dwell on them here.

In this paper, we employ for electrons the kinetic approach originally employed by Gurevich (1978) and later developed further by Dimant and Sudan (1995a) and Dimant et al. (2021). This approach assumes that due to the strong magnetic field and intense collisional angular scattering in the velocity space, the electron distribution function (EDF), $f_e(\mathbf{V}_e)$, is close to an isotropic one, $f_e(\mathbf{V}_e) \approx f_e(V_e)$, along with all its wave perturbations. We will assume that the major isotropic part of $f_e(V_e)$, at least within the thermal bulk of the electron energy distribution, is reasonably close to a Maxwellian distribution,

$$f_0(V_e) \approx n_e \left(\frac{m_e}{2\pi T_e}\right)^{3/2} \exp\left(-\frac{m_e V_e^2}{2T_e}\right), \tag{17}$$

where, under sufficiently strong electric field $\mathbf{E}_0$, the background electron temperature $T_e$ may differ dramatically from the undisturbed neutral temperature $T_n$. Indeed, the space-averaged electron temperature in the uniformly dense weakly ionized plasma is de-

termined by the mean-energy balance equation given by

$$\frac{3}{2}\,\frac{dT_e}{dt} \approx m_e \langle \nu_{en} \rangle \mathbf{V}_e^2 - \frac{3}{2} \langle \delta_{en} \nu_{en} \rangle (T_e - T_n)\,, \tag{18}$$

where $\mathbf{V}_e$ is the background electron drift velocity and the angular brackets denote averaging over the entire electron velocity distribution. Under stationary conditions, assuming only the regular electron frictional heating, we have $\mathbf{V}_e \approx \mathbf{V}_0$, where $\mathbf{V}_0 = \mathbf{E}_0 \times \mathbf{B}/B^2$, so that equation (18) yields

$$T_e(E_0) \approx T_n + \frac{2 m_e \langle \nu_{en} \rangle E_0^2}{3 \langle \delta_{en} \nu_{en} \rangle B^2}. \tag{19}$$

For sufficiently strong $\mathbf{E}_0$, the contribution of the electron frictional heating, i.e., the second term in the RHS of Eq. (19) can be comparable to the first one.

Equation (19) represents only the minimal electron temperature in the presence of the DC electric field $\mathbf{E}_0$. During the nonlinearly saturated stage of the E-region instability driven by this field there is additional heating caused by turbulent electric fields, especially those parallel to $\mathbf{B}$. This effect of anomalous electron heating (AEH) has been discussed in many papers, e.g., Foster and Erickson (2000); Dimant and Milikh (2003); Milikh and Dimant (2003); St-Maurice and Goodwin (2021); Liang et al. (2021), and references therein. For a sufficiently strong driving field, this anomalous (i.e., turbulent) heating exceeds significantly the normal (i.e., laminar) heating produced exclusively by $\mathbf{E}_0$. This paper deals with the linear dispersion relation which disregards such nonlinear processes as mode coupling, wave-frequency broadening, etc. We can assume, however, that the short-wavelength waves under study are not strongly affected by such nonlinear processes, except the spatially averaged, i.e., macroscopic, effect of AEH. We can include the latter parametrically by replacing $T_e$ given by equation (19) with the much larger actual electron temperature $T_e$ determined empirically or by using reliable heating models (Dimant & Milikh, 2003; Milikh & Dimant, 2003; Hysell et al., 2013; St-Maurice & Goodwin, 2021; Liang et al., 2021).

The main implication of equation (19), especially if we assume the full electron heating that includes AEH, is as follows. Whatever the driving field $E_0$ (reaching sometimes hundreds mV/m), the drift speed, $V_0 = E_0/B$, always remains small compared to the modified thermal speed $V_{Te} = (T_e/m_e)^{1/2}$. This takes place because within the eV range of the electron energies we always have $\langle \nu_{en} \rangle \,/\, \langle \delta_{en} \nu_{en} \rangle \gg 1$ (see above). This means that, unlike the unmagnetized ions whose Pedersen drift, $|\langle \mathbf{V}_i \rangle| \sim eE_0/(m_i \nu_{in} B)$, can

be comparable to, or even exceed, the electron bulk thermal speed $V_{Ti} = (T_i/m_i)^{1/2}$, see Sec. 2.1, electrons usually allow one to neglect any velocity-space shifts of their background Maxwellian EDF, including those connected with the neutral wind (the latter is two-to-three orders of magnitude smaller than the electron thermal speed, even for the originally cold ionospheric electrons).

For the linear wave analysis, we will start from Dimant and Sudan (1995a) equation (38),

$$\left(i\Delta_{\omega\mathbf{k}} + \hat{D}_{\omega\mathbf{k}}\right) f_{\omega\mathbf{k}}(V_e) = \hat{B}_{\omega\mathbf{k}} f_0(V_e)\, \phi_{\omega\mathbf{k}}\,, \tag{20}$$

where $f_0(V_e)$ is the background electron distribution function given by equation (17) with the electron temperature $T_e$ elevated by the AEH and often much larger than that given by equation (19); $f_{\omega\mathbf{k}}(V_e)$ is the single-harmonic linear perturbation of the electron distribution function, $\Delta_{\omega\mathbf{k}} \equiv \mathbf{k}\cdot\mathbf{V}_{e0} - \omega \approx \mathbf{k}\cdot\mathbf{V}_0 - \omega$, where $\mathbf{V}_0 = (\mathbf{E}_0 \times \mathbf{B})/B^2$; and the two differential operators $\hat{D}_{\omega\mathbf{k}}$ and $\hat{B}_{\omega\mathbf{k}}$ are given by

$$\hat{D}_{\omega\mathbf{k}} = \frac{1}{3V_e^2}\left[\left(\mathbf{k}_\perp V_e + i\mathbf{a}_{e0}\,\frac{d}{dV_e}\right)\frac{\nu_{en}(V_e)V_e^2}{\Omega_e^2}\cdot\left(\mathbf{k}_\perp V_e + i\mathbf{a}_{e0}\,\frac{d}{dV_e}\right)\right] + \frac{k_x^2 V_e^2}{3\nu_{en}(V_e)}, \tag{21}$$

where $\mathbf{a}_{e0} = e\mathbf{E}_0/m_e$ and the speed derivatives $d/dV_e$ are assumed to apply (from right to left) to the expressions right of them, including all $V_e$-dependent multipliers and $f_{\omega\mathbf{k}}(V_e)$,

$$\hat{B}_{\omega\mathbf{k}} = -\,\frac{V_{Te}^2}{3}\left[V_e\left(\frac{k_y^2\nu_{en}(V_e)}{\Omega_e^2} + \frac{k_x^2}{\nu_{en}(V_e)}\right)\frac{d}{dV_e} + \frac{2i\,(\mathbf{k}_\perp\cdot\mathbf{a}_{e0})}{\Omega_e^2 V_e^2}\,\frac{d}{dV_e}\left(V_e^2\nu_{en}(V_e)\frac{d}{dV_e}\right)\right]. \tag{22}$$

Compared to equation (38) from Dimant and Sudan (1995a), we have neglected here two kinetic operators, $\widehat{\delta\nu}_{en}$ and $\hat{\nu}_{ee}$, which describe the local electron collisional cooling and electron-electron ($e$-$e$) collisions, respectively. The former is crucial for the ETI (Dimant & Sudan, 1995b), while the latter may play a role under conditions of sufficiently dense and cold plasma, as discussed in Dimant and Sudan (1995a). The ETI exists almost exclusively in the "super-long"-wavelength limit of $|\omega| \lesssim \langle\delta_{en}\nu_{en}\rangle \ll \nu_{in}$ (Dimant & Sudan, 1995c, 1997; Dimant et al., 2023). In this paper, however, we are mostly interested in the medium-to-short wavelength range of $|\omega| \gtrsim \nu_{in} \gg \langle\delta_{en}\nu_{en}\rangle$, where the slow local $e$-$n$ collisional cooling plays no significant role. At the same time, the frictional heating may still be of importance, so that we should retain the terms $\propto \mathbf{a}_{e0}$. Also, since we are interested in strong driving fields that usually lead to significant electron heating, we can safely neglect the $e$-$e$ collisions, described by the quadratically-nonlinear operator of Coulomb collisions $\hat{\nu}_{ee}$, because the efficiency of Coulomb collisions decreases dramatically with increasing energies of the colliding particles.

### 3.1 Non-thermal approximation for electrons

Before we present the solution for the full set of electron equations, we will develop the linear kinetic theory for electrons in the limit of $\mathbf{a}_{e0} \to 0$. Unlike the well-known no-field approximation for ions (reviewed in Appendix A, Sec. A2), the results of this subsection have never been published before. We will use the $\mathbf{a}_{e0} \to 0$ results for future references, but these results may also be sufficient for a moderately strong electric field $\mathbf{E}_0$.

Unlike in the $\mathbf{a}_{i0} \to 0$ limit for ions, we will refer to the $\mathbf{a}_{e0} \to 0$ limit as the non-thermal approximation (NTA) for electrons. For magnetized electrons, the driving field $\mathbf{E}_0$ has been implicitly included in $\Delta_{\omega\mathbf{k}}$ through the mean electron drift velocity, $\mathbf{V}_{e0}$ (without this drift, no E-region plasma instability would even be possible). With this drift, but no the $\mathbf{a}_{e0}$-related terms explicitly included, all possible electron thermal effects are totally ignored.

For the Maxwellian background electron distribution function $f_0(V_e)$, when neglecting all terms $\propto \mathbf{a}_{e0}$, the above differential operators $\hat{D}_{\omega\mathbf{k}}$ and $\hat{B}_{\omega\mathbf{k}}$ reduce to a common algebraic function of the electron speed $V_e$,

$$\hat{D}_{\omega\mathbf{k}} \approx \hat{B}_{\omega\mathbf{k}} \approx S_{\omega\mathbf{k}} \equiv \frac{V_e^2}{3}\left(\frac{k_y^2\nu_{en}(V_e)}{\Omega_e^2} + \frac{k_x^2}{\nu_{en}(V_e)}\right). \tag{23}$$

This reduction is valid for sufficiently short-wavelength waves satisfying the conditions (expressed here in terms of the ion variables which are more relevant for the TFBI)

$$\frac{(kV_{Te})^2\langle\nu_{en}\rangle}{3\Omega_e^2} = \frac{(kV_{Ti})^2}{\nu_{in}}\frac{\langle\psi_\perp\rangle}{3\beta_T} \gg \langle\delta_{en}\nu_{en}\rangle\,,\ \frac{kV_0\langle\nu_{en}\rangle}{3\Omega_e}, \tag{24}$$
$$\frac{(kV_{Ti})^2}{\beta_{Ti}} \gg kV_0\Omega_i = ka_{i0}$$

where $\beta_{Ti} = T_i/T_e$ and $\langle\psi_\perp\rangle = \langle\nu_{en}\rangle\,\nu_{in}/(\Omega_e\Omega_i)$. These conditions mean that the electron diffusion described by the LHS of inequality (24) dominates over the mean electron energy loss rate, $\langle\delta_{en}\nu_{en}\rangle$, and Pedersen conductivity, $kV_0\langle\nu_{en}\rangle/(3\Omega_e)$ (these order-of-magnitude conditions are true in general, but more accurate conditions can be established after analyzing the solution of the general case for arbitrary $\mathbf{a}_{e0}$). Under these conditions, equations (20) and (23) yield a simple algebraic relation between $f_{\omega\mathbf{k}}(V_e)$ and $\phi_{\omega\mathbf{k}}$:

$$f_{\omega\mathbf{k}}(V_e) = \frac{S_{\omega\mathbf{k}}}{S_{\omega\mathbf{k}} + i\Delta_{\omega\mathbf{k}}}\ f_0(V_e)\,\phi_{\omega\mathbf{k}}\,.$$

Using $\eta^e_{\omega\mathbf{k}} = \langle f^e_{\omega\mathbf{k}}(V_e)\rangle = 4\pi\int_0^\infty f^e_{\omega\mathbf{k}}(V_e)V_e^2 dV_e/n_0$, we obtain

$$\eta^e_{\omega\mathbf{k}} = A_e\phi_{\omega\mathbf{k}}\,, \qquad A_e = \left\langle\frac{S_{\omega\mathbf{k}}}{S_{\omega\mathbf{k}} + i\Delta_{\omega\mathbf{k}}}\right\rangle, \tag{25}$$

or, equivalently,

$$A_e \equiv \frac{\eta^e_{\omega\mathbf{k}}}{\phi_{\omega\mathbf{k}}} = \int_0^\infty \frac{S_{\omega\mathbf{k}}}{S_{\omega\mathbf{k}} + i\Delta_{\omega\mathbf{k}}} \exp\left(-\frac{m_e V_e^2}{2T_e}\right) V_e^2 dV_e \Big/ \int_0^\infty \exp\left(-\frac{m_e V_e^2}{2T_e}\right) V_e^2 dV_e \,. \tag{26}$$

In the long-wavelength limit of $|\Delta_{\omega\mathbf{k}}| \gg S_{\omega\mathbf{k}}$ (roughly corresponding to $\psi_\perp|\omega| \ll \nu_{in}$), this yields an expression

$$\frac{1}{A_e} \approx \frac{\langle S^2_{\omega\mathbf{k}}\rangle}{\langle S_{\omega\mathbf{k}}\rangle^2} + i\,\frac{\Delta_{\omega\mathbf{k}}}{\langle S_{\omega\mathbf{k}}\rangle},$$

which coincides with the RHS of Eq. (60) from Dimant and Sudan (1995a). In the more general case of larger $|\omega|$ and $k$, we have to explicitly calculate the RHS of equation (26).

The above relations are valid for the general velocity dependence of $\nu_{en}(V_e)$, but elsewhere in this paper, we will assume the approximate power-law dependence of $\nu_{en}(V_e)$ employed by Dimant and Sudan (1995a):

$$\nu_{en}(V_e) = \nu_0\,\frac{V_e^2}{V_{Te}^2}, \qquad V_{Te}^2 \equiv \frac{T_e}{m_e}. \tag{27}$$

Notice that equation (62) from Dimant and Sudan (1995a) includes an additional factor 2 in the denominator of the expression for $\nu_{en}(V_e)$, so that their notation $\nu_0$ equals twice the parameter $\nu_0$ in this paper. Equation (27) yields

$$\langle \nu_{en}(V_e)\rangle = 3\nu_0 \tag{28}$$

and

$$S_{\omega\mathbf{k}} = \frac{\nu_0}{3}\left(\frac{k_y V_{Te}}{\Omega_e}\right)^2 \left(\frac{V_e^4}{V_{Te}^4} + \frac{k_x^2}{k_y^2}\frac{\Omega_e^2}{\nu_0^2}\right). \tag{29}$$

Temporarily introducing a dimensionless variable $z = V_e/(\sqrt{2}\,V_{Te})$, we obtain

$$A_e = \frac{4}{\sqrt{\pi}}\int_0^\infty \frac{z^4+\tau}{z^4+\tau+i\beta}\; z^2 e^{-z^2} dz \equiv 1 - i\,\frac{\beta}{\sqrt{\pi}}\,K(q), \tag{30a}$$

$$K(q) = 4\int_0^\infty \frac{z^2 e^{-z^2}}{z^4 - q^4}\,dz\,, \qquad q^4 = -(\tau+i\beta) = Qe^{i\Psi}, \tag{30b}$$

$$\tau = \frac{k_x^2}{4k_y^2}\frac{\Omega_e^2}{\nu_0^2}\,, \qquad \beta = \frac{3\Delta_{\omega\mathbf{k}}}{4\nu_0}\left(\frac{\Omega_e}{k_y V_{Te}}\right)^2, \tag{30c}$$

where $Q = |\tau + i\beta|$ and $\Psi = \arg\left[-(\tau+i\beta)\right]$. Usually, the real parameters $\tau$ and $\mathrm{Re}\,\beta$ are positive ($\omega_r < \mathbf{k}\cdot\mathbf{V}_0$ and hence $\mathrm{Re}\,\Delta_{\omega\mathbf{k}} > 0$), so that without loss of generality we may set the complex phase $\Psi = \arctan(\beta/\tau)$ to vary within the domain

$$\pi \le \Psi \le \frac{3\pi}{2}, \tag{31}$$

and the complex angle $\Psi/4$ belongs to the first quadrant. In this convention, all four independent roots of the 4-th order equation $q^4 = Qe^{i\Psi}$, see (30b), are given by $q = q_n$,

$$q_n = Q^{1/4} \exp\left[i\left(\frac{\Psi}{4} + \frac{n\pi}{2}\right)\right], \qquad \frac{\pi}{4} \le \frac{\Psi}{4} \le \frac{3\pi}{8}, \tag{32}$$

$$q_0 = -iq_1 = -\,q_2 = iq_3\,, \qquad n = 0,\, 1,\, 2,\, 3.$$

Using these representations, we can express in equation (30b) the function $K(q)$ as

$$K(q) = \sum_{n=0}^{3} \frac{K_n}{q_n}\,, \qquad K_n = \int_0^\infty \frac{e^{-z^2}}{z - q_n}\, dz\,. \tag{33}$$

In the summation, we consider first the term with $n = 2$. By using the above relations between the roots and making a change of the integration variable, we can express this term as

$$\frac{K_2}{q_2} = \frac{1}{q_2}\int_0^\infty \frac{e^{-z^2}}{z - q_2}\, dz = \frac{1}{q_0}\int_{-\infty}^0 \frac{e^{-z^2}}{z - q_0}\, dz$$

and combine it with the $n = 0$ term. Similarly, we proceed with the $n = 1$ and $n = 3$ terms. As a result, we obtain

$$\begin{aligned} K(q) &= \frac{1}{q_0}\int_{-\infty}^\infty \frac{e^{-z^2}}{z - q_0}\, dz + \frac{1}{q_1}\int_{-\infty}^\infty \frac{e^{-z^2}}{z - q_1}\, dz \\ &= \frac{1}{q_0}\left(\int_{-\infty}^\infty \frac{e^{-z^2}}{z - q_0}\, dz - i\int_{-\infty}^\infty \frac{e^{-z^2}}{z - iq_0}\, dz\right) \end{aligned}$$

Using equations (A27) and (A30) from Appendix A, we obtain

$$K(q) = \frac{i\pi}{q_0}\left[w(q_0) - iw(iq_0)\right] = \frac{\sqrt{\pi}}{q_0}\left[Z(q_0) - iZ(iq_0)\right]. \tag{34}$$

Alternatively, the same result could be obtained using an approach which will become helpful when obtaining the general solution (see Sec. 3.2). We implement this approach by representing from the very beginning the function $K(q)$ in equation (30b) as

$$\begin{aligned} K(q) &= 2\left(\int_0^\infty \frac{e^{-z^2}}{z^2 - q^2}\, dz + \int_0^\infty \frac{e^{-z^2}}{z^2 + q^2}\, dz\right) \\ &= \int_{-\infty}^\infty \frac{e^{-z^2}}{z^2 - q^2}\, dz + \int_{-\infty}^\infty \frac{e^{-z^2}}{z^2 + q^2}\, dz. \end{aligned} \tag{35}$$

Equation (35) is invariant with respect to the choice of the root of the quartic equation $q^4 = -(\tau + i\beta)$, see equation (32), but we will choose $q = q_0$, so that both the real and imaginary parts of $q$ are positive. Temporarily introducing auxiliary functions

$$K^{\pm}(\gamma, q) = \int_{-\infty}^\infty \frac{e^{-\gamma z^2}}{z^2 \pm q^2}\, dz, \tag{36}$$

where $\mathrm{Re}\,\gamma > 0$ and the subscripts $\pm$ correspond to the two different signs in the denominator of the integrand, we express $K(q)$ in equation (35) as

$$K(q) = K^+(1,q) + K^-(1,q). \tag{37}$$

We can calculate the integrals $K^\pm(\gamma,q)$ by differentiating them with respect to the parameter $\gamma$. This yields ordinary differential equations for the auxiliary parameter $\gamma$:

$$\frac{dK^+}{d\gamma} - q^2K^+ = -\sqrt{\frac{\pi}{\gamma}}, \qquad \frac{dK^-}{d\gamma} + q^2K^- = -\sqrt{\frac{\pi}{\gamma}}. \tag{38}$$

The general solutions of these differential equations are given by

$$\begin{aligned} K^+(\gamma,q) &= \exp\left(q^2\gamma\right)\left(C^+ - \sqrt{\pi}\int_0^\gamma \frac{\exp\left(-q^2t\right)}{\sqrt{t}}\,dt\right) \\ &= \exp\left(q^2\gamma\right)\left(C^+ - \frac{\pi}{q}\,\mathrm{erf}\left(\sqrt{\gamma}q\right)\right), \\ K^-(\gamma,q) &= \exp\left(-q^2\gamma\right)\left(C^- - \sqrt{\pi}\int_0^\gamma \frac{\exp\left(q^2t\right)}{\sqrt{t}}\,dt\right) \\ &= \exp\left(-q^2\gamma\right)\left(C^- + \frac{i\sqrt{\pi}}{q}\,\mathrm{erf}\left(i\sqrt{\gamma}q\right)\right), \end{aligned}$$

where the constants of integration, $C^\pm$, can be found from the boundary conditions at $\gamma = 0$. We will determine these boundary conditions by recalling the fact that due to our choice of the quartic root $q$ both $\mathrm{Re}\,q$ and $\mathrm{Im}\,q$ are positive. Using the residual theorem, we obtain

$$\begin{aligned} C^+ = K^+(0,q) &= \int_{-\infty}^{\infty} \frac{dz}{z^2+q^2} = \frac{\pi}{q}, \\ C^- = K^-(0,q) &= \int_{-\infty}^{\infty} \frac{dz}{z^2-q^2} = \frac{i\pi}{q}. \end{aligned}$$

Bearing in mind the fact that the error-function $\mathrm{erf}(x)$ is an odd function of its argument $x$, recalling the definitions of the functions $w$ and $Z$, and setting $\gamma = 1$, we finally obtain relations

$$K^+(1,q) = \int_{-\infty}^{\infty} \frac{e^{-z^2}}{z^2+q^2}\,dz = \frac{\pi}{q}\,w(iq) = -\frac{i\sqrt{\pi}}{q}\,Z(iq), \tag{39a}$$

$$K^-(1,q) = \int_{-\infty}^{\infty} \frac{e^{-z^2}}{z^2-q^2}\,dz = \frac{i\pi}{q}\,w(q) = \frac{\sqrt{\pi}}{q}\,Z(q), \tag{39b}$$

which also lead to equation (34). We will use relations similar to equation (39) when obtaining the approximate solution for the general case of arbitrarily large $\mathbf{a}_{e0}$.

Using equation (34), for $A_3$ given by equation (30a), we obtain

$$A_e \equiv \frac{\eta^e_{\omega\mathbf{k}}}{\phi_{\omega\mathbf{k}}} = 1 + \frac{\beta\sqrt{\pi}}{q}\,\left[w(q) - iw(iq)\right] = 1 - i\,\frac{\beta}{q}\,\left[Z(q) - iZ(iq)\right]. \tag{40}$$

where we recall again that $q(\tau,\beta) = q_0$ is the only root of equation $q^4 + \tau + i\beta = 0$ with both $\mathrm{Re}(q) > 0$ and $\mathrm{Im}(q) \geq 0$.

Note that under conditions of the drift approximation, $k\rho_e \ll 1$, and collisional regime, $\omega \ll \nu_{en}$, the conventional BGK approach for electrons, e.g., Lee et al. (1971), yields a relation that coincides with the fluid one,

$$\frac{1}{A_e} = 1 - i\beta_T \frac{\nu_{in}(\omega - \mathbf{k}\cdot\mathbf{V}_0)}{(kv_{Ti})^2\psi_\perp} = 1 + i\beta_T \frac{\nu_{in}\Delta_{\omega\mathbf{k}}}{(kv_{Ti})^2\psi_\perp}.$$

The discussed non-thermal approximation for electrons, along with the no-field approximation for unmagnetized ions, describe the pure Farley-Buneman instability with the improved kinetic description of electrons. Recall that unlike the NFA for ions, see Sec. A2, the electron part of the non-thermal wave dispersion relation for relatively short-wavelength waves, $|\omega| \gtrsim \nu_{in}$, and resulting in equation (40), had never been published before.

### 3.2 General case

In the general case, we should solve full equation (20), where the operators $\hat{D}_{\omega\mathbf{k}}$ and $\hat{B}_{\omega\mathbf{k}}$ have the differential form of equations (21) and (22) with the external electric field explicitly included via $\mathbf{a}_{e0} = e\mathbf{E}_0/m_e$. All terms proportional to $\mathbf{a}_{e0}$ describe electron thermal effects responsible for the ETI (Dimant & Sudan, 1995b, 1997). According to the fluid analysis of Dimant et al. (2023), in the short-wavelength kinetic range defined by equation (24) we might expect no ETI driving. However, the thermal terms $\propto \mathbf{a}_{e0}$ could result in the opposite effect of additional wave damping or may merely modify the wave phase velocity. Regardless of whether these effects will be significant, a quantitatively accurate analysis requires that we include all terms proportional to $\mathbf{a}_{e0}$.

Assuming for the background electron distribution function the same Maxwellian function given by equation (17) with the given electron temperature $T_e$, from equation (22) we obtain

$$\hat{B}_{\omega\mathbf{k}} f_0 = \frac{n_0}{3\,(2\pi)^{3/2}\,V_{Te}^3} \exp\left(-\frac{V_e^2}{2V_{Te}^2}\right) \times \left[\frac{\nu_0 k_y^2 V_e^4}{\Omega_e^2 V_{Te}^2} + \frac{k_x^2 V_{Te}^2}{\nu_0} + \frac{2i\,(\mathbf{k}_\perp\cdot\mathbf{a}_{e0})\,\nu_0 V_e^2\left(5V_{Te}^2 - V_e^2\right)}{\Omega_e^2 V_{Te}^4}\right] \tag{41}$$

For the further analysis, it is convenient to introduce a dimensionless variable

$$\xi \equiv z^2 = \frac{V_e^2}{2V_{Te}^2}. \tag{42}$$

In terms of this variable, equation (20) can be recast as

$$\frac{d^2 f^e_{\omega\mathbf{k}}}{d\xi^2} - \left(\chi - \frac{5}{2\xi}\right)\frac{df^e_{\omega\mathbf{k}}}{d\xi} - \left(C + \frac{5\chi}{4\xi} + \frac{\mathcal{P}}{\xi^2}\right) f^e_{\omega\mathbf{k}}$$
$$= -\left(N_0 + \frac{N_1}{\xi} + \frac{N_2}{\xi^2}\right) e^{-\xi}\phi_{\omega\mathbf{k}} \tag{43}$$

with the parameters given by

$$\chi = \frac{2ik_y V_{Te}^2 \cos\varphi}{a_{e0}}, \qquad \mathcal{P} = \frac{3\Omega_e^2 V_{Te}^2}{4\nu_0 a_{e0}^2}\left(i\Delta_{\omega\mathbf{k}} + \frac{k_x^2 V_{Te}^2}{3\nu_0}\right),$$
$$C = \frac{k_y^2 V_{Te}^4}{a_{e0}^2}, \qquad N_0 = \frac{\left(k_y^2 V_{Te}^2 - 2ik_y a_{e0}\cos\varphi\right) n_0}{(2\pi)^{3/2} a_{e0}^2 V_{Te}},$$
$$N_1 = \frac{5ik_y n_0 \cos\varphi}{(2\pi)^{3/2} a_{e0} V_{Te}}, \qquad N_2 = \frac{k_x^2 V_{Te}^2 \Omega_e^2 n_0}{4(2\pi)^{3/2} a_{e0}^2 \nu_0^2 V_{Te}}. \tag{44}$$

Here $\varphi$ is the angle between $\mathbf{k}_\perp$ and $\mathbf{a}_{e0}$: $\mathbf{k}_\perp \cdot \mathbf{a}_{e0} = k_y a_{e0}\cos\varphi$.

Before proceeding with solving equation (43) in the general case, we note that the NTA limit (corresponding to $a_{e0} \to 0$, see Sec. 3.1) can be obtained from equation (43) by neglecting both $\xi$ derivatives with respect to $\xi$ (compared to the remaining terms) and all terms whose $a_{e0} \to 0$ singularity is lower then $1/a_{e0}^2$ (in particular, by neglecting $\chi$ and $N_1$). As a result, all factors $1/a_{e0}^2$ in the LHS and RHS of equation (43) can be mutually canceled out, so that the NTA solution for $f^e_{\omega\mathbf{k}}(V_e)$ becomes

$$\lim_{a_{e0}\to 0} f^e_{\omega\mathbf{k}}(V_e) = \frac{\left(N_0 + N_2/\xi^2\right) e^{-\xi}\phi_{\omega\mathbf{k}}}{C + \mathcal{P}/\xi^2}$$
$$= \frac{n_0\left(4\nu_0 k_y^2 V_{Te}^2 \xi^2/\Omega_e^2 + k_x^2 V_{Te}^2/\nu_0\right) e^{-\xi}}{(2\pi)^{3/2} V_{Te}^3\left(4\nu_0 k_y^2 V_{Te}^2 \xi^2/\Omega_e^2 + k_x^2 V_{Te}^2/\nu_0 + 3i\Delta_{\omega\mathbf{k}}\right)}\,\phi_{\omega\mathbf{k}}. \tag{45}$$

This solution agrees with the NTA solution for $f^e_{\omega\mathbf{k}}(V_e)$ that led to equation (30a). As mentioned above, equation (45) with no thermal mechanisms explicitly included can be a reasonably good approximation for a moderately strong driving electric field. We will refer to the NTA limit when discussing the general solution for electrons.

Generally, for electrons (as well as for the ions), instead of a simple algebraic relation between $f^e_{\omega\mathbf{k}}$ and $\phi_{\omega\mathbf{k}}$, we have much less tractable differential equation (43). This equation is a linear ordinary differential equation with respect to the variable $\xi$. Unlike the first-order equation (11) for ions, equation (43) for electrons is a second-order differential equation. Such equations with non-constant coefficients can be solved exactly only on rare occasions. Besides, for electrons, similarly to ions, of a special difficulty becomes the integration of the found solution (with the relevant weight function) in order to determine the wave perturbations of the electron density.

We start by obtaining the exact solution of equation (43) in terms of integrals of special functions. This exact solution is inconvenient for practical applications, so that later we will employ reasonably accurate approximations based on relations between the actual physical parameters.

The LHS of second-order ODEs like equation (43), by using the change of the variable $\xi$ to a new one, $\xi = \lambda x$, and the substitution

$$f^e_{\omega\mathbf{k}}(\xi) = e^{\alpha x} x^{\gamma} \bar{w}(x), \qquad x = \frac{\xi}{\lambda}, \tag{46}$$

can be recast in standard Whittaker's form (Abramowitz & Stegun, 1972):

$$\text{LHS} \propto \frac{d^2\bar{w}}{dx^2} + \left( -\frac{1}{4} + \frac{\kappa}{x} + \frac{1/4 - \mu^2}{x^2} \right) \bar{w}. \tag{47}$$

Specifically for equation (43), this procedure requires the following values of the parameters $\gamma$, $\lambda$, and $\alpha$:

$$\gamma = -\frac{5}{4}, \qquad \lambda = \frac{a_{e0}}{2k_y V_{Te}^2 \sin\varphi}, \qquad \alpha = \frac{\chi\lambda}{2} = \frac{i\cot\varphi}{2}. \tag{48}$$

This automatically yields $\kappa = 0$, while the parameter $\mu$ is given by

$$\mu = \sqrt{\mathcal{P} + \frac{9}{16}} = \left[ \frac{3\Omega_e^2 V_{Te}^2}{4\nu_0 a_{e0}^2} \left( i\Delta_{\omega\mathbf{k}} + \frac{k_x^2 V_{Te}^2}{3\nu_0} \right) + \frac{9}{16} \right]^{1/2}. \tag{49}$$

Here (as well as anywhere else), we choose the branch of the square root with the positive real part.

The fact that in our case Whittaker's parameter $\kappa$ automatically becomes zero makes the recast equation (47) simpler than general Whittaker's equation. In terms of the new variable $x = \xi/\lambda$, the full recast differential equation is given by

$$\frac{d^2\bar{w}}{dx^2} + \left( -\frac{1}{4} + \frac{1/4 - \mu^2}{x^2} \right) \bar{w} = L(x), \tag{50a}$$

$$L(x) = -\lambda^2 e^{-(\alpha+\lambda)x} x^{\frac{5}{4}} \left( N_0 + \frac{N_1}{\lambda x} + \frac{N_2}{\lambda^2 x^2} \right) \phi_{\omega,\kappa}, \tag{50b}$$

where the parameters $N_{0,1,2}$ are defined by equation (44).

We start solving equation (50a) by considering first the homogeneous equation, i.e., this equation without the RHS:

$$\frac{d^2\bar{w}}{dx^2} + \left( -\frac{1}{4} + \frac{1/4 - \mu^2}{x^2} \right) \bar{w} = 0. \tag{51}$$

Equation (51) has exact analytic solutions in terms of special functions. As the two linearly independent solutions of this equation, we can choose $\sqrt{x}\ \mathrm{I}_\mu(x/2)$ and $\sqrt{x}\ \mathrm{K}_\mu(x/2)$,

where $\mathrm{I}_\mu(z)$ and $\mathrm{K}_\mu(z)$ are the modified Bessel functions of the first and second kind, respectively (Erdelyi, 1955; Abramowitz & Stegun, 1972; Arfken et al., 2013).

The modified Bessel functions of the integer or half-integer order (Abramowitz & Stegun, 1972) arise in a significant number of scientific and engineering problems, but such functions with the general complex order $\mu$ are used less frequently. For these functions, we will use relevant relations that can be found in Erdelyi (1955); Arfken et al. (2013), or can be easily deduced from some expressions given there. Since the arguments of the modified Bessel functions in our case are $x/2$, in order to avoid confusion, we will give all relations for the functions of this argument, rather than of $x$.

First, we note that the function $\mathrm{K}_\mu(x/2)$ can be expressed in terms of $\mathrm{I}_\mu(x/2)$ as

$$\mathrm{K}_\mu\left(\frac{x}{2}\right) = \frac{\pi}{2}\,\frac{\mathrm{I}_{-\mu}(x/2) - \mathrm{I}_\mu(x/2)}{\sin(\mu\pi)}, \tag{52}$$

Further, the function $\mathrm{I}_\mu(x/2)$ has the following exact series expansion convergent for arbitrarily large arguments,

$$\mathrm{I}_\mu\left(\frac{x}{2}\right) = \sum_{s=0}^{\infty} \frac{1}{s!\Gamma(s+\mu+1)}\left(\frac{x}{4}\right)^{\mu+2s}, \tag{53}$$

where $\Gamma(x)$ is the standard gamma function. Using equations (52) and (53), we can write a similar exact representation for $\mathrm{K}_\mu$,

$$\mathrm{K}_\mu\left(\frac{x}{2}\right) = \frac{\pi}{2\sin(\mu\pi)} \times \left[\sum_{s=0}^{\infty} \frac{1}{s!\Gamma(s-\mu+1)}\left(\frac{x}{4}\right)^{-\mu+2s} - \sum_{s=0}^{\infty} \frac{1}{s!\Gamma(s+\mu+1)}\left(\frac{x}{4}\right)^{\mu+2s}\right]. \tag{54}$$

For large $|x|$, according to Arfken et al. (2013), equations (14.134–14.143), we have

$$\sqrt{x}\;\mathrm{I}_\mu\left(\frac{x}{2}\right) = \frac{e^{x/2}}{\sqrt{2\pi}}\left[P_\mu\left(\frac{ix}{2}\right) - iQ_\mu\left(\frac{ix}{2}\right)\right], \tag{55a}$$

$$\sqrt{x}\;\mathrm{K}_\mu(\frac{x}{2}) = \sqrt{\frac{\pi}{2}}\;e^{-x/2}\left[P_\mu\left(\frac{ix}{2}\right) + iQ_\mu\left(\frac{ix}{2}\right)\right], \tag{55b}$$

where $P_\mu$ and $Q_\mu$ are non-convergent asymptotic series

$$P_\mu(z) \approx 1 - \frac{\left(4\mu^2-1\right)\left(4\mu^2-9\right)}{2!\,(8z)^2} + \frac{\left(4\mu^2-1\right)\left(4\mu^2-9\right)\left(4\mu^2-25\right)\left(4\mu^2-49\right)}{4!\,(8z)^4} + \ldots, \tag{56a}$$

$$Q_\mu(z) \approx \frac{4\mu^2-1}{1!\,(8z)} - \frac{\left(4\mu^2-1\right)\left(4\mu^2-9\right)\left(4\mu^2-25\right)}{3!\,(8z)^3} + \ldots \tag{56b}$$

(be cautioned that our notation $\mu$ corresponds to Arfken et al. (2013) notation $\eta$, while Arfken et al. (2013) notation $\mu$ corresponds to our $4\mu^2$).

Note that $\sqrt{x}\,\mathrm{I}_{\mu}$ and $\sqrt{x}\,\mathrm{I}_{-\mu}$ for the non-integer order $\mu$ could also form a set of two linearly independent solutions of equation (51). However, picking them as the two independent solutions would be impractical for the following reason. Since $\mu^2 = (-\mu)^2$, equation (56a) yields

$$\begin{aligned}\sqrt{x}\,\mathrm{I}_{-\mu}\left(\frac{x}{2}\right) &\approx \frac{e^{\frac{x}{2}}}{\sqrt{\pi}}\left(P_{-\mu}\left(\frac{ix}{2}\right) - iQ_{-\mu}\left(\frac{ix}{2}\right)\right)\\ &= \frac{e^{\frac{x}{2}}}{\sqrt{\pi}}\left(P_{\mu}\left(\frac{ix}{2}\right) - iQ_{\mu}\left(\frac{ix}{2}\right)\right) \approx \sqrt{x}\,\mathrm{I}_{\mu}\left(\frac{x}{2}\right),\end{aligned}$$

so that for large $x$ the functions $\sqrt{x}\,\mathrm{I}_{\mu}$ and $\sqrt{x}\,\mathrm{I}_{-\mu}$ merge asymptotically into the same exponentially growing function and hence are unsuitable to serve as the two linearly independent solutions. On the other hand, the functions $\sqrt{x}\,\mathrm{I}_{\mu}$ and $\sqrt{x}\,\mathrm{K}_{\mu}$ have a very distinct asymptotic behavior at both ends of the domain $0 \le x < \infty$. As a result, the choice of $\sqrt{x}\,\mathrm{I}_{\mu}$ and $\sqrt{x}\,\mathrm{K}_{\mu}$ as the two linearly independent solutions will allow us to obtain the unique solution of full equation (50a) more easily than if we chose the functions $\sqrt{x}\,\mathrm{I}_{\mu}$ and $\sqrt{x}\,\mathrm{I}_{-\mu}$ as the two independent solutions.

Furthermore, the exact Wronskian of the two linearly independent solutions $\mathrm{I}_{\mu}$ and $\mathrm{K}_{\mu}$ is given by

$$W \equiv \sqrt{x}\,\mathrm{K}_{\mu}\left(\frac{x}{2}\right)\frac{d}{dx}\left[\sqrt{x}\,\mathrm{I}_{\mu}\left(\frac{x}{2}\right)\right] - \sqrt{x}\,\mathrm{I}_{\mu}\left(\frac{x}{2}\right)\frac{d}{dx}\left[\sqrt{x}\,\mathrm{K}_{\mu}\left(\frac{x}{2}\right)\right] = 1. \tag{57}$$

Below, we will use this relation for finding the unique solution of full equation (50a). Also, combining equation (57) with standard properties of the 2nd-order differential equations (Abramowitz & Stegun, 1972; Arfken et al., 2013), one can easily find two exact integral relations that express these linearly independent solutions in terms of each other:

$$\sqrt{x}\,\mathrm{K}_{\mu}\left(\frac{x}{2}\right) = \sqrt{x}\,\mathrm{I}_{\mu}\left(\frac{x}{2}\right)\int_{x}^{\infty}\frac{dt}{\left[\sqrt{t}\,\mathrm{I}_{\mu}\left(t/2\right)\right]^2}, \tag{58a}$$

$$\sqrt{x}\,\mathrm{I}_{\mu}\left(\frac{x}{2}\right) = \sqrt{x}\,\mathrm{K}_{\mu}\left(\frac{x}{2}\right)\int_{0}^{x}\frac{dt}{\left[\sqrt{t}\,\mathrm{K}_{\mu}\left(t/2\right)\right]^2}. \tag{58b}$$

These relations can be used for additional testing the accuracy of some approximations implemented in Appendix B.

Now we proceed to solving equation (50a) with the parameters given by equations (44) and (48). According to equations (53)–(55), the function $\mathrm{I}_{\mu}\left(x/2\right)$ behaves regularly at $x \to 0$, but exponentially diverges as $x \to \infty$. On the contrary, the function $\mathrm{K}_{\mu}\left(x/2\right)$ diverges at $x \to 0$, but exponentially tends to zero as $x \to \infty$. Using these properties, along with equation (57) and standard methods of solving 2nd-order inhomogeneous

equations, we arrive at the following exact solution of equation (50a):

$$\bar{w}(x) = -\sqrt{x}\left[\mathrm{K}_\mu\left(\frac{x}{2}\right)\int_0^x \sqrt{z}\,\mathrm{I}_\mu\left(\frac{z}{2}\right)L(z)dz + \mathrm{I}_\mu\left(\frac{x}{2}\right)\int_x^\infty \sqrt{z}\,\mathrm{K}_\mu\left(\frac{z}{2}\right)L(z)dz\right]. \quad (59)$$

This exact solution is unique because it is the only solution which behaves regularly within the entire domain of $0 \le x < \infty$. In addition, the function $\bar{w}(x) \to 0$ as $x \to \infty$. Equation (59) is our last exact expression. From this point on, we will start making approximations.

According to the explicit expression for the function $L$ defined by equation (50b), equation (59) involves convolutions of the modified Bessel functions with power-law and exponential functions. We cannot evaluate such integrals exactly, so that it is hard to avoid approximations. Using the fact that the parameter $|\mu|$ is sufficiently large (as explained in the following paragraph), we will construct reasonably accurate approximations to the special functions $\mathrm{I}_\mu$ and $\mathrm{K}_\mu$ in terms of elementary functions. Then, within the framework of the same accuracy, we will perform all necessary integrations in the RHS of equation (59).

Before proceeding, we will make simple estimates regarding the crucial parameter $\mu$ given by equation (49). Our goal is to demonstrate that under real physical conditions in the E-region ionosphere, and for the wave frequency band of interest to us, the absolute value of $\mu$ is always large, regardless of the driving field magnitude. This fact is crucial for all approximations that will follow.

The main contribution to $\mu$ is given by the quantity

$$\Theta \equiv \frac{3\Omega_e^2 V_{Te}^2}{4\nu_0 a_{e0}^2}|\Delta_{\omega\mathbf{k}}|. \quad (60)$$

By assuming sufficiently short-wavelength waves which are most relevant to the implied kinetic regime,

$$|\Delta_{\omega\mathbf{k}}| = |\omega - \mathbf{k}\cdot\mathbf{V}_{e0}| \gtrsim \nu_{in}, \quad (61)$$

we consider first the lowest driving field of interest, $E_0 = m_e a_{e0}/e$, corresponding to the minimum FBI threshold, $E_0 = BC_s$, $C_s \sim V_{Ti}$ (for the high-latitude electrojet, typically $E_0 \simeq 20$ mV/m). This lowest field results in the maximum value of $\Theta = \Theta_{\max}$. Using equations (28) and (61), we estimate $\Theta_{\max}$ as

$$\Theta_{\max} \sim \frac{\Omega_e^2 V_{Te}^2|\Delta_{\omega\mathbf{k}}|}{\langle\nu_{en}\rangle\Omega_e^2 V_{Tn}^2} \sim \frac{m_i|\Delta_{\omega\mathbf{k}}|}{m_e\langle\nu_{en}\rangle} \gtrsim \frac{m_i\nu_{in}}{m_e\langle\nu_{en}\rangle} \simeq 5000. \quad (62)$$

Since the lowest field of interest usually modifies the E-region plasma only weakly, here we have used a simple relation $\langle\nu_{en}\rangle \simeq 10\,\nu_{in}$, which for the undisturbed conditions is approximately valid across the entire lower part of the E-region ionosphere (Gurevich, 1978; Kelley, 2009). Thus, we obtain that the maximum value of $|\mu|$ is quite large, $|\mu|_{\max} \sim \sqrt{\Theta_{\max}} \gtrsim 70$.

The parameter $\Theta$ decreases with increasing $a_{e0}$, so that for sufficiently large $a_{e0}$ it might in principle become so small that $|\mu|$ would approach its absolute minimum value of 3/4, see equation (49). However, this is never the case because as $E_0$ increases, $V_{Te}$ also increases, largely in proportion with $E_0$. Thus, even for the infinitely large electric field corresponding to $a_{e0} \to \infty$, the parameter $|\mu|$ should saturate at a sufficiently large value. Indeed, for very large $a_{e0}$, according to (19) where we assume $\Delta T_e \approx T_e \gg T_{e0}$, we obtain $V_{Te}^2 \simeq 2a_{e0}^2/(3\langle\delta_{en}\rangle\Omega_e^2)$, where $\langle\delta_{en}\rangle \simeq \langle\delta_{en}\nu_{en}\rangle/\langle\nu_{en}\rangle$. As a result, assuming $|\Delta_{\omega\mathbf{k}}| \gtrsim \nu_{in}$, we obtain for the corresponding saturated value of $\Theta$, $\Theta_{\min} \gtrsim \nu_{in}/(2\delta_{en}\nu_0)$. Using $\langle\delta_{en}\rangle \sim 3\times10^{-3}$, $\nu_0 = (1/3)\langle\nu_{en}\rangle$, and $\langle\nu_{en}\rangle \simeq 10\,\nu_{in}$ (see above), we obtain $\Theta_{\min} \gtrsim 50$, so that even the minimum value of $|\mu|$ should be sufficiently large, $|\mu|_{\min} \sim \sqrt{\Theta_{\min}} \gtrsim 7$. In this estimate, however, we have not included the anomalous electron heating (see above). If we did that, we would obtain noticeably larger values of $|\mu|_{\min}$. Even for the extreme, but still possible value of $E_0 \sim 1\,\mathrm{V/m}$, by applying currently available models of the AEH (Dimant & Milikh, 2003; Liang et al., 2021), we may assume that the temperature will be an order of magnitude larger that the $\mathbf{E}_0$-heated value, so that the actual minimum value of $|\mu|$ will be at least $|\mu|_{\min} \gtrsim 20$. For more realistic field values of $E_0 \lesssim 200$ mV/m, the value of $|\mu|$ will be somewhat larger, $|\mu| \gtrsim 25$. All these estimated values of $|\mu|$ are large enough to provide the high accuracy of all approximate expressions that will be further used.

We describe the entire procedure of simplifying the solution given by equation (43) in Appendix B. Under condition of $|\mu^2| \approx \mathcal{P} \gtrsim \Theta \gg 1$, but for an arbitrary value of $a_{e0}$, we use a series of approximations based on the same large parameter $\mathcal{P}$, so that each further approximation keeps the resultant expressions largely within the same level of accuracy. The final approximate solution of the full differential equation is given by equation (B19). Structurally, this equation is similar to equation (45) obtained for $a_{e0} \to 0$. This might suggest that the finite-$a_{e0}$ solution could also have been obtained by neglecting in equation (43) all derivatives of $f^e_{\omega\mathbf{k}}$, i.e., bypassing the direct solution of the second-order differential equation. However, comparison of equation (B19) with such an

alternative solution shows that the two solutions differ, although in the $a_{e0} \to 0$ limit they asymptotically reduce to same equation (45). Thus, in the general case of arbitrary $a_{e0}$, one cannot avoid solving second-order differential equation (43).

Finding the solution of equation (43), however, is not enough. To complete the kinetic treatment of electrons, we need to find the electron density perturbation by integrating the wave perturbation of the electron velocity distribution (very close to its dominant isotropic part) over the electron speeds,

$$n^e_{\omega\mathbf{k}} \approx 4\pi \int_0^\infty f^e_{\omega\mathbf{k}}(V_e) V_e^2 dV_e. \tag{63}$$

Using a series of carefully validated approximations, the expression for $f^e_{\omega\mathbf{k}}$ has been found in Appendix B in terms of the dimensionless variable $\xi$, see equation (B19). We rewrite equation (B19) in the following form:

$$f^e_{\omega\mathbf{k}}(z) = \frac{z^4 + \alpha_1 z^2 + \alpha_0}{z^4 - q^4} \; \frac{e^{-z^2}(k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_0) n_0 \phi_{\omega\mathbf{k}}}{(2\pi)^{3/2} V_{Te}^3 (k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_0 - a_{e0}^2/V_{Te}^2)}, \tag{64}$$

where $z = V_e/(\sqrt{2}\, V_{Te}) = \sqrt{\xi}$ and

$$\alpha_1 = \frac{5i(\mathbf{k}\cdot\mathbf{a}_{e0})}{k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_0}, \tag{65a}$$

$$\alpha_0 = \frac{k_x^2 V_{Te}^2 \Omega_e^2}{4\nu_0^2 (k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0})}, \tag{65b}$$

$$q = \left[ - \frac{(3i\nu_0 \Delta_{\omega,\mathbf{k}} + k_x^2 V_{Te}^2)\Omega_e^2}{4\nu_0^2 (k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0} - a_{e0}^2/V_{Te}^2)} \right]^{1/4}. \tag{65c}$$

The parameter $q$ generalizes the same parameter introduced in Sec. 3.1 to the case of arbitrary $\mathbf{a}_{e0}$. Bearing in mind that the integrand of equation (63) contains an additional factor $V_{e0}^2 \propto z^2$, we express the combined pre-exponential $z$-dependent factor as

$$\frac{(z^4 + \alpha_1 z^2 + \alpha_0) z^2}{(z^2 - q^2)(z^2 + q^2)} = z^2 + \alpha_1 + \frac{q^4 + \alpha_0 - \alpha_1 q^2}{2(z^2 + q^2)} + \frac{q^4 + \alpha_0 + \alpha_1 q^2}{2(z^2 - q^2)}. \tag{66}$$

Though equation (66) is invariant with respect to the choice of the quartic-root branch for the parameter $q$, we will always imply the only root with $\mathrm{Re}\, q > 0$ and $\mathrm{Im}\, q > 0$.

Using equations (66) and (39), we implement the speed integration in equation (63), yielding

$$\begin{aligned} \frac{\eta^e_{\omega\mathbf{k}}}{\phi_{\omega\mathbf{k}}} &= \frac{4(k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0})}{\sqrt{\pi}(k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0} - a_{e0}^2/V_{Te}^2)} \int_0^\infty \frac{\left(z^4 + \alpha_1 z^2 + \alpha_0\right) z^2}{z^4 - q^4}\, e^{-z^2} dz \\ &= \frac{k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0}}{k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0} - a_{e0}^2/V_{Te}^2} \\ &\times \left[ 1 + 2\alpha_1 + \frac{q^4 + \alpha_0 + \alpha_1 q^2}{q}\, Z(q) - i\, \frac{(q^4 + \alpha_0 - \alpha_1 q^2)}{q}\, Z(iq) \right], \end{aligned} \tag{67}$$

expressed here only in terms of the plasma dispersion function, $Z(x) = i\sqrt{\pi}\, w(x)$ (the final dispersion relation will also be given in terms of the $Z$-function).

In these expressions, we also have the relation

$$q^4 + \alpha_0 = - \frac{\Omega_e^2 \left[k_x^2 a_{e0}^2 + 3i(k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0})\Delta_{\omega,\mathbf{k}}\nu_0\right]}{4\nu_0^2(k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0})(k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0} - a_{e0}^2/V_{Te}^2)}, \tag{68}$$

which simplifies the reduction to the $\mathbf{a}_{e0} \to 0$ limit. As $\mathbf{a}_{e0} \to 0$, the quantity $q$ reduces to the same-notation quantity in equation (30b); and we also have $\alpha_1 \to 0$, $\alpha_0 \to \tau$, so that, according to equation (68), $q^4 + \alpha_0 \to q^4 + \tau \to -i\beta$. As a result, equation (67) reduces to NTA equation (40).

## 4 The TFBI Wave Dispersion Relation

To close the derivations of the full TFBI wave dispersion relation, we have to relate ions and electrons by applying Poisson's equation, $\nabla^2\Phi = (n_e - n_i)/\epsilon_0$, where $\epsilon_0$ is the permittivity of the free space. In our notations, for a given harmonic perturbation this equation reduces to

$$\eta^i_{\omega\mathbf{k}} - \eta^e_{\omega\mathbf{k}} = k^2\lambda_D^2\phi_{\omega\mathbf{k}}\ , \tag{69}$$

where $\lambda_D = [\epsilon_0 T_e/(e^2 n_e)]^{1/2}$ is the "electron" Debye length [the electron temperature stems from the definition of $\phi_{\omega\mathbf{k}}$ given by equation (8)]. Substituting $\eta^i_{\omega\mathbf{k}}$ from equation (15) and $\eta^e_{\omega\mathbf{k}}$ from equation (67), and then canceling $\phi_{\omega\mathbf{k}}$ from both sides of equation (69), we obtain the sought-for fully kinetic linear-wave dispersion relation:

$$I_{\rm ion} + I_{\rm electron} + k^2\lambda_D^2 = 0. \tag{70}$$

The first two terms in the LHS represent the ion and electron contributions to the TFBI wave dispersion relation, while the last term takes into account the charge separation.

The ion contribution, $I_{\rm ion}$, is given by

$$I_{\rm ion} \equiv \frac{k^2 V_{Tn}^2}{\beta_T(k^2 V_{Tn}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}\left[1 + \frac{i\nu_{in}}{\sqrt{2(k^2 V_{Tn}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}}\,\tilde{Z}\left(\frac{\omega_i}{\sqrt{2(k^2 V_{Tn}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}}\right)\right]^{-1}$$
$$\times\left[1 + \frac{\omega_i - \mathbf{k}\cdot\mathbf{U}_0}{\sqrt{2(k_\parallel^2 V_{Tx}^2 + k_y^2 V_{Ty}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}}\,\tilde{Z}\left(\frac{\omega_i - \mathbf{k}\cdot\mathbf{U}_0}{\sqrt{2(k_x^2 V_{Tx}^2 + k_y^2 V_{Ty}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}}\right)\right], \tag{71}$$

where $\tilde{Z}$ is a piece-wise modification of the regular $Z$-function:

$$\tilde{Z}(z) = \begin{cases} Z(z) & \text{if} \quad \mathrm{Re}(R_i) > 0 \quad (\text{i.e., } \gamma > -\nu_{in}), \\ -Z(-z) & \text{if} \quad \mathrm{Re}(R_i) < 0 \quad (\text{i.e., } \gamma < -\nu_{in}). \end{cases} \tag{72}$$

Here "$-Z(-z)$" can be alternatively represented as

$$-Z(-z) = i\sqrt{\pi}\, e^{-z^2}(-1 + \mathrm{erf}(iz)) = Z(z) - 2i\sqrt{\pi}\, e^{-z^2}.$$

The electron contribution, $I_{\mathrm{electron}}$, is given by

$$I_{\mathrm{electron}} = \frac{k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0}}{k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0} - a_{e0}^2/V_{Te}^2} \times \left(1 + 2\alpha_1 + \frac{q_0^4 + \alpha_0 + \alpha_1 q_0^2}{q_0}\, Z(q_0) - i\,\frac{q_0^4 + \alpha_0 - \alpha_1 q_0^2}{q_0}\, Z(iq_0)\right), \qquad (73)$$

where $q_0$ is the only quartic root ($q$) of equation (65c) with $\mathrm{Re}\, q > 0$ and $\mathrm{Im}\, q > 0$.

We are presenting all these relations in terms of the $Z$-functions only; to express them in terms of $w(x)$ one can use the simple conversion formula $Z(x) = i\sqrt{\pi}\, w(x)$, see equation (16). Recall that here $\beta_T = T_n/T_e$, $\lambda_D^2 = \epsilon_0 T_e/(e^2 n_e)$, $k^2 = k_y^2 + k_x^2$, $\omega_i = \omega + i\nu_{in} = \omega_r + i(\gamma + \nu_{in})$, $\mathbf{a}_{i0} = e\mathbf{E}_0/m_i$, $\mathbf{a}_{e0} = e\mathbf{E}_0/m_e$, $a_{e,i0} = |\mathbf{a}_{e,i0}|$, and the parameters $V_{\parallel,\perp}$ and $\mathbf{U}_0 = \mathbf{a}_{i0}/\nu_{in}$ describe the parallel/perpendicular thermal velocities and the mean drift velocity in the assumed bi-Maxwellian ion velocity distribution given by equation (5), respectively. The parameters $\alpha_0$, $\alpha_1$, and $q$ are given by equation (65),

$$\alpha_1 = \frac{5i(\mathbf{k}\cdot\mathbf{a}_{e0})}{k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_0},$$

$$\alpha_0 = \frac{k_x^2 V_{Te}^2 \Omega_e^2}{4\nu_0^2 (k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0})},$$

$$q = \left[-\frac{(3i\nu_0 \Delta_{\omega,\mathbf{k}} + k_x^2 V_{Te}^2)\Omega_e^2}{4\nu_0^2 (k_y^2 V_{Te}^2 - 2i\mathbf{k}\cdot\mathbf{a}_{e0} - a_{e0}^2/V_{Te}^2)}\right]^{1/4},$$

where $\Delta_{\omega\mathbf{k}} = \mathbf{k}\cdot\mathbf{V}_{e0} - \omega \approx \mathbf{k}\cdot\mathbf{V}_0 - \omega$ and $\mathbf{V}_0 = e\mathbf{E_0} \times \mathbf{B}/B^2$.

Without terms proportional to $\mathbf{a}_{i0}$ and $\mathbf{a}_{e0}$, equations (71) and (73) reduce to those determined by the "classical" NFA solution for ions, see equation (A20), and the NTA solution for electrons, see equations (25) and (40). These two approximations describe the pure FBI driving/damping. The terms $\propto \mathbf{a}_{i0}$ in equation (71) describe the ion thermal effects leading to the ITI, while the terms $\propto \mathbf{a}_{e0}$ in equations (73) describe the corresponding electron thermal effects. Thus, equation (70) makes it possible to trace the effects of different physical mechanisms.

Given the wavevector $\mathbf{k}$ and various plasma parameters, wave dispersion equation (70) should be solved for the complex wave frequency, $\omega = \omega_r + i\gamma$. The ion contribution to the wave dispersion relation includes the unknown $\omega$ only through the the combination $R_i = \omega + i\nu_{in}$, while the electron contribution includes $\omega$ only through the Doppler-

shifted wave frequency $\Delta_{\omega\mathbf{k}} = \mathbf{k}\cdot\mathbf{V}_{e0} - \omega$, which in turn is involved only through the quantity $q$, see equation (65c). Equation (70) contains no other $\omega$-dependent quantities.

Unlike the long-wavelength limit of the fluid-model TFBI dispersion relation described in Sec. V of Dimant et al. (2023), where the instability growth/damping rate, $\gamma$, turns out to be automatically small compared to the real wave frequency, $\omega_r$, and in the former the different instability mechanisms can be linearly separated from each other, equation (70) is much more involved. The growth or damping rate can be comparable to the real frequency, especially if $\omega_r \sim \nu_{in}$. While in the long-wavelength range the wave phase velocity is largely unaffected by the instability mechanisms, in the kinetic regime all instability mechanisms can affect the wave phase velocity. Equation (70) should serve as a basis for such analyses.

## 5 Discussion

In this section, we discuss the limitations of our analytic theory, which has far fewer constraints than the previous kinetic studies. The present fully kinetic theory of the combined TFBI is a serious improvement and a significant step forward in the analytical study of the electrojet instabilities.

The major limitation of this analysis is that we neglect finite ion magnetization, which means that it applies to ionospheric altitudes below 110 km. Including the ion magnetization in the analytic theory is reserved for the future, but the electron-related part of this dispersion relation remains valid throughout the entire E region (albeit with a caveat discussed below). If a future kinetic theory includes the ion magnetization then there should be no need to upgrade the electron-related part.

The next important issue is that for the background velocity distributions of ions and electrons we assumed modified Maxwellian velocity distributions: a shifted bi-Maxwellian distribution for ions and an almost isotropic, but strongly heated Maxwellian distribution for electrons. Note that all previous kinetic studies of the E-region instabilities assumed even much simpler, fully undisturbed and isothermal, Maxwellian velocity distributions for both ions and electrons. In our theory, when choosing the parameters for the modified background velocity distributions, such as the mean particle densities, temperatures, and the average velocity drift (for ions), we require these parameters to be consistent with the imposed electric field and plasma parameters. However, some actual

characteristics of the background velocity distributions cannot be fully mimicked by the modified Maxwellian velocity distributions. Indeed, according to recent results by Koontaweepunya et al. (2024) (these results have been partially reviewed in this paper's Appendix A1), under strong DC electric field, the PIC-determined ion velocity distribution becomes strongly shifted, has an anisotropic effective temperature, and is skewed in the $\mathbf{E}_0$-direction. The simple BGK-based analytic theory accurately predicts the mean ion drift and the overall mean effective temperature, but it greatly exaggerates the temperature anisotropy and skewness. The employed bi-Maxwellian distribution ansatz can accurately mimic the ion drift and anisotropic temperatures, provided they are taken from PIC simulations or from a more accurate analytical model, but the bi-Maxwellian ansatz cannot mimic the velocity-distribution skewness. How consequential might be this inaccuracy is not clear yet, but the shifted bi-Maxwellian ion function with the consistently chosen parameters seems to be the closest to reality, while remaining still analytically tractable.

Under the action of a strong electric field, the nearly isotropic electron velocity distribution can deviate significantly from any heated Maxwellian distribution, but this usually takes place outside of the thermal bulk, i.e., closer to the higher-energy tail of the electron velocity distribution, see, e.g., Gurevich (1978); Milikh and Dimant (2003). The major contributions to all integrals that resulted in our wave dispersion relation have been made by the thermal bulk of the electron velocity distribution, $V_e \lesssim V_{Te}$, so that this inaccuracy does not seem to be crucial. As for the ion background velocity distribution, there is hardly a better tractable model of the background electron velocity distribution than the Maxwellian distribution with an appropriately chosen temperature.

For electrons, however, there is one more, potentially serious, limitation associated with the applicability of the entire electron kinetic approach based on equation (20). This equation had been originally derived based on the assumption of strong isotropization of the electron distribution function (Gurevich, 1978; Dimant & Sudan, 1995a), along with all its wave perturbations. In the perpendicular to $\mathbf{B}$ direction, this isotropization is provided by fast Larmor rotation of electrons, compared to the wave period, $\Omega_e \gg |\omega|$. At the E region, this condition is easily fulfilled. However, on any plane that includes the direction parallel to $\mathbf{B}$, this isotropization usually requires sufficiently frequent collisions, $\nu_{en} \gg |\omega|$ (or, at least, $\nu_{en} \gtrsim |\omega|$). This represents a much stronger restriction than that in the perpendicular to $\mathbf{B}$ plane. The electron kinetic effects are mostly relevant for $|\omega| \gtrsim \nu_{in}$. Under undisturbed conditions of the cold E-region ionosphere,

the electron-to-ion collision rate ratio, $\nu_{en}/\nu_{in}$, is only about ten (see above), so that there may not be enough room for the applicability of our kinetic approach for electrons. However, two favorable circumstances help.

To illustrate the point, we do some estimates for a typical VHF radar, e.g., for the high-latitude radar ICEBEAR ($f_{\rm Radar} = 49.5$ MHz) (St-Maurice et al., 2023; Ivarsen et al., 2024). The corresponding scatter from ionospheric irregularities with the wavelength $\lambda \approx 3$ m and the corresponding wavevector $k = 2\pi/\lambda \simeq 2$ m$^{-1}$ (along the radar LoS). Assuming first the electric field at the instability threshold, $E_0 = 20$ mV/m, we have $V_0 \approx C_s \approx 400$ m/s. Then the corresponding wave frequency will be $\omega \sim kV_0 \approx 800$ s$^{-1}$. Consider first the worst case of the top boundary for the instability, which is slightly above the ion magnetization boundary, $\nu_{in} = \Omega_i$ (Dimant & Oppenheim, 2004). At high latitudes, where $B_0 \simeq 5 \times 10^4$ nT, we have $\nu_{in} \simeq \Omega_i \approx 160$ s$^{-1}$, so that $\nu_{en} \simeq 10\nu_{in} \simeq 1,600$ s$^{-1}$. Thus, at the ion magnetization boundary (typically around 120 km of altitude), the wave frequency $\omega \sim 800$ s$^{-1}$ is deeply inside the kinetic regime, $\omega \gg \nu_{in}$, while the same $\omega$ is only about a half of $\nu_{en} \simeq 1,600$ s$^{-1}$. This means that $\omega$ is not much less than $\nu_{en}$, so that the applicability condition is fulfilled only marginally. However, if the driving field $E_0$ is much stronger, $E_0 \gg E_{\rm Thr}$ (the case much more relevant to our theory), then the ion-neutral collision frequency $\nu_{in}$ stays the same, while $\omega \sim kV_0$ increases roughly in proportion with $E_0$, and the electron-neutral collision frequency, $\nu_{en}(T_e)$, increases even faster, $\nu_{en} \propto T_e \propto E_0^2$. This improves the applicability condition dramatically. On the other hand, at altitudes well below the magnetization boundary, where the ions can be treated as essentially unmagnetized (in consistency with the ion kinetic treatment of this paper), the two collision frequencies exponentially grow with decreasing height (in linear proportion with the neutral density). As a result, while $\nu_{in}$ becomes closer to $\omega$, $\nu_{en}$ goes farther away from it. This also improves the applicability of our electron kinetic approach, especially for large $E_0$.

There are also limitations associated with the employed kinetic approach to the ion-neutral collisions (i.e., the popular BGK collisional model with constant $\nu_{in}$), the simplified analytical choice of the velocity dependence of the electron-neutral collision frequency, $\nu_{en}(V_e) \propto V_e^2$, and some other approximations. We have already discussed the BGK model for ions, so will not repeat this discussion here. We will only mention that the biggest BGK model flaw, i.e., the absence of the collisional scattering that should redistribute the particles over angles in the ion-velocity space, is worst when describing

a stationary ion velocity distribution (Koontaweepunya et al., 2024). For the time-varying wave perturbations with $|\omega| \gtrsim \nu_{in}$, the error introduced by this flaw is less important than using the BGK model for the stationary distribution (though even for the latter the BGK collisional approach can be considered as a reasonably satisfactory approximation). As far as the velocity dependence of $\nu_{en}(V_e)$ is concerned, it is an average fitting model; the actual velocity dependence is more complicated and not necessarily well known for all conditions. Gurevich (1978) suggested a slightly different power-law dependence, $\nu_{en}(V_e) \propto V_e^{5/3}$. Direct comparison of figures in Gurevich (1978) shows, however, that our approximation of $\nu_{en}(V_e) \propto V_e^2$ works even better. For our specific analytical approach and results, this power-law velocity dependence is vital.

Finally, the new kinetic theory of this paper does not describe the ETI in full measure because, bearing in mind sufficiently short-wavelength waves, $|\omega| \gtrsim \nu_{en} \gg \delta_{en}\nu_{en}$, we have neglected local electron collisional cooling compared to the wave-related electron frictional heating and thermal transport. For longer-wavelength waves, $|\omega| \lesssim \delta_{en}\nu_{en}$, we can use the previously obtained kinetic or fluid-model results (Dimant & Sudan, 1995b, 1995c, 1997). Inclusion of the local electron collisional cooling into the present kinetic theory would require solving a more general second-order differential equation, which is even more complicated than equation (43). If needed, the general electron kinetic theory that covers the entire range of wave frequencies can be developed in the future.

## 6 Summary and Conclusions

In this paper, we have developed a fully kinetic analytical theory of the thermal Farley-Buneman instability (TFBI). This combined instability includes all known short-scale E-region instabilities, except the spatially inhomogeneous gradient drift instability (GDI). The latter usually produces relatively long-wavelength turbulence, typically tens or hundreds meters, for which the kinetic theory is not urgently needed, though such theories exist, e.g., Schlegel and St.-Maurice (1983); Basu (1995).

For the first time in the E-region instability theoretical research, the driving electric field $\mathbf{E}_0$ has been included in the kinetic description of ions. As a result, the ion thermal instability (ITE) driving has been automatically included in the kinetic theory. For electrons, in addition to the standard inclusion of $\mathbf{E}_0$ via the $\mathbf{E}_0 \times \mathbf{B}$-drift velocity, the driving electric field was included previously in the collisional kinetic theory developed

by Dimant and Sudan (1995a, 1995b). That theory had allowed the authors to predict the new effect of the electron thermal instability (ETI). Compared to that earlier kinetic theory, this paper considers a shorter-wavelength range, which is especially relevant to the kinetic treatment. We do not expect this short-wavelength range to be linearly unstable due to the ETI, but the reverse effect of increased wave stabilization and possible phase-velocity modifications caused by electron thermal effects associated with the explicitly included field $\mathbf{E}_0$ can affect dramatically the entire instability process and may be of importance for the accurate interpretation of radar observations.

The principal outcome of our TFBI-related kinetic theory is the linear wave dispersion relation given by equation (70). Structurally similar to the wave dispersion relations in the earlier kinetic theories for ions (Farley, 1963a; Lee et al., 1971; Schmidt & Gary, 1973; Ossakow et al., 1975; Schlegel & St.-Maurice, 1983; Basu, 1995), this much more complicated fully kinetic dispersion relation also includes only elementary functions combined with the plasma dispersion functions, $Z(x)$ (Fried & Conte, 1961). Unlike the earlier FBI-related theories, however, our equation (70) includes $Z$-functions of different arguments in both the ion and electron parts of the dispersion equation.

The results of our theory can be of importance for accurate interpretation of radar signals scattered from relatively high E-region altitudes where the ionospheric density irregularities with the wavelength corresponding to the given radar frequency are within the short-wavelength kinetic range. We should bear in mind, however, that this theory is still limited to altitudes where ions are essentially unmagnetized (below 110 km). It may also predict the power structure of the dissipation region of the spectra measured by rockets, though nonlinear effects may also contribute.

Using the general kinetic wave dispersion relation given by equation (70), one can determine the instability threshold for the given short-wavelength waves and establish whether the observed radar signals were scattered from linearly unstable or stable waves. For a sufficiently strong driving electric field, this equation will also help find the fastest growing unstable waves over the entire wavevector $\mathbf{k}$ space, provided the wave frequency $\omega$ satisfies the condition $\delta_{en}\nu_{en} \ll |\omega| \lesssim \nu_{en}$. The right-hand restriction is not as severe as it might look because the $e$-$n$ collision frequency $\nu_{en}$ increases with the electron temperature, which in turn sharply increases with the strong driving field. Finally, this linear dispersion equation will provide the wave phase velocities which may approximate

the observed radar signal Doppler shifts, even if the plasma turbulence is in the highly nonlinear state. To accurately predict the full irregularity spectrum and nonlinearly saturated wave amplitudes, fully kinetic computer simulations or much more complicated nonlinear models should be employed. Nevertheless, this theory enables one to validate these codes and make predictions of where the codes will prove useful.

## Appendix A Ion Kinetic Theory: Mathematical Details

In this appendix, we review some aspects of the ion kinetic theory and provide mathematical details of the derivations.

### A1 The ion background velocity distribution in the BGK collision-model approximation

Here we briefly reproduce, and give more details to, the analytical model of Koontaweepunya et al. (2024) regarding the background ion distribution function (BIDF) modified by a strong electric field $\mathbf{E}_0$. This simplified analytic model is based on the ion kinetic equation with the BGK collision model, see equation (4).

Introducing

$$F_0(u) \equiv \iint_{-\infty}^{\infty} f_{i0}(V_{ix}, V_{iy}, V_{iz}) dV_{iy} dV_{iz} \tag{A1}$$

and

$$\mathbf{a}_{i0} = \frac{e\mathbf{E}_0}{m_i}, \qquad u \equiv \frac{\nu_{in} V_{ix}}{a_{i0}}, \qquad u_T \equiv \frac{\nu_{in} V_{Tn}}{a_{i0}}, \qquad V_{Tn} = \left(\frac{T_n}{m_i}\right)^{1/2} \tag{A2}$$

(here $a_{i0} \equiv |\mathbf{a}_{i0}|$), we have

$$f_{i0}(\mathbf{V}_i) = \frac{F_0(u)}{2\pi V_{Tn}^2} \exp\left(-\frac{V_{iy}^2 + V_{iz}^2}{2V_{Tn}^2}\right), \tag{A3}$$

$$f_{i0}^{\mathrm{Coll}} = \frac{F_0^{\mathrm{Coll}}(u)}{2\pi V_{Tn}^2} \exp\left(-\frac{V_{iy}^2 + V_{iz}^2}{2V_{Tn}^2}\right),$$

$$F_0^{\mathrm{Coll}}(u) = \frac{n_0}{\sqrt{2\pi} V_{Tn}} \exp\left(-\frac{V_{ix}^2}{2V_{Tn}^2}\right) = \frac{n_0 \nu_{in}}{a_{i0}\sqrt{2\pi} u_T} \exp\left(-\frac{u^2}{2u_T^2}\right).$$

Then, integrating equation (4) over the entire domain of the perpendicular velocities $V_{iy}$ and $V_{iz}$, we reduce it to

$$\frac{dF_0}{du} = -\left(F_0 - F_0^{\mathrm{Coll}}(u)\right) \tag{A4}$$

or, equivalently,

$$\frac{dG_0}{du} = -G_0(u) + \frac{1}{\sqrt{2\pi} u_T} \exp\left(-\frac{u^2}{2u_T^2}\right), \tag{A5}$$

where

$$G_0(u) = \frac{a_{i0}}{n_0 \nu_{in}}\ F_0(V_x). \tag{A6}$$

The solution of equation (A5), under the asymptotic boundary conditions of $\lim_{u\to\pm\infty} G_0(u) = 0$, is given by

$$\begin{aligned} G_0(u) &= \frac{1}{2}\exp\left(-u + \frac{u_T^2}{2}\right)\left[1 + \mathrm{erf}\left(\frac{u - u_T^2}{\sqrt{2}u_T}\right)\right] \\ &= \frac{1}{2}\exp\left(-\frac{u^2}{2u_T^2}\right) w\left(-i\,\frac{u - u_T^2}{\sqrt{2}u_T}\right), \end{aligned} \tag{A7}$$

where $\mathrm{erf}(x)$ and $w(\zeta)$,

$$\begin{aligned} \mathrm{erf}(x) &= \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt, \\ w(\zeta) &= e^{-\zeta^2}\left(1 + \frac{2i}{\sqrt{\pi}}\int_0^\zeta e^{t^2}dt\right) = e^{-\zeta^2}\left[1 + \mathrm{erf}(i\zeta)\right] \end{aligned} \tag{A8}$$

are standard error functions (Abramowitz & Stegun, 1972).

Firstly, we discuss the small-field limit of $a_{i0} \to 0$. In this limit, the LHS of equation (A5) should also be small, resulting in

$$G_0(u) \to \frac{1}{\sqrt{2\pi}\,u_T}\ \exp\left(-\frac{u^2}{2u_T^2}\right), \tag{A9}$$

corresponding to $f_{i0} \to\ f_{i0}^{\mathrm{Coll}}$. According to the definitions given by equation (A2), the limit of $a_{i0} \to 0$ corresponds to $u_T \gg 1$. The asymptotic analysis of the solution (A7) shows that equation (A9) is valid not for the entire ion velocity distribution but only for the bulk of the ion velocity distribution constrained by the condition

$$u \ll u_T^2. \tag{A10}$$

Indeed, at large values of $|x|$, the error function has the following asymptotic behavior (Abramowitz & Stegun, 1972):

$$\mathrm{erf}\,(x) \approx \begin{cases} 1 - \frac{\exp\left(-x^2\right)}{x\sqrt{\pi}} & \text{if} \quad x \gg 1 \\ -1 + \frac{\exp\left(-x^2\right)}{(-x)\sqrt{\pi}} & \text{if} \quad (-x) \gg 1 \end{cases}. \tag{A11}$$

According to equation (A10), we have $u - u_T^2 < 0$ and $u_T^2 - u \approx u_T^2 \gg 1$, so that for the error function in equation (A7) we should use the bottom asymptotic expression (A11),

$$1 + \mathrm{erf}\left(\frac{u - u_T^2}{\sqrt{2}u_T}\right) \approx \frac{1}{u_T}\sqrt{\frac{2}{\pi}}\exp\left(-\frac{u^2}{2u_T^2} + u - \frac{u_T^2}{2}\right).$$

Then $G_0(u)$ does reduce to the asymptotic function given by equation (A9). We should bear in mind, however, that for the ion distribution outside the range of (A10), i.e., for $u \gtrsim u_T^2$, the effect of even small electric field cannot be neglected.

Figure 1 from Koontaweepunya et al. (2024) shows a few examples of the zeroth-order velocity distributions of $G_0(u)$. It reduces to the undisturbed Maxwellian (with the neutral temperature) under conditions $u_T \gg 1$ and $u_T^2 \gg u$, or, in the original variables,

$$a_{i0} = \frac{eE_0}{m_i} \ll \nu_{in}\sqrt{\frac{T_n}{m_i}}, \qquad V_{ix} \ll \frac{\nu_{in}V_{Tn}^2}{a_{i0}} = \frac{\nu_{in}T_n}{eE_0} \tag{A12}$$

As the electric field exceeds a critical value,

$$E_{\rm cr} = \frac{\nu_{in}\sqrt{m_i T_n}}{e} = \frac{B}{\kappa_i}\sqrt{\frac{T_n}{m_i}} \approx \frac{14.4}{\kappa_i}\left(\frac{T_n}{300{\rm K}}\right)^{\frac{1}{2}}\left(\frac{B}{0.5{\rm G}}\right)\frac{\rm mV}{\rm m}, \tag{A13}$$

the ion distribution function becomes noticeably distorted by the ion heating. Here we assumed $m_i \simeq 30\, m_p$, where $m_p$ is the proton mass, and $\kappa_i = \Omega_i/\nu_{in}$ is the ion magnetization parameter. The latter increases exponentially with the altitude because the collision frequency $\nu_{in}$ is proportional to the neutral density obeying mostly the barometric law. In this paper, we assume unmagnetized ions, $\kappa_i \ll 1$. At higher latitudes, this condition is marginally fulfilled near 110 km ($\kappa_i \lesssim 0.3$) but much better around 100 km ($\kappa_i \simeq 0.06$). The critical field, $E_{\rm cr}$, in this altitude range varies from $\simeq 50$ mV/m at 110 km to $\simeq 250$ mV/m at 100 km. During big geomagnetic storms, strong electric fields $E_0$ occur, often $E_0 \sim 100$ mV/m, but sometimes $E_0$ can even reach 250 mV/m and more. In such extreme events, one can expect significant distortions of the background ion distribution function (BIDF). Figure 1 from Koontaweepunya et al. (2024) shows that the BIDF can be significantly skewed, growing a prominent tail in the $\mathbf{E}_0$-direction. In the framework of the BGK approximation, in the two perpendicular to $\mathbf{E}_0$ directions $f_{i0}(\mathbf{V}_i)$ the BIDF remains close to the undisturbed Maxwellian function $f_{i0}^{\rm Coll}(\mathbf{V}_i)$. We should remind, however, that collisional angular scattering (not included in the BGK approximation) smears to some degree the extended velocity distribution towards those directions. This is perfectly clear from the PIC simulations also performed in Koontaweepunya et al. (2024).

In the original velocity variables, according to the above definitions, the entire BIDF is given by

$$\begin{aligned} f_{i0}(\mathbf{V}_i) &= \frac{n_0\nu_{in}}{2\pi a_{i0}V_{Ti}^2}\exp\left(-\frac{V_{iy}^2+V_{iz}^2}{2V_{Ti}^2}\right)G_0\left(u\right) \\ &= \frac{n_0\nu_{in}}{4\pi a_{i0}V_{Ti}^2}\exp\left[-\frac{\nu_{in}V_{ix}}{a_{i0}}+\frac{1}{2}\left(\frac{\nu_{in}V_{Ti}}{a_{i0}}\right)^2-\frac{V_{iy}^2+V_{iz}^2}{2V_{Ti}^2}\right] \\ &\times\left[1+{\rm erf}\left(\frac{V_{ix}}{\sqrt{2}V_{Ti}}-\frac{\nu_{in}V_{Ti}}{\sqrt{2}a_{i0}}\right)\right]. \end{aligned} \tag{A14}$$

Directly from equations (A5) and (A7), we can easily deduce the following expressions for the three first moments of the BIDF:

$$\int_{-\infty}^{\infty} G_0(u)du = 1, \qquad \int_{-\infty}^{+\infty} uG_0(u)du = 1, \qquad \int_{-\infty}^{\infty} u^2 G_0(u)du = u_T^2 + 2. \tag{A15}$$

The first one expresses the particle density conservation, $n_i = \int f_i d^3V_i$; the second one provides the expression for the $x$-component of the mean ion drift velocity, $\langle V_{ix}\rangle = a_{i0}/\nu_{in}$, where $\langle\cdots\rangle$ denotes the velocity average, $\langle\cdots\rangle = \int(\cdots)f_i d^3V_i / \int f_i d^3V_i$; and the third one provides the effective ion temperature, $T_{\text{eff}} = (m_i/3)(\mathbf{V}_i - \langle\mathbf{V}_i\rangle)^2$, established as the equilibrium balance between the $\mathbf{E}_0$-field frictional heating and collisional cooling, $T_{\text{eff}} = T_n + (1/3)m_i U_0^2$, $U_0 = |\langle\mathbf{V}_i\rangle| = eE_0/(m_i\nu_{in})$. One can also deduce all higher-order moments of the BIDF. They characterize the BIDF skewness, but this is of no importance for our purposes.

The exact BGK solution for the distorted ion background distribution function $f_{i0}(\mathbf{V}_i)$ given by equation (A73) is hard for further analytical treatment that involves three velocity integrations. In Sec. 2.1 we have constructed a shifted and heated bi-Maxwellian function, see equation (5), that accurately describes the exact Pedersen drift and total frictional heating by the electric field $\mathbf{E}_0$. This analytic construct, however, lacks the skewness of $f_{i0}$ described by the above BGK analytic solution or, to a much lesser degree, by that observed in the PIC simulations (Koontaweepunya et al., 2024), as we discuss below.

The exact BGK solution is not exact in the real physical sense, as we already discussed in Sec. 2.1. The BGK operator has a major flaw that it does not include collisional angular scattering from the main direction parallel to the driving $\mathbf{E}_0$-field (i.e., the $x$ direction) to the perpendicular, $y$ and $z$, directions. This leads to the fact that the BGK-described frictional heating of ions is distributed extremely uneven between different ion velocity components: in this approximation, the entire ion heating occurs strictly along the $x$ direction (purely parallel heating) with zero heating in the two other directions. In real situations, however, during an ion-neutral collision the effective rates of the momentum and energy losses are close in value, so that some parallel heating should inevitably be transferred to the other directions through the angular scattering. Such scattered heating has been observed in our PIC simulations (Koontaweepunya et al., 2024). Assuming the Gaussian approximation of $f_{i0}$, we can mimic a more realistic anisotropic frictional heating by choosing two different temperatures for the parallel to $\mathbf{E}_0$ and per-

pendicular to $\mathbf{E}_0$ directions ($T_x$ and $T_y$, respectively). In this paper, these two different temperatures are just free input parameters that could be specified, e.g., by comparison with PIC simulations. We should bear in mind, however, that these two parameters are not totally independent because they are constrained by the total balance of $\mathbf{E}_0$-field frictional heating,

$$\frac{T_x}{3}+\frac{2T_y}{3}=T_{\rm eff}=T_n+\frac{m_iU_0^2}{3}, \tag{A16}$$

see Sec. 2.1), so that only the ratio $T_x/T_y$ is really a free parameter (it should vary between 0 and 1, with a possible $E_0$-dependence).

### A2 Linear perturbations: No-field approximation for ions

Here we discuss linear wave perturbations of the ion velocity distribution in the no-field approximation (NFA). This approximation produced the well-known ion part of the pure FBI kinetic dispersion relation (Farley, 1963a; Lee et al., 1971; Schmidt & Gary, 1973; Ossakow et al., 1975; Schlegel & St.-Maurice, 1983; Basu, 1995). In the NFA approximation, the total electric field $\mathbf{E}$ in the ion kinetic equation includes only the electrostatic wave component, $\mathbf{E}=-\nabla\Phi$, but not the driving external DC component, $\mathbf{E}_0$.

In this paper, we use the NFA ion kinetic solution for comparisons and references, but the main reason why we review it here is that a small twist in its derivation produces an unexpected analytic evaluation of two non-conventional integrals. The evaluation of these integrals, depicted below in Sec. A21, becomes a crucial step for us when we solve in Sec. 2 the general case of kinetic ions under an arbitrarily strong electric field $\mathbf{E}_0$.

Now we start the NFA wave kinetic analysis. Linearizing equation (1) with respect to small wave perturbations, for a separate harmonic wave $\propto\exp[i(\mathbf{k}\cdot\mathbf{r}-\omega t)]$, we obtain

$$(\mathbf{k}\cdot\mathbf{V}_i-\omega_i)f^i_{\omega\mathbf{k}}=-i\nu_{in}f^{\rm Coll}_{i0}\eta_{\omega\mathbf{k}}+\frac{V_{Ti}^2}{\beta_T}\,\phi_{\omega\mathbf{k}}\left(\mathbf{k}\cdot\frac{\partial f_{i0}}{\partial\mathbf{V}_i}\right). \tag{A17}$$

Here

$$\omega_i\equiv\omega+i\nu_{in}=\omega_r+i(\gamma+\nu_{in}),\qquad \beta_T\equiv\frac{T_i}{T_e},\qquad V_{Ti}^2=\frac{T_i}{m_i}, \tag{A18}$$

$f_{i0}(\mathbf{V}_i)$ is the undisturbed uniform and constant BIDF, see equation (5), and $f^i_{\omega\mathbf{k}}$ is the corresponding wave perturbation. The parameter $T_i$ can be any constant temperature (e.g., the ion mean temperature $T_i=(2T_\perp+T_\parallel)/3$) because, in accord with the definitions of $\phi_{\omega\mathbf{k}}$, see equation (8), and parameters $\beta_T$, $V_{Ti}^2$, equation (A17) includes no actual $T_{i,e}$-dependencies (we have included the inconsequential parameter $T_i$ for convenience).

Without $\mathbf{E}_0$, the LHS of equation (A17) includes no $\mathbf{V}_i$-derivatives of $f^i_{\omega\mathbf{k}}$. This means neglecting the mean ion Pedersen drift and frictional heating. This simplifies dramatically the linear-wave analysis of the ion velocity distribution. The total neglect of the driving DC field $\mathbf{E}_0$ also yields $f_{i0} = f^{\rm Coll}_{i0}$, where $f^{\rm Coll}_{i0}$ is given by equation (2) (here we assume $T_i = T_n$). As a result, the corresponding purely algebraic equation (A17) yields

$$f^i_{\omega\mathbf{k}} = -\frac{i\nu_{in} f^{\rm Coll}_{i0}}{\mathbf{k}\cdot\mathbf{V}-\omega_i}\,\eta^i_{\omega\mathbf{k}} - \frac{(\mathbf{k}\cdot\mathbf{V}_i) f^{\rm Coll}_{i0}}{\beta_T(\mathbf{k}\cdot\mathbf{V}_i-\omega_i)}\,R_{\omega\mathbf{k}}. \tag{A19}$$

Using the expression for $\eta_{\omega\mathbf{k}}$ in terms of $f^i_{\omega\mathbf{k}}$, see equation (8), after the corresponding integration of the both sides of equation (A19), we obtain the sought-for linear ion relationship,

$$A_1\phi_{\omega\mathbf{k}} = \beta_T A_2 \eta^i_{\omega\mathbf{k}}, \tag{A20}$$

where

$$A_1 = -\left\langle \frac{\mathbf{k}\cdot\mathbf{V}_i}{\mathbf{k}\cdot\mathbf{V}_i-\omega_i}\right\rangle = -\left(1+\omega_i\left\langle\frac{1}{\mathbf{k}\cdot\mathbf{V}_i-\omega_i}\right\rangle\right), \tag{A21a}$$

$$A_2 = \left\langle \frac{\mathbf{k}\cdot\mathbf{V}_i-\omega}{\mathbf{k}\cdot\mathbf{V}_i-\omega_i}\right\rangle = 1+i\nu_{in}\left\langle\frac{1}{\mathbf{k}\cdot\mathbf{V}_i-\omega_i}\right\rangle. \tag{A21b}$$

Here the angular brackets denote the velocity average over the Maxwellian function $f^{\rm Coll}_{i0}(\mathbf{V}_i)$,

$$\begin{aligned}\langle\cdots\rangle &= \frac{\iiint(\cdots)\exp\left(-\frac{V_i^2}{2V_{Ti}^2}\right)d^3V_i}{\iiint\exp\left(-\frac{V_i^2}{2V_{Ti}^2}\right)d^3V_i}\\ &= \frac{1}{(2\pi)^{3/2}V_{Ti}^3}\iiint(\cdots)\exp\left(-\frac{V_i^2}{2V_{Ti}^2}\right)d^3V_i.\end{aligned} \tag{A22}$$

In accord with absence of the ion frictional heating, we have $T_i = T_\parallel = T_\perp = T_n$, so that $V^2_{Ti} = V^2_{Tx} = V^2_n$.

Before going any further, we check the long-wavelength (low-frequency) limit, loosely defined by the conditions $kV_{Ti} \ll \nu_{in}$, $|\omega| \ll \nu_{in}$, where $k \equiv |\mathbf{k}|$. The linear theory of such waves is usually successfully treated using much simpler fluid-model equations. It is straightforward to verify that in the long-wavelength limit the kinetic relationship given by equation (A20) reduces to the corresponding fluid-model relationship,

$$\phi_{\omega\mathbf{k}} \approx \beta_T\left[\frac{\omega\,(\omega+i\nu_{in})}{k^2V_{Ti}^2}-1\right]\eta^i_{\omega\mathbf{k}}. \tag{A23}$$

Now we proceed with the more general NFA kinetic relationship (A20) for arbitrary wavelengths. Temporarily, we set up the coordinate system in such a way that the wavevector $\mathbf{k}$ is directed along the $x$-axis: $k = k_x$ (so that $\mathbf{k}\cdot\mathbf{V}_i = kV_{ix}$). From equations

(A21) and (A22) we see that in order to calculate both multipliers, $A_1$ and $A_2$, we need to evaluate just a single integral

$$I(p) = \int_{-\infty}^{\infty} \frac{e^{-\xi^2}}{\xi - p}\, d\xi, \qquad p \equiv \frac{\omega_i}{\sqrt{2} k V_{Ti}}, \tag{A24}$$

($\omega_i \equiv \omega + i\nu_{in}$) so that

$$A_1 = -\left(1 + \frac{p}{\sqrt{\pi}}\, I(p)\right), \qquad A_2 = 1 + \frac{i\nu_{in}}{\omega + i\nu_{in}}\, \frac{p}{\sqrt{\pi}}\, I(p). \tag{A25}$$

Treating $\xi$ as a complex variable and temporarily making a change of the variable from $\xi$ to $\zeta = \xi - p$, we obtain

$$I(p) = e^{-p^2} J(p), \qquad J(p) \equiv \int_C \frac{e^{-\zeta^2 - 2p\zeta}}{\zeta}\, d\zeta, \tag{A26}$$

where the corresponding contour of integration in the complex plane, $C$, represents a straight line parallel to the real axis and shifted downward or upward in the complex plane by $i\,\mathrm{Im}\, p$, depending on the sign of $\mathrm{Im}\, p$. We can deform this contour to another contour $C'$ which is composed of the two semi-infinite segments of the horizontal $\zeta$ axis with $\mathrm{Im}\,\zeta = 0$ closed from below by an infinitely small semi-circle of the radius $\epsilon \to 0$ around $\zeta = 0$ if $\mathrm{Im}\, p > 0$ (or from above if $\mathrm{Im}\, p < 0$). Differentiating $J(p)$ with respect to the parameter $p$, we obtain

$$\frac{dJ}{dp} = -2 \int_{-i\epsilon - \infty}^{-i\epsilon + \infty} e^{-\zeta^2 - 2p\zeta}\, d\zeta = -2\sqrt{\pi} e^{p^2}.$$

Integrating this equation, we obtain $J(p) = \mathrm{const} - 2\sqrt{\pi} \int_0^p e^{t^2} dt$, where 'const' is determined by the solution at $p = 0$. Directly from the definition of both $J(p)$ for $p = 0$ and the contour $C'$, we see that the odd integrand reduces the entire integration to a semi-residual $i\pi$ for $\mathrm{Im}\, p > 0$ (anti-clockwise) or $-i\pi$ for $\mathrm{Im}\, p < 0$ (clockwise). As a result, we obtain

$$I(p) = \int_{-\infty}^{\infty} \frac{e^{-\xi^2}}{\xi - p}\, d\xi = i\pi w(p) \qquad \text{if} \quad \mathrm{Im}(p) > 0, \tag{A27a}$$

$$I(p) = \int_{-\infty}^{\infty} \frac{e^{-\xi^2}}{\xi - p}\, d\xi = -i\pi w(-p) \qquad \text{if} \quad \mathrm{Im}(p) < 0, \tag{A27b}$$

where the $w$-function, $w(x) = e^{-x^2}[1 + \mathrm{erf}(ix)]$, see equation (A8), can be also written as (Abramowitz & Stegun, 1972)

$$w(\zeta) = e^{-\zeta^2} \mathrm{erfc}(-i\zeta) = \frac{2}{\sqrt{\pi}}\, e^{-\zeta^2} \int_{-i\zeta}^{\infty \times e^{i\varphi}} e^{-t^2} dt. \tag{A28}$$

Here the phase $\varphi$ in the complex plane lies within the sector $-\pi/4 < \varphi < \pi/4$.

The $w$-function can be also represented in terms of the plasma dispersion function, $Z(z)$,

$$w(\zeta) \equiv -\frac{i}{\sqrt{\pi}}\, Z(\zeta)\,, \qquad Z(\zeta) = i\int_0^\infty \exp\left(i\zeta t - \frac{t^2}{4}\right) dt = i\sqrt{\pi}\, w(\zeta). \tag{A29}$$

In terms of the $Z$-functions, equation (A27) becomes

$$I(p) = \int_{-\infty}^{\infty} \frac{e^{-\xi^2}}{\xi - p}\, d\xi = \sqrt{\pi}\, Z(p) \qquad \text{if} \;\; \mathrm{Im}(p) > 0, \tag{A30a}$$

$$I(p) = \int_{-\infty}^{\infty} \frac{e^{-\xi^2}}{\xi - p}\, d\xi = -\sqrt{\pi}\, Z(-p) \qquad \text{if} \;\; \mathrm{Im}(p) < 0. \tag{A30b}$$

Using an easily provable relation $-Z(-z) = (Z(z^*))^*$, the RHS of equation (A30b) can be also written as $\sqrt{\pi}\,(Z(p^*))^*$.

The earlier kinetic theories of ions were restricted to the case of $\mathrm{Im}\, p > 0$ ($\gamma > -\nu_{in}$), because the opposite case of $\mathrm{Im}\, p < 0$ corresponds to the linearly stable, highly wave-damping regime of $\gamma < -\nu_{in}$. In the general case, however, one should bear in mind both cases because the major output of radar observations is the wave phase velocity, without definite knowledge of whether the observed plasma wave is linearly stable or unstable.

To conclude this section, we note that for sufficiently large $|\zeta|$ the function $Z(\zeta)$ has the following asymptotic series (Fried & Conte, 1961):

$$\begin{aligned} Z(\zeta) &\approx i\sqrt{\pi}\,\sigma(\zeta) e^{-\zeta^2} - \sum_{n=0}^{\infty} \frac{(n-1/2)!}{\sqrt{\pi}\,\zeta^{2n+1}} \\ &= i\sqrt{\pi}\,\sigma(\zeta) e^{-\zeta^2} - \frac{1}{\zeta}\left(1 + \frac{1}{2\zeta^2} + \frac{3}{4\zeta^4} + \frac{15}{8\zeta^6} + \ldots\right), \end{aligned} \tag{A31}$$

where

$$\sigma(\zeta) = \begin{cases} 0 & \text{if} \quad \mathrm{Im}\,\zeta > 0, \\ 1 & \text{if} \quad \mathrm{Im}\,\zeta = 0, \\ 2 & \text{if} \quad \mathrm{Im}\,\zeta < 0, \end{cases}$$

and the gamma function of a half-integer argument, $(n-1/2)! \equiv \Gamma(n+1/2)$, can be expressed in terms of the integer factorials as $\sqrt{\pi}\,(2n)!/(2^{2n} n!)$. Using equation (A31), one can verify that under conditions of $kV_{Ti}, |\omega| \ll \nu_{in}$, leading to $|R_i| \approx \nu_{in}$ and $|p| \gg 1$, the above expressions agree with fluid-model equation (A23).

### *A21 Useful integrals*

Now we slightly modify the above derivation by using a more general coordinate system. Instead of directing the wavevector $\mathbf{k}$ strictly along the $x$-axis, we will rotate the

coordinate axes around the $z$-axis, so that the wavevector $\mathbf{k}$ has two non-zero components: $k_x$ and $k_y$ (but still $k_z = 0$). In such coordinate system, one might use, e.g., the polar coordinates in the velocity space, $V_i$, $\theta_i$, so that $\mathbf{k} \cdot \mathbf{V}_i = kV_i \cos\theta_i$, $k = (k_x^2 + k_y^2)^{1/2}$, and obtain the above result in a more symmetric way with respect to the $\mathbf{k}$ components. But such derivation would produce nothing new and useful. If, however, we keep using the Cartesian coordinate system, then by comparing the results we will arrive at a non-trivial integral whose exact evaluation, exclusively in terms of the Gaussian and error functions, will help us obtain the wave dispersion relation in the general case of arbitrarily large electric field $\mathbf{E}_0$ (see below).

Now we proceed with the corresponding treatment. We are no longer interested in obtaining the relationship between the electric potential and density wave perturbations because we already know the result (see above). What we are pursuing here is the mutual consistency of two slightly different mathematical approaches, so that they give the same result. To this end, we restrict ourselves to the calculation of the following double integral:

$$R = \iint_{-\infty}^{\infty} \frac{\exp\left(-u^2\right) d^2u}{\mathbf{k} \cdot \mathbf{u} - p}. \tag{A32}$$

assuming $\operatorname{Im} p > 0$.

For the quantity $R$, the simple first approach with $k_y = 0$ and positive $k_x = k$, yields

$$\begin{aligned} R &= \int_{-\infty}^{\infty} \exp\left(-u_y^2\right) du_y \int_{-\infty}^{\infty} \frac{\exp\left(-u_x^2\right) du_x}{k_x \left(u_x - p\right)} = \frac{\sqrt{\pi}}{k_x}\, I(p) \\ &= \frac{i\pi^{3/2}}{k}\, w\left(p\right) = \frac{i\pi^{3/2}}{k}\, \exp\left(-p^2\right) \left[1 + \operatorname{erf}\left(ip\right)\right]. \end{aligned} \tag{A33}$$

The second approach with two non-zero vector components $k_x$ and $k_y$ yields

$$R = \iint_{-\infty}^{\infty} \frac{\exp\left(-u^2\right) d^2u}{\mathbf{k} \cdot \mathbf{u} - p} = \int_{-\infty}^{\infty} \exp\left(-u_y^2\right) du_y \int_{-\infty}^{\infty} \frac{\exp\left(-u_x^2\right) du_x}{k_x u_x + k_y u_y - kp}, \tag{A34}$$

where the internal integral has the same structure as that in equation (A24) with $p$ replaced by $p - k_y u_y/k$ (the restriction $\operatorname{Im} p > 0$ easily transfers to $\operatorname{Im}(p - k_y u_y/k) > 0$). Applying (A27), the internal integration yields

$$\begin{aligned} &\int_{-\infty}^{\infty} \frac{\exp\left(-u_x^2\right) du_x}{k_x u_x + k_y u_y - kp} = \frac{i\pi}{k_x}\, w\left(\frac{kp - k_y u_y}{k_x}\right) \\ &= \frac{i\pi^{3/2}}{k}\, \exp\left[-\left(\frac{kp - k_y u_y}{k_x}\right)^2\right] \left[1 + \operatorname{erf}\left(i\, \frac{kp - k_y u_y}{k_x}\right)\right]. \end{aligned} \tag{A35}$$

In the remaining integral of equation (A34), we change the variable $u_y$ to a new complex variable $\tau$,

$$\tau = \frac{kp - k_y u_y}{k_x}, \qquad u_y = \frac{kp - \tau k_x}{k_y}. \tag{A36}$$

In terms of the new variable $\tau$, we combine $\exp\left(-u_y^2\right)$ with the exponential function in equation (A35), using $\left(k_x^2 + k_y^2\right)^{1/2} = k$, and then deform the corresponding contour of integration in the complex $\tau$-plane to make this integration along the new real axis (the required conditions will be discussed later). This yields

$$R = \frac{i\pi^{3/2}}{k} \exp\left(-p^2\right) \int_{-\infty}^{\infty} \exp\left[-\frac{\left(k\tau - pk_x\right)^2}{k_y^2}\right] \left[1 + \operatorname{erf}\left(i\tau\right)\right] d\tau. \tag{A37}$$

Equating this $R$ to the same function given by equation (A33), we arrive at an unexpected and fairly simple integral relationship:

$$\int_{-\infty}^{\infty} \exp\left[-\frac{\left(k\tau - pk_x\right)^2}{k_y^2}\right] \left[1 + \operatorname{erf}\left(i\tau\right)\right] d\tau = 1 + \operatorname{erf}\left(ip\right). \tag{A38}$$

In this form, equation (A38) contains a clear asymmetry with respect to $k_x$ and $k_y$, which should not be the case. Besides, by its derivation, it is clear that this relationship may have serious restrictions associated with the fact that $k_x$ and $k_y$ have been implied to be real. With these restrictions and asymmetry, the usefulness of this relationship may be limited. It turns out, however, that after a simple generalization the applicability of a more general relation covers a much broader domain of complex parameters.

We demonstrate this by introducing complex parameters $A$, $B$, and $C$, which for equation (A38) reduce to $A = k/k_y$, $B = pk_x/k_y$, $C = i$, but in general are not limited to these specific values. In terms of these parameters, equation (A38) acquires a more general form:

$$\int_{-\infty}^{\infty} \exp\left[-\left(Az - B\right)^2\right] \left[1 + \operatorname{erf}\left(Cz\right)\right] dz = \frac{\sqrt{\pi}}{A} \left[\operatorname{sgn}(\operatorname{Re} \mathrm{A}) + \operatorname{erf}\left(\frac{BC}{\sqrt{A^2 + C^2}}\right)\right], \tag{A39}$$

where $\operatorname{sgn}(x)$ is the standard signum function of a real argument $x$: $\operatorname{sgn}(x) = x/|x|$ if $x \neq 0$ and $\operatorname{sgn}(x) = 0$ if $x = 0$. Below we prove equation (A39) directly, without reference to preceding equation (A38).

Before performing this proof, we note that equation (A39) should hold separately for the two separate parts of $[1 + \operatorname{erf}(Cz)]$: those proportional to "1" on both sides of this equation,

$$\int_{-\infty}^{\infty} \exp\left[-\left(Az - B\right)^2\right] dz = \operatorname{sgn}(\operatorname{Re} A)\, \frac{\sqrt{\pi}}{A}. \tag{A40}$$

and those proportional to the error function,

$$\int_{-\infty}^{\infty} \exp\left[-(Az-B)^2\right] \operatorname{erf}(Cz)\, dz = \frac{\sqrt{\pi}}{A} \operatorname{erf}\left(\frac{BC}{\sqrt{A^2+C^2}}\right) \tag{A41}$$

(recall that by the square root we always mean the branch with the positive real part). Equation (A40) presents a well-known integral whose convergence is restricted by $|\arg(A)| < \pi/4$ or, equivalently,

$$\operatorname{Re}(A^2) > 0. \tag{A42}$$

Equation (A41), however, is non-standard and less trivial. Popular tables of integrals, such as Dwight (1961); Jeffrey (1995); Gradshteyn and Ryzhik (2007) do not present the exact analytic result of such integration. Prudnikov et al. (1986) do this, but they present the result of such integration in terms of more complicated special functions, which are of no practical use.

To proceed with our integral, we temporarily denote the LHS of equation (A41) as

$$K(C) = \int_{-\infty}^{\infty} \exp\left[-(Az-B)^2\right] \operatorname{erf}(Cz)\, dz \tag{A43}$$

and differentiate the function $K(C)$ with respect to the complex parameter $C$. This yields

$$\frac{dK}{dC} = \frac{2}{\sqrt{\pi}} \exp\left(-\frac{B^2C^2}{A^2+C^2}\right) \int_{-\infty}^{\infty} \exp\left\{-\frac{\left[\left(A^2+C^2\right)z - AB\right]^2}{A^2+C^2}\right\} z dz.$$

Transforming the integral in the RHS as follows,

$$\begin{aligned}
&\int_{-\infty}^{\infty} \exp\left\{-\frac{\left[\left(A^2+C^2\right)z - AB\right]^2}{A^2+C^2}\right\} z dz \\
&= \int_{-\infty}^{\infty} \exp\left[-\left(A^2+C^2\right)\left(z - \frac{AB}{A^2+C^2}\right)^2\right]\left(z - \frac{AB}{A^2+C^2}\right) dz \\
&+ \frac{AB}{A^2+C^2} \int_{-\infty}^{\infty} \exp\left[-\left(A^2+C^2\right)\left(z - \frac{AB}{A^2+C^2}\right)^2\right] dz,
\end{aligned} \tag{A44}$$

we see that the contribution of the first integral goes away due to the oddity of the integrand, while the second integral, under restriction

$$\operatorname{Re}\left(A^2+C^2\right) > 0, \tag{A45}$$

can be evaluated in a standard way by deforming the integration path in the complex plane using, yielding $\sqrt{\pi}\left(A^2+C^2\right)^{-1/2}$. As a result, we obtain

$$\frac{dK}{dC} = \frac{2AB}{\left(A^2+C^2\right)^{3/2}} \exp\left(-\frac{B^2C^2}{A^2+C^2}\right). \tag{A46}$$

The definition of the error function is such that $\mathrm{erf}(0) = 0$, so that, according to equation (A43), we have $K(0) = 0$. Using this as the boundary condition, we integrate equation (A46) and obtain

$$\begin{aligned} K(C) &= \int_0^C \frac{2AB}{\left(A^2 + C^2\right)^{3/2}} \exp\left(-\frac{B^2C^2}{A^2 + C^2}\right) dC \\ &= \frac{\sqrt{\pi}}{A} \,\mathrm{erf}\left(\frac{BC}{\sqrt{A^2 + C^2}}\right). \end{aligned} \tag{A47}$$

This result can be easily verified by differentiating the RHS of equation (A47) with respect to $C$.

Equation (A47) closes the proof of equation (A41) and hence of fuller equation (A39) for complex parameters $A$, $B$, and $C$, under constraints imposed by equations (A42) and (A45). In the general case, reduction of the original integral to this form may involve a linear transform of the original variable in the complex plane with the corresponding deformation of the integration path to the real axis of $z$. This makes the total number of parameters (three) somewhat redundant. To avoid this redundancy and simplify the result, we assign for the parameter $C$ a specific value $C = i$, so that the two remaining parameters ($A$ and $B$) could be determined from the original integral. Then equation (A39) reduces to a more compact relationship:

$$\int_{-\infty}^{\infty} \exp\left[-\left(Az - B\right)^2\right] \left(1 + \mathrm{erf}\left(iz\right)\right) dz = \frac{\sqrt{\pi}}{A} \left[\mathrm{sgn}(\mathrm{Re}\, A) + \mathrm{erf}\left(\frac{iB}{\sqrt{A^2 - 1}}\right)\right]. \tag{A48}$$

The corresponding integral convergence conditions given by separate equations (A42) and (A45) reduce now to a single condition:

$$\mathrm{Re}(A^2 - 1) > 0. \tag{A49}$$

Recalling the definition of the standard function $w(x)$ given by equation (A28) and using transformations similar to those in equation (A44), we rewrite equation (A48) as

$$\begin{aligned} &\int_{-\infty}^{\infty} \exp\left[-\left(A^2 - 1\right)\left(z - \frac{AB}{A^2 - 1}\right)^2\right] w\left(z\right) dz \\ &= \frac{\sqrt{\pi}}{A}\, \tilde{w}\left(\frac{B}{\sqrt{A^2 - 1}}\right), \end{aligned} \tag{A50}$$

where the extended $w$-function, $\tilde{w}(z)$, is given by

$$\tilde{w}\left(\frac{B}{\sqrt{A^2 - 1}}\right) = \begin{cases} w\left(B/\sqrt{A^2 - 1}\right) & \text{if} \quad \mathrm{Re}\, A > 0, \\ -w\left(-B/\sqrt{A^2 - 1}\right) & \text{if} \quad \mathrm{Re}\, A < 0 \end{cases} \tag{A51}$$

and we ignore the degenerate case of $\mathrm{Re}\, A = 0$. Here "$-w(-z)$" can be identically represented as

$$-w(-z) = e^{-z^2}(-1 + \mathrm{erf}(iz)) = w(z) - 2e^{-z^2},$$

which means that $-w(-z)$ is shifted from $w(z)$ by a simple Gaussian function.

Due to the proportionality of the $w$ and $Z$-functions, the two $w$-functions on both sides of equation (A50) can be simultaneously replaced by the corresponding $Z$-functions:

$$\begin{aligned}&\int_{-\infty}^{\infty} \exp\left(-(A^2-1)\left(z - \frac{AB}{A^2-1}\right)^2\right) Z(z)dz \\ &= \frac{\sqrt{\pi}}{A}\, \tilde{Z}\left(\frac{B}{\sqrt{A^2-1}}\right),\end{aligned} \tag{A52}$$

where

$$\tilde{Z}\left(\frac{B}{\sqrt{A^2-1}}\right) = \begin{cases} Z\left(B/\sqrt{A^2-1}\right) & \text{if} \quad \mathrm{Re}\, A > 0, \\ -Z\left(-B/\sqrt{A^2-1}\right) & \text{if} \quad \mathrm{Re}\, A < 0 \end{cases} \tag{A53}$$

and "$-Z(-z)$" can be alternatively represented as

$$-Z(-z) = i\sqrt{\pi}\, e^{-z^2}(-1 + \mathrm{erf}(iz)) = Z(z) - 2i\sqrt{\pi}\, e^{-z^2}.$$

We emphasize that using proper linear transformation of the integration variable and contour deformation, any integrable convolution of the Gaussian and $w$ or $Z$-functions with two different complex arguments can always be reduced to the form given by equations (A48), (A50), or (A52) with only two adjustable parameters $A$ and $B$.

Now we evaluate another useful integral,

$$I_1 \equiv \int_{-\infty}^{\infty} z \exp\left[-(A^2-1)\left(z - \frac{AB}{A^2-1}\right)^2\right] w(z)dz, \tag{A54}$$

which differs from that in equation (A50) by the additional multiplier $z$ in the integrand. To proceed with this integral evaluation, we temporarily introduce an auxiliary integral,

$$I_2 \equiv (A^2-1)\int_{-\infty}^{\infty} \left(z - \frac{AB}{A^2-1}\right) \exp\left[-(A^2-1)\left(z - \frac{AB}{A^2-1}\right)^2\right] w(z)dz$$

On the one hand, this definition combined with equation (A50) gives

$$I_2 = (A^2-1)I_1 - B\sqrt{\pi}\, w\left(\frac{B}{\sqrt{A^2-1}}\right). \tag{A55}$$

On the other hand, we can evaluate $I_2$ directly, using integration by parts,

$$\begin{aligned} I_2 &= -\frac{1}{2}\int_{-\infty}^{\infty} \frac{d}{dz} \exp\left[-(A^2-1)\left(z - \frac{AB}{A^2-1}\right)^2\right] w\,(z)\, dz \\ &= \frac{1}{2}\int_{-\infty}^{\infty} \exp\left[-(A^2-1)\left(z - \frac{AB}{A^2-1}\right)^2\right] \frac{dw}{dz}\, dz, \end{aligned}$$

where, as usual, we imply a good behavior of the integrand at $\pm\infty$. Then, using the easily verifiable differential equation for the $w$-function,

$$\frac{dw}{dz} = 2\left(\frac{i}{\sqrt{\pi}} - zw\left(z\right)\right),$$

we obtain

$$I_2 = \int_{-\infty}^{\infty} \exp\left[-(A^2-1)\left(z - \frac{AB}{A^2-1}\right)^2\right] \times \left(\frac{i}{\sqrt{\pi}} - zw\left(z\right)\right) dz = \frac{i}{\sqrt{A^2-1}} - I_1. \tag{A56}$$

Comparison of equations (A55) and (A56) yields the explicit expression for $I_1$, i.e., another useful integral relation:

$$\int_{-\infty}^{\infty} z \exp\left[-(A^2-1)\left(z - \frac{AB}{A^2-1}\right)^2\right] w(z)dz = \frac{B\sqrt{\pi}}{A^2}\,\tilde{w}\left(\frac{B}{\sqrt{A^2-1}}\right) + \frac{i}{A^2\sqrt{A^2-1}}. \tag{A57}$$

In terms of the $Z$-functions, the equivalent integral relation is given by

$$\int_{-\infty}^{\infty} z \exp\left(-(A^2-1)\left(z - \frac{AB}{A^2-1}\right)^2\right) Z(z)dz = \frac{B\sqrt{\pi}}{A^2}\,\tilde{Z}\left(\frac{B}{\sqrt{A^2-1}}\right) - \frac{\sqrt{\pi}}{A^2\sqrt{A^2-1}}. \tag{A58}$$

One can easily verify that these integrals remain invariant with respect to the simultaneous change of $A$, $B$ to $-A$, $-B$, respectively. In doing so, we should bear in mind that this change toggles the piece-wise conditions of the extended function $\tilde{w}$, as defined in equation (A51), because $\mathrm{Re}(-A) > 0$ becomes $\mathrm{Re}\,A < 0$ and vice versa.

Thus, any integral convolution of a Gaussian function with an error function (or the related $w$ and $Z$-functions) of linearly independent complex arguments, either with or without an additional weight function $z$, can always be reduced to the two non-conventional integrals given by equations (A40), (A41), (A50) and (A57) (or equivalent equations (A52) and (A58) in terms the $Z$-function). This reduction may require a simple linear transformation of the integration variable $z$, followed by the standard deformation of the integration contour in the new complex plane to make it the real axis. Since the integrand functions are analytical, the main restriction is the good behavior of the integrand at $|z| \to \infty$ (the integrand there should tend to zero faster than $1/|z|$, providing the convergence of the integral with no contribution from the infinitely large contour). It is important that the results of these integrations are also represented in terms of the $w$ or $Z$ func-

tions, albeit with simple piece-wise conditions. Equations (A50) and (A57), or equivalent equations (A52) and (A58), play a crucial part in our calculations of the ion contribution to the general kinetic wave dispersion relation (see below).

### A3 Solving for $F^{(1)}_{\omega\mathbf{k}}$ and evaluating its contribution to $\eta^{i}_{\omega\mathbf{k}}$

Here we find the solution for $F^{(1)}_{\omega\mathbf{k}}$ and then evaluate the corresponding double integral. Equation (13a) can be rewritten as

$$\frac{\partial F^{(1)}_{\omega\mathbf{k}}}{\partial V_{ix}} + \frac{i}{a_{i0}}(k_x V_{ix} + k_y V_{iy} - \omega_i) F^{(1)}_{\omega\mathbf{k}} = \frac{P}{a_{i0}} \exp\left(-\frac{V_{ix}^2 + V_{iy}^2}{2V_{Tn}^2}\right), \tag{A59}$$

where

$$P \equiv \frac{\nu_{in} n_0 \eta^{i}_{\omega\mathbf{k}}}{2\pi V_{Tn}^2}. \tag{A60}$$

Before proceeding with solving general differential equation (A59), we consider its no-field limit of $a_{i0} \to 0$. In this limit, the term $\partial F^{(1)}_{\omega\mathbf{k}}/\partial V_{ix}$, can be neglected compared to the other, infinitely large terms, so that $F^{(1)}_{\omega\mathbf{k}}$ reduces to

$$F^{(1)}_{\omega\mathbf{k}} = -\frac{iP}{k_x V_{ix} + k_y V_{iy} - \omega_i} \exp\left(-\frac{V_{ix}^2 + V_{iy}^2}{2V_{Tn}^2}\right).$$

Assuming $k_y = 0$ and using equations (A24)–(A28) leading to

$$\int_{-\infty}^{\infty} \frac{e^{-\xi^2}}{\xi - p}\, d\xi = i\pi \tilde{w}(p),$$

we easily obtain

$$\begin{aligned} \iint_{-\infty}^{\infty} F^{(1)}_{\omega\mathbf{k}} dV_{ix} dV_{iy} &= \frac{\sqrt{2}\,\pi^{3/2} P V_{Tn}}{k}\; \tilde{w}\left(\frac{\omega_i}{\sqrt{2}\,kV_{Tn}}\right) \\ &= \sqrt{\frac{\pi}{2}}\,\frac{\nu_{in} n_0 \eta^{i}_{\omega\mathbf{k}}}{kV_{Tn}}\; \tilde{w}\left(\frac{\omega_i}{\sqrt{2}\,kV_{Tn}}\right) \end{aligned} \tag{A61}$$

We will use this equation for comparison with the corresponding limit following from the solution of general equation (A59).

Now we proceed with the general case of arbitrary $a_{i0}$. We start from the most important case of $\mathrm{Re}(R_i) > 0$. After a long and detailed description of the corresponding solution, in the end of this section, we will outline the opposite case of $\mathrm{Re}(R_i) < 0$.

The case of $\mathrm{Re}(R_i) > 0$ corresponds to $\gamma > -\nu_{in}$. This case covers all linearly unstable waves and probably the most of (if not all) other linear waves. In this case, a simple analysis of linear differential equation (A59) suggests choice of the asymptotic bound-

ary condition of $F^{(1)}_{\omega\mathbf{k}} \to 0$ as $V_x \to -\infty$, resulting in

$$F^{(1)}_{\omega\mathbf{k}}(V_{ix}, V_{iy}) = \frac{P}{a_{i0}} \int_{-\infty}^{V_{ix}} \exp\left[ - \frac{U_x^2 + V_{iy}^2}{2V_{Tn}^2} + \frac{ik_x \left(U_x^2 - V_{ix}^2\right)}{2a_{i0}} + \frac{i(k_y V_{iy} - \omega_i)\,(U_x - V_{ix})}{a_{i0}} \right] dU_x. \quad \text{(A62)}$$

To obtain the corresponding contribution to $\eta^i_{\omega\mathbf{k}}$ in equation (14), we should integrate $F^{(1)}_{\omega\mathbf{k}}$ with respect to both $V_{ix}$ and $V_{iy}$ over the entire 2-D velocity domain. Integrating first with respect to $V_{iy}$, for the corresponding $V_{iy}$-dependent multiplier we obtain

$$\int_{-\infty}^{\infty} \exp\left[ - \frac{V_{iy}^2}{2V_{Tn}^2} + \frac{ik_y V_{iy}\,(U_x - V_{ix})}{a_{i0}} \right] dV_{iy} = \sqrt{2\pi}\, V_{Tn} \exp\left[ - \frac{k_y^2 V_{Tn}^2\,(U_x - V_{ix})^2}{2a_{i0}^2} \right].$$

To evaluate this integral, we have changed the integration variable to a new complex variable in order to reduce the integrand to an easily integrable Gaussian function. Then, in the new complex plane we have deformed the corresponding integration path using a standard approach by making the final integration path to lie fully along the new real axis. As usual, this has been done by using the good analytical properties of the integrand, combined with the proper behavior of the Gaussian function at an infinite contour within the allowed domains of the complex plane that provide the integral convergence. We will do similar manipulations in a few other places without dwelling on such standard and well-known details.

Further, we have to integrate over $V_{ix}$. When doing this, it is more beneficial to swap the order of integrations between $V_{ix}$ and $U_x$. Adjusting the internal integration limits accordingly, we obtain

$$\iint_{-\infty}^{\infty} F^{(1)}_{\omega\mathbf{k}}(V_{ix}, V_{iy}) dV_{iy} dV_{ix} = \frac{PV_{Tn}\sqrt{2\pi}}{a_{i0}} \int_{-\infty}^{\infty} dU_x \exp\left( - \frac{U_x^2}{2V_{Tn}^2} \right) \times \int_{U_x}^{\infty} \exp\left[ - \frac{k_y^2 V_{Tn}^2\,(V_{ix} - U_x)^2}{2a_{i0}^2} - \frac{ik_x \left(V_{ix}^2 - U_x^2\right)}{2a_{i0}} + \frac{i\,(V_{ix} - U_x)\,\omega_i}{a_{i0}} \right] dV_{ix}. \quad \text{(A63)}$$

Redistributing terms within the exponents as follows,

$$- \frac{k_y^2 V_{Tn}^2\,(V_{ix} - U_x)^2}{2a_{i0}^2} - \frac{ik_x \left(V_{ix}^2 - U_x^2\right)}{2a_{i0}} + \frac{i\,(V_{ix} - U_x)\,\omega_i}{a_{i0}} = - \frac{k_y^2 V_{Tn}^2 + ik_x a_{i0}}{2a_{i0}^2} \left( V_{ix} - \frac{k_y^2 V_{Tn}^2 U_x + i\omega_i a_{i0}}{k_y^2 V_{Tn}^2 + ik_x a_{i0}} \right)^2 - \frac{(\omega_i - k_x U_x)^2}{2\left(k_y^2 V_{Tn}^2 + ik_x a_{i0}\right)}$$

we reduce equation (A63) to

$$\iint_{-\infty}^{\infty} F^{(1)}_{\omega\mathbf{k}}(V_{ix},V_{iy})dV_{iy}dV_{ix} = \frac{PV_{Tn}\sqrt{2\pi}}{a_{i0}} \times \int_{-\infty}^{\infty} \exp\left[-\frac{U_x^2}{2V_{Tn}^2} - \frac{(\omega_i - k_xU_x)^2}{2\left(k_y^2V_{Tn}^2 + ik_xa_{i0}\right)}\right] dU_x \times \int_{U_x}^{\infty} \exp\left[-\frac{k_y^2V_{Tn}^2 + ik_xa_{i0}}{2a_{i0}^2}\left(V_{ix} - \frac{k_y^2V_{Tn}^2U_x + i\omega_ia_{i0}}{k_y^2V_{Tn}^2 + ik_xa_{i0}}\right)^2\right] dV_{ix}, \tag{A64}$$

Then, in accord with the definition of the $w$-function given by equation (A8), the internal integral gives

$$\int_{U_x}^{+\infty} \exp\left[-\frac{k_y^2V_{Tn}^2 + ik_xa_{i0}}{2a_{i0}^2}\left(V_{ix} - \frac{k_y^2V_T^2U_x + iR_ia_{i0}}{k_y^2V_{Tn}^2 + ik_xa_{i0}}\right)^2\right] dV_{ix} = a_{i0}\sqrt{\frac{\pi}{2(k_y^2V_{Tn}^2 + ik_xa_{i0})}} \exp\left[\frac{(\omega_i - k_xU_x)^2}{2(k_y^2V_{Tn}^2 + ik_xa_{i0})}\right] w\left[\frac{\omega_i - k_xU_x}{\sqrt{2(k_y^2V_{Tn}^2 + ik_xa_{i0})}}\right].$$

Here, and elsewhere across the text, we choose the branch of the square root with the positive real part. As a result, equation (A64) yields

$$\iint_{-\infty}^{\infty} F^{(1)}_{\omega\mathbf{k}}(V_{ix},V_{iy})dV_{iy}dV_{ix} = \frac{\pi PV_{Tn}}{\sqrt{k_y^2V_{Tn}^2 + ik_xa_{i0}}} \times \int_{-\infty}^{\infty} \exp\left(-\frac{U_x^2}{2V_{Tn}^2}\right) w\left[\frac{\omega_i - k_xU_x}{\sqrt{2(k_y^2V_{Tn}^2 + ik_xa_{i0})}}\right] dU_x \tag{A65}$$

To evaluate the remaining integral, we make the argument of the error function a new integration variable

$$z = \frac{\omega_i - k_xU_x}{\sqrt{2(k_y^2V_{Tn}^2 + ik_xa_{i0})}} \tag{A66}$$

and then deform the integration path is such a way that it becomes the real axis in the complex $z$-plane. Then equation (A65) becomes

$$\iint_{-\infty}^{\infty} F^{(1)}_{\omega\mathbf{k}}(V_{ix},V_{iy})dV_{iy}dV_{ix} = \frac{\sqrt{2}\,\pi PV_{Tn}}{k_x} \int_{-\infty}^{\infty} \exp\left[-\frac{k_y^2V_{Tn}^2 + ik_xa_{i0}}{k_x^2V_{Tn}^2}\left(z - \frac{\omega_i}{\sqrt{2(k_y^2V_{Tn}^2 + ik_xa_{i0})}}\right)^2\right] w(z)dz. \tag{A67}$$

Designating

$$\frac{k_y^2V_{Tn}^2 + ik_xa_{i0}}{k_x^2V_{Tn}^2} = A^2 - 1, \qquad \frac{\omega_i}{\sqrt{2(k_y^2V_{Tn}^2 + ik_xa_{i0})}} = \frac{AB}{A^2-1},$$

we obtain

$$A = \frac{\sqrt{k^2V_{Tn}^2 + ik_xa_{i0}}}{|k_x|V_{Tn}}, \tag{A68a}$$

$$B = \frac{\omega_i}{|k_x|V_{Tn}}\sqrt{\frac{k_y^2V_{Tn}^2 + ik_xa_{i0}}{2(k^2V_{Tn}^2 + ik_xa_{i0})}}, \tag{A68b}$$

where $k^2 = k_x^2 + k_y^2$ (recall that in the assumed coordinate system $k_z = 0$). Now we can apply to equation (A67) the integral relation given by equation (A50). Then, recalling the definition of $P$ given by equation (A60), we obtain

$$\iint_{-\infty}^{\infty} F_{\omega\mathbf{k}}^{(1)} dV_{ix} dV_{iy} = \frac{\sqrt{\pi}\,\nu_{in} n_0 \eta_{\omega\mathbf{k}}^i}{\sqrt{2(k^2 V_{Tn}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}}\; w\left(\frac{\omega_i}{\sqrt{2(k^2 V_{Tn}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}}\right), \tag{A69}$$

where we have returned to the coordinate-invariant format, $k_x a_{i0} = \mathbf{k}\cdot\mathbf{a}_{i0}$. In the limit of $\mathbf{a}_{i0} \to 0$, equation (A69) reduces to equation (A61).

Now we outline the solution for $F_{\omega\mathbf{k}}^{(1)}$ in the opposite case of $\mathrm{Re}(R_i) < 0$, corresponding to $\gamma < -\nu_{in}$. Although this case may never be realized for any linear waves, we need to include it for completeness of the wave dispersion relation.

The major difference from the $\mathrm{Re}(R_i) > 0$ case is that if $\mathrm{Re}(R_i) < 0$ then equation (A59) should be solved using a different asymptotic boundary condition, namely $F_{\omega\mathbf{k}}^{(1)} \to 0$ as $V_x \to \infty$. After making the same steps that lead to equation (A63), we arrive now at a slightly modified solution,

$$\iint_{-\infty}^{\infty} F_{\omega\mathbf{k}}^{(1)}(V_{ix}, V_{iy}) dV_{iy} dV_{ix} = -\frac{P V_{Tn}\sqrt{2\pi}}{a_{i0}} \int_{-\infty}^{\infty} dU_x \exp\left(-\frac{U_x^2}{2V_{Tn}^2}\right)$$
$$\times \int_{-\infty}^{U_x} \exp\left[-\frac{k_y^2 V_{Tn}^2 \left(V_{ix} - U_x\right)^2}{2a_{i0}^2} - \frac{ik_x\left(V_{ix}^2 - U_x^2\right)}{2a_{i0}} + \frac{i\left(V_{ix} - U_x\right)\omega_i}{a_{i0}}\right] dV_{ix}. \tag{A70}$$

In this equation, the integral $\int_{U_x}^{\infty}(\cdots)dV_{ix}$ from equation (A63) is replaced by the same integral, but with different integration limits, $-\int_{-\infty}^{U_x}(\cdots)dV_{ix}$. Making the same steps as described above for the $\mathrm{Re}(R_i) > 0$ case, we arrive eventually at an expression

$$\iint_{-\infty}^{\infty} F_{\omega\mathbf{k}}^{(1)} dV_{ix} dV_{iy} = -\frac{\sqrt{\pi}\,\nu_{in} n_0 \eta_{\omega\mathbf{k}}^i}{\sqrt{2(k^2 V_{Tn}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}}\; w\left(-\frac{\omega_i}{\sqrt{2(k^2 V_{Tn}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}}\right) \tag{A71}$$

where the function $w(z)$ in the RHS of equation (A69) is replaced by $-w(-z)$; recall that $-w(-z) = w(z) - 2\exp(-z^2)$. The existence of two different functional expressions, depending on the sign of $\mathrm{Re}(R_i)$, is the same as in equations (A27) and (A51), although the latter has a different parameter condition that separates them.

### A4 Solving for $F_{\omega\mathbf{k}}^{(2)}$ and evaluating its contribution to $\eta_{\omega\mathbf{k}}^i$

Here we find the solution for $F_{\omega\mathbf{k}}^{(2)}$ and evaluate the corresponding contribution to $\eta_{\omega\mathbf{k}}^i$. Similarly to the above treatment for $F_{\omega\mathbf{k}}^{(1)}$, we provide detailed calculations for the major case of $\mathrm{Re}(R_i) > 0$ (i.e., $\gamma > -\nu_{in}$), and then present the result for the heavily damped waves of $\mathrm{Re}(R_i) < 0$ ($\gamma < -\nu_{in}$).

Equation (13b) can be rewritten as

$$\frac{\partial F^{(2)}_{\omega\mathbf{k}}}{\partial V_{ix}} + \frac{i}{a_{i0}}(k_x V_{ix} + k_y V_{iy} - \omega_i) F^{(2)}_{\omega\mathbf{k}} = \frac{iV_{Tn}^2}{\beta_T a_{i0}}\, \phi_{\omega\mathbf{k}} \left( k_x \frac{\partial}{\partial V_{ix}} + k_y \frac{\partial}{\partial V_{iy}} \right) \int_{-\infty}^{\infty} f_{i0} dV_{iz}, \tag{A72}$$

where, according to Eq. (5),

$$\int_{-\infty}^{\infty} f_{i0} dV_{iz} = \frac{n_0}{2\pi V_{Tx} V_\perp} \exp\left( - \frac{(V_{ix} - U_0)^2}{2V_{Tx}^2} - \frac{V_{iy}^2}{2V_{Ty}^2} \right). \tag{A73}$$

Temporarily denoting the RHS of Eq. (A72) by $R(V_{ix})$,

$$R(V_{ix}) \equiv \frac{iV_{Tn}^2}{\beta_T a_{i0}}\, \phi_{\omega\mathbf{k}} \left( k_x \frac{\partial}{\partial V_{ix}} + k_y \frac{\partial}{\partial V_{iy}} \right) \int_{-\infty}^{\infty} f_{i0} dV_{iz},$$

we obtain the solution for $F^{(2)}_{\omega\mathbf{k}}$,

$$F^{(2)}_{\omega\mathbf{k}}(V_{ix}) = \exp\left\{ - \frac{i}{a_{i0}} \left[ \frac{k_x V_{ix}^2}{2} + (k_y V_{iy} - \omega_i) V_{ix} \right] \right\} \times \int_{-\infty}^{V_{ix}} R(U_x) \exp\left\{ \frac{i}{a_{i0}} \left[ \frac{k_x U_x^2}{2} + (k_y V_{iy} - \omega_i) U_x \right] \right\} dU_x,$$

that satisfies the boundary condition of $\lim_{V_{ix}\to-\infty} F^{(2)}_{\omega\mathbf{k}}(V_{ix}) = 0$.

In order to find $F^{(2)}_{\omega\mathbf{k}}$, we need to solve differential equation (13b), where $\int_{-\infty}^{\infty} f_{i0} dV_{iz}$ is given by equation (6). Under assumption of $\lim_{V_{ix}\to-\infty} F^{(2)}_{\omega\mathbf{k}} = 0$, the formal solution can be written as

$$F^{(2)}_{\omega\mathbf{k}} = Q \int_{-\infty}^{V_{ix}} dU_x \left( \frac{k_x (U_x - U_0)}{V_{Tx}^2} + \frac{k_y V_{iy}}{V_{Ty}^2} \right) \times \exp\left\{ - \frac{(U_x - U_0)^2}{2V_{Tx}^2} - \frac{V_{iy}^2}{2V_{Ty}^2} + \frac{i}{a_{i0}} \left[ \frac{k_x \left( U_x^2 - V_{ix}^2 \right)}{2} + (k_y V_{iy} - \omega_i) (U_x - V_{ix}) \right] \right\}, \tag{A74}$$

where

$$Q \equiv - \frac{iV_{Tn}^2 n_0 \phi_{\omega\mathbf{k}}}{2\pi \beta_T a_{i0} V_{Tx} V_{Ty}}. \tag{A75}$$

To obtain the ion contribution to the linear dispersion relations, we need to integrate $F^{(2)}_{\omega\mathbf{k}}$ over the two remaining components in the ion velocity space ($V_{ix}$ and $V_{iy}$). First, we integrate $F^{(2)}_{\omega\mathbf{k}}$ over $V_{iy}$. This straightforward integration yields

$$\int_{-\infty}^{\infty} F^{(2)}_{\omega\mathbf{k}} dV_{iy} = \sqrt{2\pi} V_T Q \int_{-\infty}^{V_{ix}} dU_x \left[ \frac{k_x (U_x - U_0)}{V_{Tx}^2} + \frac{ik_y^2 (U_x - V_{ix})}{a_{i0}} \right] \times \exp\left\{ - \frac{(U_x - U_0)^2}{2V_{Tx}^2} + \frac{i}{a_{i0}} \left[ \frac{k_x \left( U_x^2 - V_{ix}^2 \right)}{2} - \omega_i (U_x - V_{ix}) \right] - \frac{k_y^2 V_{Ty}^2 (U_x - V_{ix})^2}{2a_{i0}^2} \right\}. \tag{A76}$$

Second, we need to integrate this expression over $V_{ix}$. This will be a multi-step procedure.

We start by introducing an intermediate quantity

$$M \equiv \int_{-\infty}^{\infty} dV_{ix} \int_{-\infty}^{V_{ix}} dU_x \left[ \frac{k_x(U_x - U_0)}{V_{Tx}^2} + \frac{ik_y^2 \left(U_x - V_{ix}\right)}{a_{i0}} \right]$$
$$\times \exp\left\{ - \frac{(U_x - U_0)^2}{2V_{Tx}^2} + \frac{i}{a_{i0}} \left[ \frac{k_x \left(U_x^2 - V_{ix}^2\right)}{2} - \omega_i \left(U_x - V_{ix}\right) \right] - \frac{k_y^2 V_{Ty}^2 \left(U_x - V_{ix}\right)^2}{2a_{i0}^2} \right\},$$

so that

$$\int_{-\infty}^{\infty} F_{\omega \mathbf{k}}^{(2)} dV_{ix} dV_{iy} = \sqrt{2\pi}\, V_T Q M. \tag{A77}$$

In order to simplify further calculations, we swap the order of integration between $V_{ix}$ and $U_x$,

$$M = \int_{-\infty}^{\infty} dU_x \exp\left[ - \frac{(U_x - U_0)^2}{2V_{Tx}^2} \right] \int_{U_x}^{\infty} dV_{ix} \left[ \frac{k_x(U_x - U_0)}{V_{Tx}^2} + \frac{ik_y^2}{a_{i0}} \left(U_x - V_{ix}\right) \right]$$
$$\times \exp\left\{ - \frac{k_y^2 V_{Ty}^2}{2a_{i0}^2} \left(U_x - V_{ix}\right)^2 + \frac{i}{a_{i0}} \left[ \frac{k_x \left(U_x^2 - V_{ix}^2\right)}{2} - \omega_i \left(U_x - V_{ix}\right) \right] \right\}. \tag{A78}$$

Then, redistributing the terms inside both the pre-exponent and the exponents, we rewrite equation (A78) as

$$M = \int_{-\infty}^{\infty} dU_x \exp\left[ - \frac{(U_x - U_0)^2}{2V_{Tx}^2} - \frac{(\omega_i - k_x U_x)^2}{2(k_y^2 V_\perp^2 + ik_x a_{i0})} \right] \int_{U_x}^{\infty} dV_{ix}$$
$$\times \left[ \frac{k_x(U_x - U_0)}{V_{Tx}^2} + \frac{k_y^2(\omega_i - k_x U_x)}{k_y^2 V_{Ty}^2 + ik_x a_{i0}} - \frac{ik_y^2}{a_{i0}} \left( V_{ix} - \frac{k_y^2 V_{Ty}^2 U_x + iR_i a_{i0}}{V_{Ty}^2 k_y^2 + ik_x a_{i0}} \right) \right]$$
$$\times \exp\left[ - \frac{k_y^2 V_{Ty}^2 + ik_x a_{i0}}{2a_{i0}^2} \left( V_{ix} - \frac{k_y^2 V_{Ty}^2 U_x + i\omega_i a_{i0}}{k_y^2 V_{Ty}^2 + ik_x a_{i0}} \right)^2 \right]. \tag{A79}$$

This simplifies evaluating the internal integral over $V_{ix}$, so that we obtain

$$M = \int_{-\infty}^{\infty} \exp\left[ - \frac{(U_x - U_0)^2}{2V_{Tx}^2} \right] K(U_x) dU_x, \tag{A80}$$

where

$$K(U_x) \equiv \frac{\sqrt{\pi}\, a_{i0}}{\sqrt{2(k_y^2 V_{Ty}^2 + ik_x a_{i0})}} \left[ \frac{k_x(U_x - U_0)}{V_{Tx}^2} + \frac{k_y^2(\omega_i - k_x U_x)}{V_{Ty}^2 k_y^2 + ik_x a_{i0}} \right]$$
$$\times \; w\left[ \frac{\omega_i - k_x U_x}{\sqrt{2(k_y^2 V_{Ty}^2 + ik_x a_{i0})}} \right] - \frac{ik_y^2 a_{i0}}{k_y^2 V_{Ty}^2 + ik_x a_{i0}}.$$

To calculate the remaining integral over $U_x$, we make the complex argument of the $w$-function a new integration variable,

$$z = \frac{\omega_i - k_x U_x}{\sqrt{2(k_y^2 V_{Ty}^2 + ik_x a_{i0})}}.$$

This transforms equation (A80) to

$$M = \frac{\sqrt{2\pi}a_{i0}}{k_x V_{Tx}^2}\int_{-\infty}^{\infty}\left[\frac{k_y^2(V_{Tx}^2 - V_{Ty}^2) - ik_x a_{i0}}{\sqrt{k_y^2 V_{Ty}^2 + ik_x a_{i0}}}\, z + \frac{R_i - k_x U_0}{\sqrt{2}}\right]$$
$$\times \exp\left\{-\frac{k_y^2 V_{Ty}^2 + ik_x a_{i0}}{k_x^2 V_{Tx}^2}\left[z - \frac{\omega_i - k_x U_0}{\sqrt{2(k_y^2 V_{Ty}^2 + ik_x a_{i0})}}\right]^2\right\} w(z)dz \qquad \text{(A81)}$$
$$- \frac{i\sqrt{2\pi}\, k_y^2 a_{i0} V_{Tx}}{k_y^2 V_{Ty}^2 + ik_x a_{i0}},$$

where, in a usual way, assuming the good analytic behavior of the integrand and its integrability at infinities, we have reduced the new integration path in the complex $z$-plane to its real axis.

Introducing temporary parameters

$$A = \frac{\sqrt{k_x^2 V_{Tx}^2 + k_y^2 V_{Ty}^2 + ik_x a_{i0}}}{k_x V_{Tx}}, \qquad \text{(A82a)}$$

$$B = \frac{\omega_i - k_x U_0}{k_x V_{Tx}}\sqrt{\frac{k_y^2 V_{Ty}^2 + ik_x a_{i0}}{2(k_x^2 V_{Tx}^2 + k_y^2 V_{Ty}^2 + ik_x a_{i0})}}, \qquad \text{(A82b)}$$

we have

$$\frac{k_y^2 V_{Ty}^2 + ik_x a_{i0}}{k_x^2 V_{Tx}^2} = A^2 - 1, \qquad \frac{\omega_i - k_x U_0}{\sqrt{2(k_y^2 V_{Ty}^2 + ik_x a_{i0})}} = \frac{AB}{A^2 - 1}.$$

so that we now can apply to (A81) equations (A50) and (A57). After a straightforward algebra and by using equation (A77), we arrive at the sought-for result

$$\iint_{-\infty}^{\infty} F_{\omega\mathbf{k}}^{(2)} dV_{ix} dV_{iy} = -\frac{k^2 V_{Tn}^2 n_0 \phi_{\omega k}}{\beta_T (k_x^2 V_{Tx}^2 + k_y^2 V_{Ty}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}$$
$$\times\left[1 + \frac{i\sqrt{\pi}(\omega_i - \mathbf{k}\cdot\mathbf{U}_0)}{\sqrt{2(k_x^2 V_{Tx}^2 + k_y^2 V_{Ty}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}}\, w\left(\frac{\omega_i - \mathbf{k}\cdot\mathbf{U}_0}{\sqrt{2(k_x^2 V_{Tx}^2 + k_y^2 V_{Ty}^2 + i\mathbf{k}\cdot\mathbf{a}_{i0})}}\right)\right], \qquad \text{(A83)}$$

where, similarly to equation (A69), we have expressed the result in to the coordinate-invariant form, $k_x = k_x$, $k_y = k_y$, and recalled that

$$\mathbf{U}_0 = \frac{\mathbf{a}_{i0}}{\nu_{in}} \qquad \text{(A84)}$$

For $V_{Tx}^2 = V_{Ty}^2 = V_{Tn}^2$, $\mathbf{a}_{i0} = 0$, $\mathbf{U}_0 = 0$, equation (A83) reduces to

$$\iint_{-\infty}^{\infty} F_{\omega\mathbf{k}}^{(2)} dV_{ix} dV_{iy} = -\frac{n_0 \phi_{\omega k}}{\beta_T}\left[1 + \sqrt{\frac{\pi}{2}}\,\frac{i\omega_i}{kV_{Tn}}\, w\left(\frac{\omega_i}{\sqrt{2}kV_{Tn}}\right)\right], \qquad \text{(A85)}$$

in full agreement with the corresponding no-field expression.

The non-standard case of heavily damped waves, $\gamma < -\nu_{in}$, i.e., of $\mathrm{Re}(R_i) < 0$, for $F^{(2)}_{\omega\mathbf{k}}$ is fully analogous to that of $F^{(1)}_{\omega\mathbf{k}}$. Using the same arguments that led us to equation (A70), instead of equation (A74) we obtain now the solution

$$F^{(2)}_{\omega\mathbf{k}} = -Q \int_{V_{ix}}^{\infty} dU_x \left( \frac{k_x (U_x - U_0)}{V_{Tx}^2} + \frac{k_y V_{iy}}{V_{Ty}^2} \right) \times \exp\left\{ - \frac{(U_x - U_0)^2}{2V_{Tx}^2} - \frac{V_{iy}^2}{2V_{Ty}^2} + \frac{i}{a_{i0}} \left[ \frac{k_x \left(U_x^2 - V_{ix}^2\right)}{2} + \left(k_y V_{iy} - \omega_i\right)\left(U_x - V_{ix}\right) \right] \right\},$$

which differs from equation (A74) by the integration limits and the sign in front of $Q$. As a result, making the same further steps that led us to equation (A83), we obtain for the integral $\int\int_{-\infty}^{\infty} F^{(2)}_{\omega\mathbf{k}} dV_{ix} dV_{iy}$ the same expression, except that in the RHS of equation (A83) the $w$-function of its complex argument $z$ is replaced by $-w(-z) = w(z) - 2\exp(-z^2)$, in total analogy with equation (A71) for $F^{(1)}_{\omega\mathbf{k}}$.

## Appendix B Simplification of the General Solution for Electrons

In this appendix, we simplify the exact general solution for $f^{e}_{\omega\mathbf{k}}(\xi)$ given by equation (59) and preceding equations (46)–(49). The goal of these simplifications is to make the solution more tractable for practical uses. We will make a series of approximations based on the fact that the absolute value of the complex parameter $\mu$ is sufficiently large. In this appendix, we use a local variable $z$ which is not to be confused with such notation from section 3.1.

Firstly, we approximate the modified Bessel functions $\mathrm{I}_\mu(x/2)$ and $\mathrm{K}_\mu(x/2)$ by using the following exact integral representations (Erdelyi, 1955; Arfken et al., 2013),

$$\mathrm{I}_\mu\left(\frac{x}{2}\right) = \frac{1}{\sqrt{\pi}\,\Gamma\left(\mu + \frac{1}{2}\right)} \left(\frac{x}{4}\right)^\mu \int_{-1}^{1} \exp\left(\frac{xt}{2}\right) \left(1 - t^2\right)^{\mu - \frac{1}{2}} dt, \tag{B1a}$$

$$\mathrm{K}_\mu\left(\frac{x}{2}\right) = \frac{\sqrt{\pi}}{\Gamma\left(\mu + \frac{1}{2}\right)} \left(\frac{x}{4}\right)^\mu \int_{1}^{\infty} \exp\left(-\frac{xt}{2}\right) \left(t^2 - 1\right)^{\mu - \frac{1}{2}} dt. \tag{B1b}$$

These relations are valid for the real positive argument $x$ and arbitrary complex order $\mu$ (whose real part is assumed positive). Analytically continuing the real domain of integrations in equations (B1a) and (B1b) into the complex plane of $t$, deforming the contours of integration in such a way that they pass through the most appropriate saddle points with the direction of passage aligned with the steepest descent, we will eventu-

ally arrive at the following approximate expressions:

$$\mathrm{I}_\mu\left(\frac{x}{2}\right) \approx \tilde{I}_\mu\left(\frac{x}{2}\right)$$
$$\equiv \frac{1}{\sqrt{\pi}\,(4\eta^2+x^2)^{\frac{1}{4}}}\left(\frac{\sqrt{4\eta^2+x^2}-2\eta}{x}\right)^{\eta+\frac{1}{2}}\exp\left(\frac{\sqrt{4\eta^2+x^2}}{2}\right), \tag{B2a}$$
$$\mathrm{K}_\mu\left(\frac{x}{2}\right) \approx \tilde{K}_\mu\left(\frac{x}{2}\right)$$
$$\equiv \frac{\sqrt{\pi}}{(4\eta^2+x^2)^{\frac{1}{4}}}\left(\frac{\sqrt{4\eta^2+x^2}+2\eta}{x}\right)^{\eta+\frac{1}{2}}\exp\left(-\frac{\sqrt{4\eta^2+x^2}}{2}\right), \tag{B2b}$$

where we temporarily introduced a new parameter

$$\eta \equiv \mu - \frac{1}{2}. \tag{B3}$$

Approximations given by equation (B2) involve only elementary functions. While assuming $|\mu| \gg 1$, the difference between $\mu$ and $\eta$ may seem negligible, but in the process of derivations it is better to retain this difference and neglect it only at the very end. From equation (B1), we can easily verify that the approximations $\tilde{I}_\mu$ and $\tilde{K}_\mu$ are related as

$$\mathrm{I}_\mu\left(\frac{x}{2}\right)\mathrm{K}_\mu\left(\frac{x}{2}\right) \approx \tilde{I}_\mu\left(\frac{x}{2}\right)\tilde{K}_\mu\left(\frac{x}{2}\right) = \frac{1}{\sqrt{4\eta^2+x^2}}. \tag{B4}$$

This approximate relation will dramatically simplify our results, as will become clear soon.

As $|\eta| \to \infty$, the approximate functions $\tilde{I}_\mu$ and $\tilde{K}_\mu$ asymptotically converge to the actual modified Bessel functions. For the finite parameter $\eta$, the accuracy of these approximations monotonically increases with both $|\eta|$ and $x$. Direct numerical comparisons show that for sufficiently large $|\eta|$ these approximations are quite accurate. Below we demonstrate analytically that the functions $\tilde{I}_\mu$ approximates $\mathrm{I}_\mu$ within the 10% accuracy for $|\eta| \gtrsim 5$ and 2% for $|\eta| \gtrsim 25$, while $\tilde{K}_\mu$ approximates $\mathrm{K}_\mu$ within even a much better accuracy of 1% for $|\eta| \gtrsim 5$ and 0.1% for $|\eta| \gtrsim 25$.

To quantify the accuracy of these approximations in a systematic way, we consider the worst-accuracy limit of $x \ll 1$. In this limit, equations (53) and (54) lead to

$$\mathrm{I}_\mu\left(\frac{x}{2}\right) \approx \frac{1}{\Gamma(\mu+1)}\left(\frac{x}{2}\right)^{\mu} = \frac{1}{(\eta+1/2)\Gamma(\eta+1/2)}\left(\frac{x}{4}\right)^{\eta+\frac{1}{2}} \tag{B5a}$$
$$\mathrm{K}_\mu\left(\frac{x}{2}\right) \approx \frac{\pi}{2\sin(\mu\pi)\,\Gamma(1-\mu)}\left(\frac{x}{2}\right)^{-\mu} = \frac{1}{2}\,\Gamma\left(\eta+\frac{1}{2}\right)\left(\frac{x}{4}\right)^{-\left(\eta+\frac{1}{2}\right)} \tag{B5b}$$

where we used $\mu = \eta + 1/2$ and the well-known gamma-function relations

$$\Gamma(z+1) = z\Gamma(z), \qquad \Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin\mu\pi}. \tag{B6}$$

In the same limit of $x \ll 1$, the approximate functions $\tilde{I}_\mu$ and $\tilde{K}_\mu$ are given by

$$\tilde{I}_\mu\left(\frac{x}{2}\right) \approx \frac{1}{\sqrt{2\pi\eta}}\left(\frac{x}{4\eta}\right)^{\eta+\frac{1}{2}} e^{\eta}, \qquad \tilde{K}_\mu\left(\frac{x}{2}\right) \approx \sqrt{\frac{\pi}{2\eta}}\left(\frac{4\eta}{x}\right)^{\eta+\frac{1}{2}} e^{-\eta}, \tag{B7}$$

so that for the ratios of the approximate functions to the actual ones we have

$$\frac{\tilde{I}_\mu(x/2)}{\mathrm{I}_\mu(x/2)} \approx \frac{1+1/(2\eta)}{\Lambda\left(\eta\right)}, \qquad \frac{\tilde{K}_\mu(x/2)}{\mathrm{K}_\mu(x/2)} \approx \Lambda\left(\eta\right), \tag{B8}$$

where

$$\Lambda\left(\eta\right) \equiv \frac{\sqrt{2\pi}}{\Gamma\left(\eta+\frac{1}{2}\right)}\left(\frac{\eta}{e}\right)^{\eta}. \tag{B9}$$

Using Stirling's formula with the first- and second-order corrections (Arfken et al., 2013),

$$\Gamma(z+1) \approx \sqrt{2\pi z}\left(\frac{z}{e}\right)^{z}\left(1+\frac{1}{12z}+\frac{1}{288z^2}\right), \tag{B10}$$

where we assume $|z| \gtrsim 1$, and expanding $\Lambda\left(\eta\right)$ in Taylor series with respect to $1/\eta$, we arrive at a simple approximation

$$\Lambda\left(\eta\right) \approx 1+\frac{1}{24\eta}+\frac{5}{1152\eta^2}. \tag{B11}$$

Then equations (B8) and (B11) yield

$$\frac{\tilde{I}_\mu(x/2)}{\mathrm{I}_\mu(x/2)} \approx 1+\frac{11}{24\eta}-\frac{3}{128\eta^2} \approx 1+\frac{0.46}{\eta}-\frac{0.023}{\eta^2}, \tag{B12a}$$

$$\frac{\tilde{K}_\mu(x/2)}{\mathrm{K}_\mu(x/2)} \approx 1+\frac{1}{24\eta}+\frac{5}{1152\eta^2} \approx 1+\frac{0.0417}{\eta}+\frac{0.0043}{\eta^2}. \tag{B12b}$$

The inaccuracy of the approximations $\tilde{I}_\mu(x/2)$ and $\tilde{K}_\mu(x/2)$ is determined by the corresponding highest-order corrections $\propto 1/\eta$. Comparing them with the second-order corrections $\propto 1/\eta^2$, we see that these expansions are valid for $|\eta| \gg 0.1$, i.e., virtually for all values of interest. We see that the function $\tilde{K}_\mu(x/2)$ approximates the actual function $\mathrm{K}_\mu(x/2)$ within 10% accuracy for $|\eta|$ even as small as $|\eta| \sim 0.5$. However, to provide the same 10% accuracy for $\tilde{I}_\mu(x/2)$, the parameter $|\eta|$ should be at least an order of magnitude larger, $|\eta| \gtrsim 5$. The physical estimates for the E-region conditions ($|\eta| \approx |\mu| \gtrsim 25$) made in Sec. 3.2 put the parameter $\eta$ well within this bound. The inaccuracy of the product given by equation (B4) is determined by

$$\frac{\tilde{I}_\mu(x/2)\tilde{K}_\mu(x/2)}{\mathrm{I}_\mu(x/2)\mathrm{K}_\mu(x/2)} \approx 1+\frac{1}{2\eta}. \tag{B13}$$

The accuracy of both approximations improves as $x$ increases, but for all $x$ the accuracy of $\tilde{K}_\mu(x/2)$ always remains significantly better than that of $\tilde{I}_\mu(x/2)$.

After substituting the original functions $\mathrm{I}_\mu$ and $\mathrm{K}_\mu$ with their approximations in terms of elementary functions, $\tilde{I}_\mu$ and $\tilde{K}_\mu$, the integrals in equation (59) still cannot be

evaluated exactly. This requires further simplifications based on the specific behavior of these functions within the integration domain. Of special importance is the fact that the real part of the parameter $\eta = \mu - 1/2$, according to equations (49) and (60) is positive and large. This leads to the fact that for sufficiently large $|\eta|$, the integrands in equation (59) mostly concentrate either near the upper integration limit [for the first term in the RHS of equation (59)] or near the lower integration limit (for the second term there).

This suggests the following procedure to further approximate the analytic evaluation of the integrals in equation (59). The entire $\delta$-function-like integrands can be locally approximated near the integration limit at $z = x$ by fictitious exponential functions that can be integrated analytically. The fictitious exponents are determined by the corresponding logarithmic derivatives of those integrands at $z = x$. Within the main bulk of each function (in close proximities to the corresponding integration limits), the integrands and the fictitious exponential functions are reasonably close to each other. Outside those bulks they may differ significantly, but the contributions of the corresponding integration domains to the total integrals are relatively small. Then for each fictitious exponential function, the other integration limit can be extended to the corresponding infinity, so that the effect of that integration limit effectively disappears. As a result of the entire approximations, we might expect that the integral of each fictitious exponential function will be reasonable close to the corresponding exact integral in equation (59). This procedure becomes asymptotically accurate as $|\eta| \to \infty$. For the finite value of $|\eta|$, the acceptable accuracy of the final results may require somewhat larger values of $|\eta|$ than needed for approximating the modified Bessel functions (see above) $|\eta| \gtrsim 5$, but mostly also well within the limits of the above physical estimates, $|\eta| \approx |\mu| \gtrsim 25$.

The described procedure of this local approximate integration leads to a remarkable consequence: the result of each integration becomes proportional to the integrand value at the upper (or lower) integration limit $x$ multiplied by a simple elementary function of $x$. As a result, if the exact pre-integral modified Bessel functions are also replaced by the corresponding approximate functions $\tilde{I}_\mu(z/2)$ or $\tilde{K}_\mu(z/2)$, then in each integration term these two functions will merge into a common factor $\tilde{I}_\mu(z/2)\,\tilde{K}_\mu(z/2)$, for which we can apply the simple relation given by equation (B4). After making other minor approximations (see below), the summation of the two terms yields the function $L(x)$ divided by a simple polynomial. This will allow us to obtain the final result in terms of the error functions (or, equivalently, the $Z$-functions).

Bearing in mind equations (B2) and (50b), we see that each integrand in the RHS of equation (59) is proportional to a fraction in the $\eta$-th power, an exponential function, the pre-exponential multiplier $(4\eta^2+z^2)^{-\frac{1}{4}}$, a factor $\sqrt{z}$ preceding the modified Bessel functions in equation (59), as well as $z^{5/4}$ and powers of $z$ stemming from $L(z)$. Within the narrow bulk of the integrands, the $\eta$-th powers and exponential functions play the major role, while the other, much slower varying, functions can be pulled out of the integrals by approximately replacing their integration variable $z$ with the corresponding integration limit $x$ (this will lead to an additional small loss of accuracy).

Temporarily denoting the essential functions in the corresponding integrals by $f(z)$ and $g(z)$, according to the procedure described above, we obtain

$$\int_0^x f\left(z\right)dz \approx \frac{f\left(x\right)}{\frac{d}{dx}\ln f\left(x\right)}, \qquad \int_x^\infty g\left(z\right)dz \approx \frac{g\left(x\right)}{-\frac{d}{dx}\ln g\left(x\right)}. \tag{B14}$$

To calculate the logarithmic derivatives, we temporarily disregard in $f_n(x)$ and $g_n(x)$ all multipliers that exert no effect on the logarithmic derivatives and leave only the essential functions:

$$f\left(x\right) = \left(\frac{\sqrt{4\eta^2+x^2}-2\eta}{x}\right)^{\eta+\frac{1}{2}} \exp\left(\frac{\sqrt{4\eta^2+x^2}}{2}-\left(\alpha+\lambda\right)x\right), \tag{B15a}$$

$$g\left(x\right) = \left(\frac{\sqrt{4\eta^2+x^2}+2\eta}{x}\right)^{\eta+\frac{1}{2}} \exp\left(-\frac{\sqrt{4\eta^2+x^2}}{2}-\left(\alpha+\lambda\right)x\right). \tag{B15b}$$

Then the corresponding logarithmic derivatives become

$$\frac{d}{dx}\ln f\left(x\right) = \frac{4\eta^2+2\eta+x^2-2\left(\alpha+\lambda\right)x\sqrt{4\eta^2+x^2}}{2x\sqrt{4\eta^2+x^2}}, \tag{B16a}$$

$$-\frac{d}{dx}\ln g\left(x\right) = \frac{4\eta^2+2\eta+x^2+2\left(\alpha+\lambda\right)x\sqrt{4\eta^2+x^2}}{2x\sqrt{4\eta^2+x^2}}. \tag{B16b}$$

Our final approximation is neglecting in the numerators of equations (B16a) and (B16b) the term $2\eta$ compared to the largest constant term $4\eta^2$. While leading to an additional relatively small loss of accuracy, this neglect will significantly simplify both equation (B16) and all expressions that will follow,

$$\frac{d}{dx}\ln f\left(x\right) \approx \frac{\sqrt{4\eta^2+x^2}-2\left(\alpha+\lambda\right)x}{2x}, \tag{B17a}$$

$$-\frac{d}{dx}\ln g\left(x\right) \approx \frac{\sqrt{4\eta^2+x^2}+2\left(\alpha+\lambda\right)x}{2x}. \tag{B17b}$$

These expressions lead to the following approximate integral evaluations:

$$\int_0^x \sqrt{z}\,\mathrm{I}_\mu\left(\frac{z}{2}\right)L(z)dz \approx \frac{2x^{\frac{3}{2}}\tilde{I}_\mu(x/2)L(x)}{\sqrt{4\eta^2+x^2}-2\left(\alpha+\lambda\right)x},$$

$$\int_x^\infty \sqrt{z}\,\mathrm{K}_\mu\left(\frac{z}{2}\right)L(z)dz \approx \frac{2x^{\frac{3}{2}}\tilde{K}_\mu(x/2)L(x)}{\sqrt{4\eta^2+x^2}+2\left(\alpha+\lambda\right)x}.$$

Now we apply these expressions to the full solution given by equation (59), where we replace the exact pre-integral functions $\mathrm{I}_\mu(x/2)$ and $\mathrm{K}_\mu(x/2)$ by their approximations $\tilde{I}_\mu(x/2)$ and $\tilde{K}_\mu(x/2)$. As a result, in each of the two terms, the two different modified Bessel functions – one in the integrand and the other in the pre-integral factor – combine together to the the common factor $\tilde{I}_\mu(x/2)\,\tilde{K}_\mu(x/2)$. After applying equation (B4), we arrive at the following approximate solution for $\tilde{w}(x)$:

$$\tilde{w}(x) \approx -\frac{x^2 L(x)}{\mu^2 + [1/4 - (\alpha+\lambda)^2]x^2} = \frac{(N_0\xi^2 + N_1\xi + N_2)e^{-\alpha x}x^{-\gamma}e^{-\xi}\phi_{\omega\mathbf{k}}}{\mu^2 + [1/4 - (\alpha+\lambda)^2]x^2}, \tag{B18}$$

where we have finally neglected the difference between $\eta$ and $\mu$. Then the original function $f^e_{\omega\mathbf{k}}(\xi)$ becomes

$$f^e_{\omega\mathbf{k}}(\xi) = e^{\alpha x}x^{\gamma}\tilde{w}(x) \approx \frac{N_0\xi^2 + N_1\xi + N_2}{\mu^2 + [1/4 - (\alpha+\lambda)^2]x^2}\; e^{-\xi}\phi_{\omega\mathbf{k}}.$$

The validity of the approaches used above, including the approximate integrations, can be additionally verified by applying them to the exact integral relations given by equation (58).

Neglecting in the expression for $\mu$ the term $9/16$ compared to the large parameter $\mathcal{P}$, see equation (49), and using the expressions of $\lambda = \xi/x$ and $\alpha$ given by equation (48), we obtain

$$\mu^2 \approx \mathcal{P} = \frac{\Omega_e^2 V_{Te}^2}{4\nu_0 a_{e0}^2}\left(3i\Delta_{\omega\mathbf{k}} + \frac{k_x^2 V_{Te}^2}{\nu_0}\right),$$
$$\alpha + \lambda = \frac{1}{2\sin\varphi}\left(i\cos\varphi + \frac{a_{e0}}{k_y V_{Te}^2}\right), \qquad x^2 = \frac{4k_y^2 V_{Te}^4 \sin^2\varphi}{a_{e0}^2}\;\xi^2,$$

so that the denominator in equation (B18) becomes

$$\mu^2 + \left[\frac{1}{4} - (\alpha+\lambda)^2\right]x^2$$
$$\approx \frac{\Omega_e^2 V_{Te}^2}{4\nu_0 a_{e0}^2}\left(3i\Delta_{\omega\mathbf{k}} + \frac{k_x^2 V_{Te}^2}{\nu_0}\right) + \frac{k_y^2 V_{Te}^4 \xi^2}{a_{e0}^2}\left(1 - \frac{2ia_{e0}\cos\varphi}{k_y V_{Te}^2} - \frac{a_{e0}^2}{k_y^2 V_{Te}^4}\right)$$

Recalling the definitions of the parameters $N_{0,1,2}$ given by equation (44), we obtain

$$f^e_{\omega\mathbf{k}}(\xi) \approx \frac{n_0 e^{-\xi}}{(2\pi)^{3/2} V_{Te}^3}$$
$$\times \frac{4\left(k_y^2 V_{Te}^2 - 2ik_y a_{e0}\cos\varphi\right)\xi^2 + 20ik_y\xi a_{e0}\cos\varphi + k_x^2 V_{Te}^2(\Omega_e^2/\nu_0^2)}{4\left(k_y^2 V_{Te}^2 - 2ik_y a_0\cos\varphi - a_0^2/V_{Te}^2\right)\xi^2 + (3i\Delta_{\omega\mathbf{k}}\nu_0 + k_x^2 V_{Te}^2)(\Omega_e^2/\nu_0^2)}\;\phi_{\omega\mathbf{k}}. \tag{B19}$$

In the $a_{e0} \to 0$ limit, this expression reduces to equation (45) which was obtained by neglecting in equation (43) all derivatives of $f^e_{\omega\mathbf{k}}(\xi)$ with respect to $\xi$.

**Acknowledgments**

At various stages of this research, it was supported by NSF Grants 1755350, 1007789, 0442075, 2301644, 9812679, 0332354, 2350366, and 2531480; NASA Grants 80NSSC21K1322, 80NSSC19K0080, and 80NSSC25K7753.

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